Multidimensional Lattice Walk Enumeration Through Coefficient Extraction Operators

by

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Abstract

In this thesis we investigate the application of an operator paradigm to the enumeration of lattice walks in multiple dimensions, in order to automatize some of the procedures involved in finding generating function Q for the walks. In particular, given a lattice walk model in some number of dimensions, with some regional constraints and interactions, and arbitrary step set, we may write a functional equation $(1 - t\Gamma S)Q = q$, where Γ is some coefficient extraction operator. We also use the operator perspective to unify and extend the Algebraic and Obstinate kernel methods. Most of the thesis will deal with lattice walk models with small steps confined to the non-negative orthant and the end of the thesis will cover the generalizations.

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Preface

This work intends to establish a new framework of notation and tools for the enumeration of random walks with or without interactions, and similar problems. The notation makes it much easier to explore the symmetries of the problem and write down summation formulas where it was not possible to do so before (with a simplified notation). The tools make the derivation of functional equations for various problems an almost trivial matter. The real power is in combining both approaches to gain insights into the generating function world of enumerating random walks on constrained regions in multidimensional spaces.

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Chapter 1

Introduction

The subject of the enumeration of lattice walk models is a classic combinatorial one, but it has seen a flurry of research in recent years, including the full classification of the generating functions for models with small steps on the quarter plane, and partial classification of their three-dimensional equivalent. There are many other formulations, including self-avoiding walks, walks with interactions with the lattice and with themselves, and so on. Furthermore, there are ideas about automatizing the derivations of equations and generating functions. It appears that we are approaching a more general theory that contains all of these disparate topics under one umbrella. In this thesis, we will provide more evidence that such a theory may exist and we will do so by recasting several methods and stages of analysis into a language of operators. Furthermore, the change of perspective allows for the quick formulation of functional equations for a broad class of lattice walk models. This makes it possible to automatically derive explicit functional equations for models that are nice enough.

Apart from being an interesting mathematical construct, lattice walk models enjoy some applications outside of mathematics. There are parts of the physical world that can be modelled via walk models — for example, various proteins and polymers, although in those cases the models can be more complicated. There is a rich theory of continuous and discrete lattice walks. Specifically, in lattice walk enumeration, we count the number of walks with certain characteristics subject to a multitude of constraints. The topic often leads to the use of generating functions as a powerful tool in the analysis. In the end, we gain insight into what makes certain lattice walk models easier and others more difficult. What follows is a brief recollection of the recent history and state of the art of the problem.

1.1 Recent History

One may start with lattice walks that do not have any constraints — the walks may go anywhere, while taking whatever steps are available. The enumeration does not even need generating functions — just construct all possible sequences of steps from a set $\mathcal{S}$ and then you will find that the number of walks of length n is just $|\mathcal{S}|^n$. In more complicated models we make use of the generating function. To help us, we define a *step polynomial* S keeping inventory of the step set $\mathcal{S}$.

The next step up in difficulty is to consider walks on half-spaces, where walks are not allowed to venture past points with one coordinate equal to zero. These cases are not as simple to solve but various methods allow us to solve them. Check [3–6] for work in one or more dimensions. It turns out that the generating functions we obtain are algebraic. Naturally, we are led to consider cases where the region is an intersection of half-spaces. Even for two dimensions, the problem has generated a rich theory with many different cases — ometimes the generating functions are not even D-finite, let alone algebraic. The main object of study in these cases is a group G of bi-rational transformations preserving the polynomial $K = 1 - tS$, called the *kernel* of the model. For a classification of two-dimensional walks with small steps in the quarter plane, see [8].

When G is finite, in the 23 non-equivalent cases, we obtain a D-finite generating function, as can be seen in the classifications [7–10] for some specific cases. The main tool in the study of the finite group cases is the *algebraic kernel* method where we sum different functional equations produced by group actions by elements of G. This is done to cancel the majority of the terms in the functional equation and allow us to find out information about the model directly. We can also look beyond small steps. For example, see [11] for a treatment of models with finite groups and repeated steps.

There are 56 non-equivalent cases of quarter plane walks with an infinite group. For 51 of them, the generating function is not D-finite [13] . The remaining five cases required a fair bit more work [14, 15]. The main tool used to tackle these five cases is the *iterated kernel* method where we use properties of the roots of the kernel K to show the generating function is not D-finite.

After the classification was completed, we were naturally led to investigate higher-dimensional variants of this problem. In three dimensions we have the recent partial classification [18] of lattice walk models with up to six steps, where the authors extend the algebraic kernel method to handle the majority of finite group cases encountered. However, we do not need to confined our analysis on the positive orthant. For example, see [16, 17] for analysis of models where the region of interest is a Weyl chamber.

Another generalization is to consider walks with interactions and/or self-avoiding walks (or SAWs). These two types of generalizations have important applications in physics, because they can be used to model polymers, proteins and other chains. For some research on SAWs with various

interactions, see [19–25]. The topic of SAWs does not lend itself as nicely to using generating functions like the ones considered so far, so we make simplifications. First, we can consider lattice models like before but with some interactions. For example, see [26, 27] for walks with interactions in an octant. See [28–30] for walks with interactions on the plane, where the authors used the *obstinate* kernel method to find relationships between walks with interactions and walks without them.

There has also been progress towards the automatization of the classification of random walk models. For example, see [34] for some classification and [35] for analysis of some of the computational challenges involved in automation. There is also automation in the derivation of asymptotics for some models, for example [37, 38]. A lot of the works mentioned so far have also discussed asymptotics.Since the enumeration of lattice walks is a combinatorial subject, there are connections with automation in combinatorics too. For examples, see [31, 32] for details on symmetric functions, and [33] for a holonomic approach to combinatorics.

Since there is such a rich theory of lattice walks, we hope that there is some kind of general theory unifying the various kernel methods and approaches to solving lattice walk models. By recasting some of the analysis in terms of operators, this thesis provides further evindence that such a theory may exist. The optimistic thing would be to suppose that a general algorithm exists that takes an arbitrary lattice walk model, in arbitrarily many dimensions, with arbitrary finite step sets and with arbitrary constraints; and outputs various data about the model: functional equation for the generating function, a solution and information about the model's D-finiteness. It is only logical that, if such an algorithm were to exist, it would not have difficulty with the particulars of the lattice walk model. While this kind of algorithm may be too strong to envision, it may be the case that there are atrocious lattice walk models that we cannot solve, the idea of the existence of such an algorithm has been a strong motivator in our pursuit of generalizations. This thesis succeeds in demonstrating that the first task is achievable. For a vast assortment of lattice walk models, we can almost-automatically write down a functional equation for its generating function counting length, position and interactions. The equation looks like

$$(1 - t\Gamma S)Q = q\,, \tag{1.1}$$

where Γ is an operator on the variables of Q, S is the Laurent step polynomial keeping inventory of the steps in polynomial form, and q is some Laurent polynomial, serving as an initial condition.

1.2 Roadmap

In this thesis we will work towards a few different goals.

1. Begin the classification of walks with small steps on the quarter plane but with interactions.

2. Extend the algebraic and obstinate kernel methods to models with interactions and higher dimensions.

3. Automatically derive functional equations for higher-dimensional models with various restrictions

In Chapter 2, we will study the enumeration of one-dimensional lattice walk models with small steps and interactions with the origin. We will derive explicit generating functions for all models and the analysis will show the first examples of the methods that we will use in later chapters. For example, we will see how operators make the derivations easier.

In Chapter 3, we will us the powerful operator perspective to derive functional equations for the generating function enumerating higher-dimensional lattice walk models with small steps and interactions with the bounding hyperplanes. In the latter part of the chapter, we will focus on the classification of two-dimensional models and some first solutions. This chapter contains most of the definitions relevant in the analysis.

In Chapter 4, we will recast some of the algebraic kernel analysis in terms of operators and extend it to handle lattice walk models with interactions. We will focus on two-dimensional models with finite groups and see that if the method extends, we will have some tighter constraints. In the end, we are able to handle highly-symmetric models with group of order four, half of the models with group of order six, and all models with group of order eight.

In Chapter 5, we will investigate the obsinate kernel method for the two-dimensional models with interactions and finite groups. Again, we will recast the method in operator notation. For the interacting cases, the algebraic and the obstinate kernel methods are very similar. They can both be understood in the language of weighted orbital operators.

In Chapter 6, we will finally investigate mode general random walks models. That is, we will find an operator functional equation for a large class of random walk models — those with general regional and marking constraints, general step sets and general starting conditions. The difficulty lies in finding a good representation for the operators involved.

Chapter 2

One-dimensional Lattice Walk Models

2.1 Derivation of the Functional Equation

Before we tackle the higher-dimensional lattice walk problems, it will be instructive to visit the one-dimensional case which will help illustrate the methods and ideas that we will employ throughout this work. We begin with one-dimensional lattice walk models with smalls steps, where the size of each step is at most one unit in each direction. Here the step set is $\mathcal{S} \subseteq \{\text{-}1, 0, 1\}$, which can also be represented via the step polynomial

$$S = S_{\text{-}1}\overline{x} + S_0 + S_1 x\,. \tag{2.1}$$

Here S_k is 1 or 0 if k is in $\mathcal{S}$ or not, respectively.

First, we will look at a basic way to generate lattice walks and then we will look at how we can encapsulate it into an algebraic method. We start with the empty walk. We can grow it by doing one of three actions: step to the left, step to the right, or stay in one spot. Stepping to the left takes us to the negative axis and we do not want any walks that go there, so we ignore it. Thus we may extend the walk by stepping to the right or staying in one spot. These new walks have length one and they can be extended by each of the three actions. We can easily keep track of the visits to the origin - each time the walks stay or step on the origin, we increment a counter. We may continue the process as long as we wish, always removing any walks that end on negative coordinates.

Ultimately, it is easier to work with polynomials that encapsulate the important features in a smaller package. Also, working with polynomials and power series allows us to use the powerful methods of generating functions to peek into the inner workings of lattice walk models. To set up for those methods, we shall keep track of each walk's length via powers of t, ending position via

powers of x, and visits to the origin with powers of a. We can perform the previous analysis in the language of monomials. We start with the empty walk of length zero. It corresponds to the monomial $1 = a^0x^0t^0$. Extending the walk corresponds to a multiplication by $\overline{x}$, x or 1, depending on whether we step to the left or to the right, or we stay on the same coordinate. The two resulting walks will have monomials xt and at, respectively. We have removed the walk that would have had a monomial $\overline{x}t$ since it ends on a negative coordinate. The next iteration has walks with monomials at^2, xt^2, x^2t^2, a^2t^2, axt^2. We can continue the procedure as long as we wish.

Of course, it would be difficult to work with individual lattice walks and it would be better to work with the whole generating function enumerating the walks. Suppose that the generating function for walks of length k is Q_kt^k. Then the full generating function is

$$Q = \sum_{k=0}^{\infty} Q_k t^k \,. \tag{2.2}$$

The key idea is to attempt to work with all the walks at the same time and extend all of them simultaneously. If we are successful, we will be able to encapsulate the process in an equation for Q and then manipulate it algebraically to find out more about the lattice walk model. We start by writing a functional equation relating Q_{k+1} and Q_k. All the walks of length k are encoded in Q_kt^k. To extend all of them by one step, we multiply the function by tS. By the distributive property, the product of the two sums wil lead to all possible multiples of monomials for lattice walks of length k and monomials for possible steps that we can take. However, some of the resulting monomials may have a negative power of x, corresponding to a walk ending on a negative coordinate. Here is another key idea. To remove the monomials with negative powers of x, we will use some kind of operator Γ, but we will worry about its representation later. For now, we have the iteration

$$Q_{k+1}t^{k+1} = \Gamma(tSQ_kt^k) = \Gamma(SQ_k)t^{k+1} \,. \tag{2.3}$$

In other words, $Q_{k+1} = \Gamma(SQ_k)$ for all $k \geqslant 0$. We may sum both sides of the equation for $k \geqslant 0$ to find

$$\sum_{k=0}^{\infty} Q_{k+1}t^{k+1} = \sum_{k=0}^{\infty} \Gamma(SQ_k)t^k \,. \tag{2.4}$$

We are working with formal series, after all, and we are looking for the easiest approach right now. It will be safe to assume that Γ does not interact with t and that we may factor Γ outside of the summation. After all, we are looking for an operator that will be able to work with the whole

generating function. With these assumptions in hand, we may manipulate the equation as follows.

$$Q - Q_0 = t\Gamma S \sum_{k=0}^{\infty} Q_k t^k \tag{2.5}$$

$$Q = 1 + t\Gamma(SQ)\,. \tag{2.6}$$

We may rewrite the functional equation as

$$(1 - t\Gamma S)Q = 1\,, \tag{2.7}$$

so as to highlight the operator nature of the equation.

We just need to find a suitable representation for the operator Γ and will have completed the analysis. The operator needs remove all monomials with negative powers of x. However, the lowest order of x we may attain in SQ_k or SQ is -1. That is, all valid walks have power of x that is greater or equal to zero, and extending them by taking one step can only go as far as -1 on the axis. It is enough to take out the $\overline{x}$ term from the generating function, and we can achieve this using coefficient extractions. Applying the operator $\overline{x}[\overline{x}]$ on a function extracts the $\overline{x}$ term from the function. The operator that eliminates such a term from a function is just

$$E = 1 - \overline{x}[\overline{x}]\,, \tag{2.8}$$

where 1 is the identity operator. We can also find an operator that marks walks ending on the origin. It is enough to multiply their monomials, which have no x, with the label a. We can be represent this action via the operator

$$L = 1 + (a-1)[x^0] = 1 + b[x^0]\,, \tag{2.9}$$

where $b = a - 1$. It takes a function and adds to it a $(a-1)$ multiple of the function's x^0 term. The final function has a x^0 term with coefficient $1 + (a-1) = a$. Finally, to find Γ it is enough to multiply out the operators L and E as

$$\begin{aligned} \Gamma = LE &= (1 - \overline{x}[\overline{x}])(1 + b[x^0]) \\ &= 1 + b[x^0] - \overline{x}[\overline{x}] - \overline{x}[\overline{x}]b[x^0] \\ &= 1 + b[x^0] - \overline{x}[\overline{x}]\,. \end{aligned} \tag{2.10}$$

We would have obtained the same result if expaned the product EL instead. The last term vanished because once we have made the first extraction using $[x^0]$, that term no longer contains a variable x,

and the $[\overline{x}]$ extraction picks out nothing. Here is an example that illustrates how the operators work in an actual problem.

Example 2.1. Suppose that $S = x + \overline{x}$. Starting with $Q_0 = 1$, we have

$$Q_1 = LES(1) = (1 + b[x^0] - \overline{x}[\overline{x}])(x + \overline{x}) = x + \overline{x} - \overline{x} = x \tag{2.11}$$

$$Q_2 = LES(Q_1) = (1 + b[x^0] - \overline{x}[\overline{x}])(x^2 + 1) = x^2 + 1 + b = a + x^2 \tag{2.12}$$

$$\begin{aligned} Q_3 = LES(Q_2) &= (1 + b[x^0] - \overline{x}[\overline{x}])(a\overline{x} + ax + x + x^3)(?) \\ &= a\overline{x} + ax + x + x^3 - a\overline{x} = (1 + a)x + x^3\,. \end{aligned} \tag{2.13}$$

Therefore,

$$\begin{aligned} Q &= Q_0 + Q_1 t + Q_2 t^2 + Q_3 t^3 + \cdots \\ &= 1 + xt + (a + x^2)t^2 + (ax + x + x^3)t^3 + \cdots\,. \end{aligned} \tag{2.14}$$

Note that if we apply Γ to tSQ, we obtain Q again but with the first term, $Q_0 = 1$, missing.

The theory so far is consistent with the present methods for finding functional equations for the generating functions for lattice walk models because in the end we reach the same functional equations. However, the perspective is different and we can go further than before. To show the consistency with the theory so far, we will expand the operator form and write down a functional equation that does not involve operators. We make a change of notation for the rest of the chapter in order to reduce clutter. Let $Q_k = [x^k]Q$. We start with

$$\begin{aligned} [x^0]SQ &= [x^0]((S_{-1}\overline{x} + S_0 + S_1 x)Q) \\ &= (S_{-1}x[x] + S_0\overline{x}[\overline{x}] + S_1\overline{x}^2[\overline{x}^2])Q \\ &= S_{-1}Q_1 + S_0 Q_0\,, \end{aligned} \tag{2.15}$$

where last equality follows since $Q_{-1} = 0$ since that would include walks ending on $\overline{x}$. A similar computation shows that $[\overline{x}]SQ = S_{-1}Q_0$ (?) and we have find the functional equation

$$\begin{aligned} 1 &= Q - tLE(SQ) \\ &= Q - t(SQ + b[x^0]SQ - \overline{x}[\overline{x}]SQ \\ &= Q - t(SQ + b(S_{-1}Q_1 + S_0 Q_0) - \overline{x}(S_{-1}Q_0))\,. \end{aligned} \tag{2.16}$$

We can rewrite it in the following form

$$(1 - tS)Q - t(bS_0 - \overline{x}S_{-1})Q_0 - btS_{-1}Q_1 = 1\,. \tag{2.17}$$

Here $Q = Q(x)$, $Q_0 = Q(0)$ and $Q_1 = Q'(0)$. It turns out that the Q_1 term is not needed. To see this, first eliminate the Q_1 term by applying $[x^0]$ to both sides of the equation.

$$\begin{aligned} 1 &= Q_1^0 - t(S_{-1}Q_1 + S_0Q_0) - tbS_0Q_0 - btS_{-1}Q_1 \\ &= (1 - atS_0)Q_0 - atS_{-1}Q_1 \,. \end{aligned} \tag{2.18}$$

We can solve this equation for Q_1, that is,

$$tS_{-1}Q_1 = \overline{a}((1 - atS_0)Q_0 - 1)\,, \tag{2.19}$$

and substitute the result into the first functional equation. For now, we do not worry about the possibility of $S_{-1} = 0$, because it leads to the same conclusion anyway. Upon simplifaction we use the fact that $1 - c = 1 - (1 - \overline{a}) = \overline{a}$ the master functional equation.

$$(1 - tS)Q - (c - t\overline{x}S_{-1})Q_0 = \overline{a}\,. \tag{2.20}$$

It turns out that the back-substitution is unnecesary. This is a good thing because in higher dimensions we would have to perform an exponential number of back-substitutions to obtain such an equation: 2^d substitutions for a d-dimensional problem. Previous approaches in [21, 30] have only dealt with two-dimensional problems that have only a few back-substitutions. It is the use of operators that allows us to circumvent the problem altogether. To see why, consider the action of the marking operator L. To mark a monomial, we just multiply it by a. Therefore, to undo the marking we only need to multiply the monomial by $1/a$. That is, L has an inverse and we can write it down immediately. We have

$$L^{-1} = 1 + \left(\frac{1}{a} - 1\right)[x^0] = 1 - c[x^0]\,. \tag{2.21}$$

If we apply this operator to the functional equation, the right-hand side is simply $L^{-1}(1) = 1 - c = \overline{a}$ and the left-hand side is

$$\begin{aligned} L^{-1}(1 - tLES)Q &= L^{-1}Q - tESQ \\ &= (Q - cQ_0) - t(SQ - \overline{x}S_{-1}Q_0) \\ &= (1 - tS)Q - (c - t\overline{x}S_{-1})Q_0\,. \end{aligned} \tag{2.22}$$

We have a great shortcut, indeed.

Proposition 2.2. *Let $S = S_{-1}\overline{x} + S_0 + S_1x$ be the Laurent step polynomial for a one-dimensional*

lattice walk model restricted to the non-negative axis. Then the generating function Q satisfies the functional equations

$$(L^{-1} - tES)Q = \overline{a}\,, \tag{2.23a}$$

$$(1 - tS)Q - (c - t\overline{x}S_{-1})Q_0 = \overline{a}\,. \tag{2.23b}$$

The first one we call an operator equation, and the second, master functional equation.

The operator notation/perspective is powerful. With it, we can derive functional equations for many lattice walk models in arbitrary dimensions with ease. Usually, derivations have employed inclusion-exclusion and careful book-keeping (Lemma 4 in [8, p. 10] ,[15, p. 58]) to write master functional equations, or their precursors, in a direct manner. Unfortunately, that approach is difficult to generalize to arbitrary dimensions and more dificult restrictions/marking.

We can also attend to coefficient extraction in a neater way. Instead of extracting $[x^k]$ from the master functional equation, it is better to first go through the operator equation, since $[x^k]$ interacts with L and E in a nice way. We have

$$[x^0]L^{-1} = [x^0] - [x^0]c[x^0] = [x^0] - c[x^0] = \overline{a}[x^0] \tag{2.24}$$

$$[x^0]E = [x^0] - [x^0]\overline{x}[\overline{x}] = [x^0]\,. \tag{2.25}$$

When we extract the x^0 term from the functional equation, we have

$$\begin{aligned}[x^0]\overline{a} = \overline{a} &= [x^0](L^{-1} - tES)Q \\ &= (\overline{a}[x^0] - t[x^0]S)Q \\ &= [x^0](\overline{a} - tS)Q\,.\end{aligned} \tag{2.26}$$

Therefore, we have a functional equation for the first coefficient

$$[x^0](1 - atS)Q = 1\,. \tag{2.27}$$

We can perform the extraction immediately and it simplifies to

$$(1 - atS_0)Q_0 - atS_{-1}Q_1 = 1\,. \tag{2.28}$$

Extracting higher coefficients is not difficult either. If $k \geqslant 1$, we have

$$[x^k]L^{-1} = [x^k] - [x^k]c[x^0] = [x^k]\,, \tag{2.29}$$

$$[x^k]E = [x^k] - [x^k]\overline{x}[\overline{x}] = [x^k]\,, \tag{2.30}$$

and

$$[x^k](1 - tS)Q = 0\,. \tag{2.31}$$

We expand this term to find

$$\begin{aligned} Q_k - t(S_{-1}Q_{k+1} + S_0Q_k + S_1Q_{k-1}) &= 0 \\ (1 - S_0t)Q_k - t(S_{-1}Q_{k+1} + S_1Q_{k-1}) &= 0\,. \end{aligned} \tag{2.32}$$

Proposition 2.3. *Let $S = S_{-1}\overline{x} + S_0 + S_1x$ be the Laurent step polynomial for a one-dimensional lattice walk model. Let $Q_k = [x^k]Q$ be the generating functions for the walks ending at coordinate k. For $k \geqslant 1$, these functions satisfy the equations*

$$(1 - atS_0)Q_0 - atS_{-1}Q_1 = 1\,, \tag{2.33}$$

$$(1 - S_0t)Q_k - t(S_{-1}Q_{k+1} + S_1Q_{k-1}) = 0\,. \tag{2.34}$$

2.2 Solving Functional Equations For Various Step Sets

If a model has steps which can never be taken, we might as well remove them, and obtain a simpler model. The generating function will be the same in each case and we can identify three major cases: $S = \delta$ or $S = \delta + x$ or $S = \delta + \overline{x} + x$. We will handle the first two cases simultaneously, and the third one, later.

2.2.1 The $S = \delta$ or $S = \delta + x$ case

The condition allows us to find Q_0 directly from the $[x^0]$ equation.

$$Q_0 = \frac{1}{1 - a\delta t}\,. \tag{2.35}$$

We can substitute Q_0 into the master functional equation.

Proposition 2.4. *Let $S = \delta$ or let $S = \delta + x$. Then the full generating function is given by*

$$Q(x, a; t) = \frac{\overline{a}}{1 - tS}\left(1 + \frac{b}{1 - a\delta t}\right) = \frac{1 - \delta t}{(1 - a\delta t)(1 - tS)}\,. \tag{2.36}$$

If we let $a = 1$ we find that

$$\widehat{Q} = \frac{1}{(1 - tS)}\,, \tag{2.37}$$

and therefore we have the linear relation

$$(1 - a\delta t)Q - (1 - \delta t)\widehat{Q} = 0\,, \tag{2.38}$$

We can write it as $B((1-a\delta t)Q) = 0$ using the operator $B = 1-[b^0]$. There is a similar relationship for Q_0. Extracting the x^0 term from the last equation leads to

$$B((1 - a\delta t)Q_0) = 0 \tag{2.39}$$

or in other words,

$$Q_0 = \frac{1 - a\delta t}{1 - \delta t}\widehat{Q}_0\,. \tag{2.40}$$

Proposition 2.5. *Let $B = 1 - [b^0]$ and let Q be the generating function for a lattice walk model with $S = \delta$ or $S = \delta + x$. Then Q and Q_0 satisfy the functional equations*

$$B((1 - a\delta t)Q) = 0 \quad \textit{and} \quad B((1 - a\delta t)Q_0) = 0\,. \tag{2.41}$$

In other words,

$$Q = \frac{1 - a\delta t}{1 - \delta t}\widehat{Q} \quad \textit{and} \quad Q_0 = \frac{1 - a\delta t}{1 - \delta t}\widehat{Q}_0\,. \tag{2.42}$$

Even such small examples already show connections between Q and $\widehat{Q}$. It will be interesting to find such relationships for higher-dimensional lattice walk models.

2.2.2 The $S = \delta + \overline{x} + x$ case

In this case, we will apply the obstinate kernel method, as is done in the two-dimensional cases in Secton 4.2 of [30, p. 12] and Section 4.1 of [21, p. 13]. It requires finding an x value that makes the kernel $K = 1 - tS$ vanish in the functional equation

$$(1 - tS)Q - (c - t\overline{x})Q_0 = \overline{a}\,. \tag{2.43}$$

The equation $1 - tS = tx^2 - (1 - \delta t)x + t = 0$ is quadratic with two roots that are functions of t. We have to pick the one that has no singularity at $t = 0$. The roots are

$$\sigma = \frac{1 - \delta t - \sqrt{(1 - \delta t)^2 - 4t^2}}{2t} \quad \text{and} \quad \sigma^* = \frac{1 - \delta t + \sqrt{(1 - \delta t)^2 - 4t^2}}{2t}\,, \tag{2.44}$$

and they multiply to 1, so $\overline{\sigma} = \sigma^*$. Next, we substitute $x = \sigma$ in the equation. For now, we have to assume that the whole term KQ vanishes and we obtain

$$-(c - t\overline{\sigma})Q_0 = \overline{a}\,. \tag{2.45}$$

Now we can solve for Q explicitly. We calculate

$$Q = \frac{\overline{a}}{1 - tS}\left(1 - \frac{t\overline{x} - c}{t\overline{\sigma} - c}\right) = \frac{t}{1 - tS}\left(\frac{\overline{x} - \overline{\sigma}}{b - a\overline{\sigma}t}\right)\,. \tag{2.46}$$

Proposition 2.6. *Let $S = \delta + \overline{x} + x$. Then the generating functions for this model are*

$$Q_0 = \frac{1}{a\overline{\sigma}t - b} \quad \textit{and} \quad Q = \frac{t(\overline{\sigma} - \overline{x})}{(1 - tS)(a\overline{\sigma}t - b)}\,, \tag{2.47}$$

where $\overline{\sigma} = (1 - \delta t + \sqrt{(1 - \delta t)^2 - 4t^2})/(2t)$.

As before, we can find a linear relationship between $Q(x, a; t)$ and $Q(x, 1; t)$. Note that $(a\overline{\sigma}t - b)Q_0$ and $(a\overline{\sigma}t - b)Q$ are functions that do not depend on a. Therefore, they are annihilated by the operator B.

Proposition 2.7. *Let $S = \delta + \overline{x} + x$ and $f = a\overline{\sigma}t - b$. Then $B(fQ) = B(fQ_0) = 0$.*

We can also look at the problem from a different angle - that of the algebraic kernel method used extensively in the quarter plane model classification [8]. We find a polynomial $f(a, t)$, for which $B(fQ_0) = c$. The analysis will serve to illustrate the group approach together with the kernel method. Note that S, and therefore K, is invariant under the map $* : x \mapsto \overline{x}$, and we can write a second functional equation to complement the first. That is,

$$(1 - tS)Q - (c - t\overline{x})Q_0 = \overline{a}\,, \tag{2.48}$$

$$(1 - tS)Q^* - (c - tx)Q_0 = \overline{a}\,. \tag{2.49}$$

To eliminate the Q_0 term ,we multiply the first equation by $c - tx$, the second by $c - t\overline{x}$, and subtract

one equation from the other. We divide everything by $1 - tS$ and the equation that remains is

$$(c - tx)Q(x) - (c - t\overline{x})Q(\overline{x}) = \overline{a}F\,, \tag{2.50}$$

where

$$F = \frac{t(\overline{x} - x)}{1 - tS} \tag{2.51}$$

and crucially, F does not depent on a. We extract the x^1 coefficient from the equation to acquire an equation relating Q_1 and Q_0

$$cQ_1 - tQ_0 = \overline{a}F_1\,. \tag{2.52}$$

We can set $a = 1$ here to find that

$$F_1 = -t\widehat{Q}_0\,, \tag{2.53}$$

and we can back-substitute this quantity into the original equation. After we multiply both sides by a, we find that

$$bQ_1 = t(aQ_0 - \widehat{Q}_0)\,. \tag{2.54}$$

Earlier we had the equation

$$(1 - a\delta t)Q_0 = 1 + atQ_1 = 1 + atQ_1\,, \tag{2.55}$$

which we can solve for Q_1 this time and insert the solution into the equation above.

$$b((1 - a\delta t)Q_0 - 1) = t^2a(aQ_0 - \widehat{Q}_0)\,. \tag{2.56}$$

Collecting coefficients and simplifying leads us to the equation

$$(c(1 - a\delta t) - at^2)Q_0 - (-t^2)\widehat{Q}_0 = c\,, \tag{2.57}$$

and therefore, $f(a, t) = c(1 - a\delta t) - at^2$.

Proposition 2.8. *Let $S = \delta + \overline{x} + x$ and $f = c(1 - a\delta t) - at^2$. Then $B(fQ_0) = c$.*

We obtained slightly different relationships upon applying the operator B.

2.2.3 Conclusion

In the one-dimensional case, we were able to find explicit solutions for Q_0 and Q various relationships for Q and $\widehat{Q}$, together with the operator B. In higher dimensions the analysis will be more difficult, but we will find some B-operator relationships for the functions enumerating lattice walks returning to the origin. Here is a table showing the resuls for the one-dimensional lattice walk models with small steps and interaction with the origin.

S	Q_0	Q	$f(a,t)$
δ	$\dfrac{1}{1-a\delta t}$	$\dfrac{1}{1-a\delta t}$	$1-a\delta t$
$\delta+x$	$\dfrac{1}{1-a\delta t}$	$\dfrac{1-\delta t}{(1-t(\delta+x))(1-a\delta t)}$	$1-a\delta t$
$\delta+\overline{x}+x$	$\dfrac{1}{a\overline{\sigma}t-b}$	$\dfrac{t(\overline{\sigma}-\overline{x})}{(1-t(\delta+\overline{x}+x))(a\overline{\sigma}t-b)}$	$a\overline{\sigma}t-b$

Table 2.1: Generating functions for walks with small steps on the non-negative half-line, where $B(fQ_0) = B(fQ) = 0$.

Chapter 3

Higher-dimensional Lattice Walk Models

We begin the exploration of higher-dimensional models by making some useful definitions. Here are the main ingredients for a model set in $\mathbb{Z}^d$.

- A finite set $\mathcal{S}$ of steps that we can take from a point in $\mathbb{Z}^n$ to another point in $\mathbb{Z}^n$. In general, this step set may differ from point to point; for example, a honeycomb lattice embedded in $\mathbb{Z}^2$. We will assume that there is one global set of steps for all points. The set may involve arbitrarily-many finitely-sized steps with multiplicities at least one. Its Laurent polynomial version is denoted by S.

- A region $\mathcal{D} \subseteq \mathbb{Z}^d$. This is the region in $\mathbb{Z}^d$ that the lattice walks can occupy. They are not allowed to venture outside. We have already seen half-spaces and orthants. Other interesting regions are semi-infinite strips, finite boxes and wedges.

- A subset $\mathcal{D}_0$ of $\mathcal{D}$. This is a sub-region in $\mathcal{D}$ where lattice walk interactions take place — we may track how many times lattice walks visit coordinates from this region. We may count visits to different points from $\mathcal{D}_0$ in different ways. This less-studied paradigm may involve marking visits to various places: the origin, the axes in two dimensions, various higher-dimensional hyperplanes, a cone, an infinite strip, the sides of a box and so on.

- A set $\mathcal{Q}$ of initial walks that we can evolve. This could be as simple as an empty walk or complicated as some infinite set of walks. The set has a corresponding polynomial or series q.

We will be growing lattice walks from an initial condition $\mathcal{Q}$, with steps $\mathcal{S}$, within $\mathcal{D}$ in $\mathbb{Z}^d$, while counting interactions with $\mathcal{D}_0$. Of course, $\mathcal{Q}$ may be a more general term, so we will not restrict it yet. The quadruple of sets should serve as a complete picture of a lattice walk model, so we encapsulate it in a definition.

Definition 3.1. *Let a lattice walk model $\mathcal{M}$ be characterized by its step set $\mathcal{S}$, region $\mathcal{D}$, region $\mathcal{D}_0$, and initial walk set $\mathcal{Q}$.*

Definition 3.2. *The CLASSIC d-dimensional lattice walk model is the one with small steps in the positive orthant in $\mathbb{Z}^d$. The interacting CLASSIC model is the CLASSIC d-dimensional model with interactions with the hyperplanes $x_k = 0$ bounding the positive orthant.*

[FIXME: Think hard of a better name than CLASSIC.]

Example 3.3. Consider one of the one-dimensional models from the previous chapter. This is an example of a CLASSIC one-dimensional model. Here $\mathcal{D}$ is the set of non-negative integers, $\mathcal{D}_0$ consists of the point at the origin, where the marking variable is a, $\mathcal{Q}$ consists of the empty walk. Finally, $\mathcal{S}$ is one of the subsets of $\{\text{-}1, 0, 1\}$.

Example 3.4. Consider the models with $d = 2$. Here $\mathcal{D} = \{(x_1, x_2) \in \mathbb{Z}^2 \mid x_1, x_2 \geqslant 0\}$ and $\mathcal{D}_0$ consists of marking points on the non-negative axes — x_1 with label a_2, since $x_2 = 0$ here, and x_2 with label a_1, since $x_1 = 0$. Note that the origin is marked with both labels. As in the previous example, $\mathcal{Q}$ consists of the empty walk and $\mathcal{S}$ is some set of small steps, where each step has coordinates of size at most 1.

Of course, we need to keep track of all the quantities tracking properties of the walks inside a generating functions, so we continue using Q for its name.

Definition 3.5. *Let $\mathcal{M}$ be a lattice walk model and Q its associated generating function counting lattice walks' lengths with t, ending locations with $x_1, \ldots, x_d$, and interactions with $\mathcal{D}_0$ with some multi-index variable a. Then we write*

$$Q = Q(x_1, \ldots, x_d, a; t) = \sum_{k=0}^{\infty} q_k(x_1, \ldots, x_d, a) t^k \,, \tag{3.1}$$

where the coefficients q_k are Laurent poynomials in the x and a variables.

Example 3.6. In this example we will examine the simplest kind of model — one where $\mathcal{D}$ is the whole space and there are no interactions, that is, $\mathcal{D}_0$ is empty, and finally, $\mathcal{Q}$ consists of the empty walk. Since the lattice walks may pick any step from the step set $\mathcal{S}$, the walks of length one are

enumerated via tS, the walks of length two, via t^2S^2, and so on, where $S = S(x_1, \ldots, x_d)$. Then the full generating function is

$$Q(x_1, \ldots, x_d; t) = 1 + St + S^2t^2 + \cdots = \frac{1}{1 - St}\,. \tag{3.2}$$

Example 3.7. Consider the two-dimensional CLASSIC model with $\mathcal{S} = \{\text{NE, NW, SE, SW}\}$, that is, $S = (\overline{x}_1 + x_1)(\overline{x}_2 + x_2)$. The only walk of length one is the one with step NW, enumerated via x_1x_2. To obtain the walks of length two, we grow this walk for an extra step. The walk may take any of the four steps, so the walks of length two are enumerated via a_1a_2, a_1x_2, a_2x_1 and $x_1^2x_2^2$. Therefore,

$$Q = 1 + x_1x_2t + (a_1 + x_1)(a_2 + x_2)t^2 + \cdots\,, \tag{3.3}$$

We may extract coefficients too. For example, the walks ending on the x_1 axis are given by setting $x_2 = 0$ or extracting the x_2^0 term. That is,

$$[x_2^0]Q = 1 + a_2(a_1 + x_1)t^2 + \cdots\,. \tag{3.4}$$

The walks returning to the origin are given by

$$[x_1^0x_2^0]Q = 1 + a_1a_2t^2 + \cdots\,. \tag{3.5}$$

The calculations will be difficult to handle without extra notation, so in the interest of generalizing the problem-solving strategies involved in lattice walk enumeration, we need powerful notation to handle all the intricacies of the problem. Curiously, it was the notation itself that paved the way towards generalization in our original research because it is suggestive and points towards the deeper connections between the generating functions and quantities involved. With this thesis we aim to make this kind of notation more common in higher-dimensional lattice walk analysis. Typically, in the multivariate setting we see multi-index notation where a bolded, vectorial or even supressed notation denotes the collection of variables involved **[AWKWARD]**. We would use this notation as well, but it is insufficient when working with restricted geometries for region and marking subregions. Therefore, we have the following notation.

Definition 3.8. *Let* $\mathbb{0} := (0, \ldots, 0)$ *and* $\pm\mathbb{1} := (\pm 1, \ldots, \pm 1)$ *where we have as many terms in these sequences as needed in the specific context. Also, let* $[n] = \{1, \ldots, n\}$.

Definition 3.9. *For sequences* $I = (i_1, \ldots, i_m)$ *and* $J = (j_1, \ldots, j_m)$*, and an indeterminate* x*, we*

have the monomial

$$x_I^J = x_{i_1}^{j_1} \cdots x_{i_m}^{j_m}\,, \tag{3.6}$$

with special cases $x_I^{\mathbb{0}} = 1$ *and* $x_I^{\mathbb{1}} = x_I$ *and* $x_I^{\text{-}\mathbb{1}} = 1/x_I = \overline{x}_I$.

Definition 3.10. *For sequences* $I = (i_1, \ldots, i_m)$ *and* $J = (j_1, \ldots, j_m)$*, and an indeterminate* x*, we have the coefficient extraction operator*

$$[x_I^J]F = [x_{i_1}^{j_1} \cdots x_{i_m}^{j_m}]F = F_I^J\,, \tag{3.7}$$

with special cases:

- $F_I^J = F$*, if* I *is empty.*
- $F_I^J = F_J$*, if* I *is the full set of variables ordered unambiguously.*

Example 3.11. In the previous example we had $[x_2^0]Q = Q_2^0$ and $[x_1^0 x_2^0]Q = Q_{1,2}^{0,0}$. Furthermore, we may write

$$S_1^{\text{-}1} = [\overline{x}_1](x_1 + \overline{x}_1)(x_2 + \overline{x}_2) = x_2 + \overline{x}_2 \tag{3.8}$$

and $S_{1,2}^{\pm 1,\pm 1} = 1$.

[AWKWARD]We will mainly use the notation for extractions with respect to the x variables in the generating functions but, sometimes, we will be using such a coefficient extraction with respect to other variables, most commonly t. We will never mix such extractions with this notation.

Definition 3.12. *For sequences* $I = (i_1, \ldots, i_m)$ *and* $J = (j_1, \ldots, j_m)$*, we define the term extraction operator as*

$$[[x_I^J]] = x_I^J[x_I^J]\,. \tag{3.9}$$

That is, $[[x_I^J]]F = x_I^J F_I^J$.

Because there are different variables involved, there may be some ambiguity in the result, depending on the order of extraction. We will always perform coefficient extractions with respect to the x or a variables by first expanding any series in t, and then applying the extraction on the coefficients **[AWKWARD]**. The following notation makes this precise.

Definition 3.13. *Suppose that* Γ *is an operator involving* x *and* a *variables but not* t*. We apply* Γ *on power series* $F = \sum_{k=0}^{\infty} f_k t^k$ *in the following way:*

$$\Gamma F = \sum_{k=0}^{\infty} \Gamma(f_k) t^k \,. \tag{3.10}$$

Example 3.14. In the past we have performed inclusion-exclusion to arrive at functional equations for the two-dimensional problems, but when it comes to generalizing to higher dimensions, the process is difficult and prone to error. For a view of difficulties involved in not using the operator notation, see [30], where the authors had to wrestle with case-by-case analysis. Instead, we use operators to derive the functional equations cleanly and without pain. First, we will illustrate the inclusion-exclusion with the non-interacting two-dimensional problem. This is the usual procedure carried out in problems so far, like in [8]. Similarly to the one-dimensional problem, we start with the generating function Q for the lattice walks on the quarter plane. Then we can extend the walks by one step via a multiplication by tS and remove those that step outside of the quarter plane. To perform the removal, we subtract the walks that end on coordinates $(\text{-}1, k)$ or $(k, \text{-}1)$ for integers k. However, we removed too many walks. We need to re-add those walks that end on $(\text{-}1, \text{-}1)$.

The walks that step on $(\text{-}1, k)$ are those coming from the boundary and stepping with steps in the negative x_1 direction. To compute the generating function term, we start with Q_1^0 and multiply by $t\overline{x}_1 S_1^{-1}$. The situation is similar for the $(k, \text{-}1)$ coordinates. Finally, we re-add those walks that step on $(\text{-}1, \text{-}1)$. Those are walks coming from the origin in a diagonal step, so we multiply $Q_{1,2}^{0,0}$ by $t\overline{x}_1\overline{x}_2 S_{1,2}^{-1,-1}$. We have obtained Q again but with the t^0 term missing.

This analysis leads to the following functional equation for Q

$$Q = 1 + tSQ - t\overline{x}_1 S_1^{-1} Q_1^0 - t\overline{x}_2 S_2^{-2} Q_2^0 + t\overline{x}_1\overline{x}_2 S_{1,2}^{-1,-1} Q_{1,2}^{0,0} \,, \tag{3.11}$$

which is rather large even for two dimensions.

Contrast this notation to the one in [8, p. 10]. The problem quickly runs into notational difficulties for d-dimensional problems, but that can be dealt with by writing the equation as a sum over subsets of $\{1, \ldots, d\}$, like it is done for the non-interacting case in Equation 5.1 in [15, p. 58]. This method runs into trouble when we start considering interactions. We take the functional equation for the non-interacting case and modify it to add interactions. This analysis has been done before in Section 3 of [30, p. 7] and Section 3.2 of [21, p. 11], where the authors add in terms like $(a_1 - 1)T$, where T is some expression involving extractions of Q and S, but we will not go into the messy details. Instead, we will use the operator notation to handle the problem in a much more efficient manner.

3.1 Using Operators to Derive the Functional Equation

If we use operators we can build the generalizations to higher dimensions in a straight-forward manner. As we have done before, we multiply Q by tS, so we can extend the lattice walks by one step. Then we apply an operator E, to be determined later, to remove all the walks venturing outside of the positive orthant (nonzero coordinates x_k). Finally, we apply a marking operator to this term that marks the walks on the boundary hyperplanes $x_k = 0$. Since we have extended all walks by one step, we no longer have walks of size 0. Adding that term to the expression leads us back to Q itself. That is,

$$Q = 1 + LE(tSQ) \tag{3.12}$$

and the only task remaining is to find L and E. Like in the one-dimensional problem, we start removing monomials with coordinates outside of the positive orthant. The important idea here is that, if we have operators that just remove monomials, then it does not matter in what order we remove them. Let $E_k = 1 - \overline{x}_k[\overline{x}_k]$ be the operator removing monomials with x_k coordinate equal to -1. Then we may write $E = E_1 \cdots E_d$. For example,

$$E_1E_2 = (1 - \overline{x}_1[\overline{x}_1])(1 - \overline{x}_2[\overline{x}_2]) = 1 - \overline{x}_1[\overline{x}_1] - \overline{x}_2[\overline{x}_2] + \overline{x}_1\overline{x}_2[\overline{x}_1\overline{x}_2]\,, \tag{3.13}$$

and therefore, the inclusion-exclusion is a natural consequence of how the operator is set up. Similarly, if we let $L_k = 1 + (a_k - 1)[x_k^0]$, then we may write $L = L_1 \cdots L_d$. Here order does not matter since we are marking monomials with variables a_k. It does not matter if we first multiply by a_1 and then by a_2, or vice-versa. As in the one-dimensional case, the operators L_k are invertible with inverses $L_k^{-1} = 1 - c_k[x_k^0]$, and thus, the operator L is also invertible. We can apply the operator L^{-1} to the functional equation

$$L^{-1}(1 - tLES)Q = (L^{-1} - tES)Q = L^{-1}(1) = \sum_{I \subseteq [d]} c_I = \overline{a}_{[d]}\,. \tag{3.14}$$

We can write down the following result.

Proposition 3.15. *The CLASSIC interacting d-dimensional lattice walk model has a full generating function Q satisfying the functional equations*

$$(1 - tLES)Q = 1\,, \tag{3.15}$$

$$(L^{-1} - tES)Q = \overline{a}_{[d]}\,, \tag{3.16}$$

where L, E, L^{-1} are of the form $\Gamma = \Gamma_1 \cdots \Gamma_d$.

Previously, we have had to cope with coefficient extractions and back-substitution to find the expanded form of the second functional equation [30, p. 9] and [21, p. 12]). We can skip this process by using operators. In general, the second functional equation has many less terms. To see this, we take a look at the operator $L_k E_k$. We have

$$L_k E_k = 1 + b_k[x_k^0] - \overline{x}_k[\overline{x}_k] \tag{3.17}$$

and if we apply the operator to an equation, we increases the number of terms in the equation by a factor of 3. We have 3^d terms relating to SQ by the end of applying LE. On the other hand, once we apply the operator L^{-1}, we introduce 2^d terms relating to Q, but since we only have E applied on SQ, we would only have at most 2^d terms. For the case of walks of step size at most 1 and labelling the hyperplane interactions, we have exactly 2^d terms $Q_I^{\mathbb{0}}$, one for each subset $I \subseteq [d]$. In this case, we can write out the functional equation explicitly. First,

$$L^{-1}Q = (1 - c_1[x_1^0]) \cdots (1 - c_d[x_d^0])Q = \sum_{I \subseteq [d]} (-1)^{|I|} c_I Q_I^{\mathbb{0}} \tag{3.18}$$

and

$$ESQ = \sum_{I \subseteq [d]} (-1)^{|I|} \overline{x}_I [\overline{x}_I^{\mathbb{1}}](SQ) = \sum_{I \subseteq [d]} (-1)^{|I|} \overline{x}_I S_I^{-\mathbb{1}} Q_I^{\mathbb{0}}\,, \tag{3.19}$$

where the last equality follows from the fact that there are no negative powers of x_k in Q for any k. Therefore, the functional equation is

$$\sum_{I \subseteq [d]} (-1)^{|I|} (c_I - t\overline{x}_I S_I^{-\mathbb{1}}) Q_I^{\mathbb{0}} = \overline{a}_{[d]}\,. \tag{3.20}$$

As in the two-dimensional case, we let $K_I = c_I - t\overline{x}_I S_I^{-\mathbb{1}}$, where, if $I = \varnothing$, we have the empty products and empty extractions, leaving us with $K_\varnothing = K$. Finally, we can write the master equation in d dimensions.

Theorem 3.16. *For subsets I of $\{1, \ldots, d\}$, let $K_I = c_I - t\overline{x}_I S_I^{-\mathbb{1}}$. Then the CLASSIC interacting d-dimensional lattice walk model, has a full generating function Q satisfying the functional equation*

$$\sum_{I \subseteq [d]} (-1)^{|I|} K_I Q_I^{\mathbb{0}} = \overline{a}_{[d]}\,. \tag{3.21}$$

Equation 5.1 of [15, p. 58] is the non-interacting version of our equation, and it can be written as

$$KQ - \sum_{\substack{I \subseteq [d] \\ I \neq \varnothing}} (-1)^{|I|} \overline{x}_I S_I^{-\mathbb{1}} Q_I^{\mathbb{0}} = 1\,. \tag{3.22}$$

3.2 Coefficient Extractions

In this section we will see how the extraction and marking operators interact with basic coefficient extraction to produce quicker coefficient extraction of the functional equation. That is, it is faster to use the operator form of the functional equation in order to extract coefficients than it is to use the explicit form. However, this section is not essential for our understanding of the enumeration of lattice walk models. **[FIXME: See if this section may be compressed a bit, since it looks horrible right now.]**

3.2.1 Using Operator Notation For Fast Coefficient Extraction

As in the two-dimensional case, we can now perform coefficient extractions with $[x_I^{\mathbb{0}}]$ to attain further functional equations relating $Q_I^{\mathbb{0}}$ to other Q terms of the same variables or fewer variables. The operator $[x_i^k]$ commutes with all operators M_j and E_j with $j \neq i$, so we only need to consider the action on the operators $M_i = L_i^{-1}$ and E_i. We have already done so in the one-dimensional case. The zeroth power extraction yields

$$[x_i^0]M_i = [x_i^0] - [x_i^0]c_i[x_i^0] = [x_i^0] - c_i[x_i^0] = \overline{a}_i[x_i^0]\,, \tag{3.23}$$

$$[x_i^0]E_i = [x_i^0] - [x_i^0]\overline{x}_i[\overline{x}_i] = [x_i^0]\,, \tag{3.24}$$

and the k-th power extraction for $k \geqslant 1$ yields

$$[x_i^k]M_i = [x_i^k] - [x_i^k]c_i[x_i^0] = [x_i^k]\,, \tag{3.25}$$

$$[x_i^k]E_i = [x_i^k] - [x_i^k]\overline{x}_i[\overline{x}_i] = [x_i^k]\,. \tag{3.26}$$

Therefore, for $k \geqslant 0$ the operator $[x_i^k]$ eliminates both M_i and E_i from the picture, while for $k = 0$ it removes E_i but leaves a residual $\overline{a}_i$ behind for M_i. Therefore, if we extract $[x_I^{\mathbb{0}}]$ from the functional

equation, we have

$$\begin{aligned}[x_I^{\mathbb{0}}]a_{[d]} = \overline{a}_{[d]} &= [x_I^{\mathbb{0}}](M_{[d]} - tE_{[d]}S)Q \\ &= \left(\overline{a}_I M_{[d]\setminus I}[x_I^{\mathbb{0}}] - tE_{[d]\setminus I}[x_I^{\mathbb{0}}]S\right)Q \\ &= [x_I^{\mathbb{0}}](\overline{a}_I M_{[d]\setminus I} - tE_{[d]\setminus I}S)Q\,,\end{aligned} \tag{3.27}$$

so that

$$\overline{a}_{[d]\setminus I} = [x_I^{\mathbb{0}}](M_{[d]\setminus I} - a_I tE_{[d]\setminus I}S)Q\,. \tag{3.28}$$

If we rewrite the equation, we have

$$\overline{a}_I = [x_{[d]\setminus I}^{\mathbb{0}}](M_I - a_{[d]\setminus I}tE_I S)Q \tag{3.29}$$

and we can use whichever notation is more useful at the time. The latter notation underscores the idea that, when we extract coefficients for walks that end at some hyperplane intersection, we are having a reduced version of the functional equation at the cost of an added a term. When we extract a general term, we can use the operator $[x_I^{\mathbb{0}}x_J^P]$, where P is a set of powers greater than 0 and $I \cap J = \varnothing$, $|P| \geqslant 1$ and $I \cup J = K$. We find that

$$\begin{aligned}[x_I^{\mathbb{0}}x_J^P]a_{[d]} = 0 &= [x_I^{\mathbb{0}}x_J^P](M_{[d]} - tE_{[d]}S)Q \\ &= [x_I^{\mathbb{0}}x_J^P](\overline{a}_I M_{[d]\setminus K} - tE_{[d]\setminus K}S)Q\,,\end{aligned} \tag{3.30}$$

which we can write as

$$[x_I^{\mathbb{0}}x_J^P](M_{[d]\setminus K} - a_I tE_{[d]\setminus K}S)Q = 0\,. \tag{3.31}$$

If I is empty,

$$[x_J^P](M_{[d]\setminus J} - tE_{[d]\setminus J}S)Q = 0\,. \tag{3.32}$$

The main role of these calculations is to provide a shortcut for manual calculations. After all, we need these to obtain functional equations for $Q_I^{\mathbb{0}}$ for $I \subseteq [d]$.

3.2.2 Further Functional Equations

We need to go further in order to find equations similar to the master functional equation. First,

$$[x_I^{\mathbb{0}}]SQ = S_I^{\mathbb{0}}Q_I^{\mathbb{0}} + R_I\,, \tag{3.33}$$

where R_I is a collection of $S_I^{-P}Q_I^P$ terms that involves sets P whose terms are not all zero. We can use this term inside the functional equation

$$\begin{aligned}\overline{a}_{[d]\setminus I} &= M_{[d]\setminus I}[x_I^{\mathbb{0}}]Q - a_I t E_{[d]\setminus I}[x_I^{\mathbb{0}}](SQ) \\ &= M_{[d]\setminus I}Q_I^{\mathbb{0}} - a_I t E_{[d]\setminus I}(S_I^{\mathbb{0}}Q_I^{\mathbb{0}} + R_I) \\ &= (M_{[d]\setminus I} - a_I t E_{[d]\setminus I}S_I^{\mathbb{0}})Q_I^{\mathbb{0}} - a_I t E_{[d]\setminus I}R_I\end{aligned} \tag{3.34}$$

so we can rewrite it as

$$(M_{[d]\setminus I} - a_I t E_{[d]\setminus I}S_I^{\mathbb{0}})Q_I^{\mathbb{0}} = a_I(\overline{a}_{[d]} + tR_I')\,, \tag{3.35}$$

where $R_I' = E_{[d]\setminus I}R_I$. We compute the following terms

$$M_{[d]\setminus I}Q_I^{\mathbb{0}} = \sum_{J\subseteq[d]\setminus I}(-1)^{|J|}c_J Q_{I,J}^{\mathbb{0},\mathbb{0}}\,, \tag{3.36}$$

$$a_I t E_{[d]\setminus I}S_I^{\mathbb{0}}Q_I^{\mathbb{0}} = \sum_{J\subseteq[d]\setminus I}(-1)^{|J|}a_I t\overline{x}_J S_{I,J}^{\mathbb{0},-\mathbb{1}}Q_{I,J}^{\mathbb{0},\mathbb{0}}\,, \tag{3.37}$$

and the functional equation can be written as

$$\sum_{J\subseteq[d]\setminus I}(-1)^{|J|}(c_J - a_I t\overline{x}_J S_{I,J}^{\mathbb{0},-\mathbb{1}})Q_{I,J}^{\mathbb{0},\mathbb{0}} = a_I(\overline{a}_{[d]} + tR_I')\,. \tag{3.38}$$

The form of this expression suggests the following notation.

Definition 3.17. *For sets I and J that do not share any elements, we write*

$$K_J(I) = c_J - a_I t\overline{x}_J S_{I,J}^{\mathbb{0},-\mathbb{1}}\,, \tag{3.39}$$

where if $I = \varnothing$, then $K_J(I) = K_J$.

With this notation, we can write the 2^d equations for $Q_I^{\mathbb{0}}$ and $I \subseteq [d]$ in simplified form.

Corollary 3.18. *Let I be a subset of $\{1,\ldots,d\}$. The generating function $Q_I^{\mathbb{0}}$ for the d-dimensional interacting CLASSIC lattice walk model satisfies the functional equation*

$$\sum_{J\subseteq[d]\setminus I}(-1)^{|J|}K_J(I)Q_{I,J}^{\mathbb{0},\mathbb{0}} = \overline{a}_{[d]\setminus I} + a_I tR_I'\,, \tag{3.40}$$

where the R_I' term involves only higher-order coefficient extractions from Q; that is, where not all the powers of variables x are zero.

This is a clean way to write down all the equations for $Q_I^{\mathbb{0}}$ in a unifying way. To compute the R'_I terms, we first determine what R_I is. It is a sum of terms $S_I^{-P}Q_I^P$ for various non-negative P with size $|I|$ whose elements are not all 0. To calculate R'_I, we apply the operator $E_{[d]\setminus I}$ on R_I. We can try calculating some of these quantities. A key observation is that the terms Q_I^P are power series in the remaining variables, so they do not have non-negative coefficients. Let $sq = S_I^{-P}Q_I^P$. Then

$$E_j sq = sq - \overline{x}_j (sq)_j^{-1} = sq - \overline{x}_j s_j^{-1} q_j^0 \tag{3.41}$$

and thus,

$$E_K sq = \sum_{J \subseteq K} (-1)^{|J|} \overline{x}_J s_J^{-\mathbb{1}} q_J^{\mathbb{0}} \,. \tag{3.42}$$

Finally, we can write

$$R'_I = \sum_{P > \mathbb{0}} \sum_{J \subseteq [d]\setminus I} (-1)^{|J|} S_{I,J}^{-P,-\mathbb{1}} Q_{I,J}^{P,\mathbb{0}} = \sum_{J \subseteq [d]\setminus I} (-1)^{|J|} \sum_{P > \mathbb{0}} S_{I,J}^{-P,-\mathbb{1}} Q_{I,J}^{P,\mathbb{0}} \,. \tag{3.43}$$

Proposition 3.19. *The functions R'_I are given by*

$$R'_I = \sum_{J \subseteq [d]\setminus I} (-1)^{|J|} \sum_{P > \mathbb{0}} S_{I,J}^{-P,-\mathbb{1}} Q_{I,J}^{P,\mathbb{0}} \,. \tag{3.44}$$

Of course, when $d =$ or $d = 2$ we do not need the full generality of the formula, but for higher dimensions, it would be quite difficult to write the equations by hand without it.

3.2.3 Kernel Forms

We can write out the functional equations in a way that highlights the kernel K in the equation. For example, the form in the operator functional equation is

$$(1 - tLES)Q = (1 - t(LE - 1 + 1)S)Q = KQ + t(1 - LE)SQ \tag{3.45}$$

Here is another way we could have expanded it.

$$(1 - tLES)Q = LEKQ + (1 - LE)Q = LEKQ + (1 - L)Q \,, \tag{3.46}$$

where $EQ = Q$ since Q has no terms that need removing. Something interesting happens when we set all the a_k's to 1 in the last equation. This is equivalent to removing the marking operator. The

equation becomes $E(KQ) = 1$ this form suggests that we use the following substitution

$$Q = \frac{W}{x_1 \cdots x_d K} \,. \tag{3.47}$$

Then we multiply both sides of the functional equation by $x_1 \cdots x_d$ and obtain

$$x_1 \cdots x_d = x_1 \cdots x_d E(KQ) = x_1 \cdots x_d E(\overline{x}_1 \cdots \overline{x}_d W) = E'W \,, \tag{3.48}$$

where $E' = E'_1 \cdots E'_d$ and

$$E'_k = x_k E_k \overline{x}_k = x_k (1 - \overline{x}_k [\overline{x}_k]) \overline{x}_k = 1 - [x_k]^0 \,. \tag{3.49}$$

Proposition 3.20. *Let $W = x_1 \cdots x_d KQ$, where Q is the generating function for walks without interactions in the positive orthant. Then*

$$E'W = \sum_{I \subseteq [d]} (-1)^{|I|} W_I^{\emptyset} = x_1 \cdots x_d \,. \tag{3.50}$$

For example, in two dimensions we have

$$W(x_1, x_2) - W_1^0(x_2) - W_2^0(x_1) + W_{1,2}^{0,0} = x_1 x_2 \,. \tag{3.51}$$

Thus, we can write the solution to the problem as

$$Q = \frac{1}{x_1 \cdots x_d K} \left(x_1 \cdots x_d - \sum_{\emptyset \neq I \subseteq [d]} (-1)^{|I|} W_I^{\emptyset} \right). \tag{3.52}$$

In two dimensions we have

$$Q(x_1, x_2) = \frac{x_1 x_2 + W_1^0(x_2) + W_2^0(x_1) - W_{1,2}^{0,0}}{x_1 x_2 (1 - tS)} \,. \tag{3.53}$$

Example 3.21. Consider the one-dimensional model on the non-negative line with $S = \overline{x} + x$. Here the solution for the interacting version of the model is given by

$$Q = \frac{t(\overline{\sigma} - \overline{x})}{(1 - tS)(a\overline{\sigma}t - b)}, \tag{3.54}$$

so when we set $a = 1$ and simplify the expression, we find that the generating function for the

non-interacting model is given by

$$\widehat{Q} = \frac{x - \sigma}{x(1 - tS)}\,, \tag{3.55}$$

as predicted by the analysis of this section. Here $W_0 = \sigma$.

Of course, each $W_I^{\mathbb{0}}$ depends on $Q_I^{\mathbb{0}}$, so we could have seen this relationship starting with the master functional equation for Q itself. Interestingly enough, the calculation points to a differential equation. Clearly, W satisfies the differential equation $\partial_1\partial_2 W = 1$ and a similar relationship exists for the d-dimensional models.

Proposition 3.22. *The full generating function Q for d-dimensional non-interacting lattice walks in the positive orthant with small steps, satisfies the inhomogeneous PDE*

$$\partial_1 \cdots \partial_d W = \partial_1 \cdots \partial_d (x_1 \cdots x_d K Q) = 1\,. \tag{3.56}$$

The result is not difficult to generalize. If we have walks with larger steps, we will have a similar partial differential equation where the orders of variables and orders of the derivatives will be higher. If we have interactions, we will also have higher order terms.

3.3 First Solutions

3.3.1 Formal Neumann Series

So far we have obtained the equation $(1 - t\Gamma S)Q = 1$, where $\Gamma = LE$. Formally, we can expand this operator as the Neumann series and express the generating functions as

$$Q = \sum_{k=0}^{\infty} t^k (\Gamma S)^k(1)\,, \tag{3.57}$$

where $(\Gamma S)^k(1)$ indicates repeated application of ΓS. This allows us to compute Q iteratively, where the next t-coefficient is the previous one with ΓS applied to it. This is a formalization of the naive brute force approach of growing lattice walks from smaller ones: by taking a set of walks of length n, extending them by 1 step, removing the ones that end outside of the quarter plane, and finally by marking the ones that interact with the hyperplanes $x_i = 0$ at their endpoint. We have not been able to find closed forms for the repeated operator applications in general. Here is an example that illustrates the use of the operators.

Example 3.23. Let $S = x_1 + \overline{x}_1 + x_2 + \overline{x}_2 = S_1(x_1) + S_2(x_2)$. We start with $(\Gamma S)^0(1) = 1$. Then

$$(\Gamma S)(1) = LE(S_1 + S_2) = L(x_1 + x_2) = a_2 x_1 + a_1 x_2 \,, \tag{3.58}$$

$$\begin{aligned}(\Gamma S)^2(1) &= LE(S_1 + S_2)(a_2 x_1 + a_1 x_2) \\ &= LE(a_2 S_1 x_1 + a_1 x_2 S_2 + a_2 x_1 S_2 + a_1 x_2 S_1) \\ &= a_1 a_2^2 + a_2^2 x_1^2 + a_1 a_2^2 + a_1^2 x_2^2 + (a_1 + a_2) x_1 x_2 \,.\end{aligned} \tag{3.59}$$

3.3.2 Hadamard Products

We take a moment to see a result that falls out of the operator notation and which was covered in more generality in Section 5 of [18, p. 14], where the authors consider a generalization of the situation $S(x_1, x_2) = S_0(x_1) + S_1(x_1)S_2(x_2)$ but for the non-interacting case. We consider the simpler case $S = S_1(x_1)S_2(x_2)$ and its generalizations to arbitrarily-many dimensions, for the interacting case.

Example 3.24. When $d = 2$ we have the special case $S = S_1(x_1)S_2(x_2)$. The operator ΓS factors:

$$\Gamma S = \Gamma_1 \Gamma_2 S_1 S_2 = (\Gamma_1 S_1)(\Gamma_2 S_2) \,. \tag{3.60}$$

This is because Γ_2 does not interact with S_1 and the operators commute. Then

$$(\Gamma S)^k = (\Gamma_1 \Gamma_2 S_1 S_2)^k = (\Gamma_1 S_1)^k (\Gamma_2 S_2)^k \tag{3.61}$$

and we use the Neumann series to find

$$Q = \sum_{k=0}^{\infty} t^k (\Gamma S)^k = \sum_{k=0}^{\infty} t^k (\Gamma_1 S_1)^k (\Gamma_2 S_2)^k = \left(\sum_{k=0}^{\infty} t^k (\Gamma_1 S_1)^k \right) \odot_t \left(\sum_{k=0}^{\infty} t^k (\Gamma_2 S_2)^k \right) \,, \tag{3.62}$$

where $\odot_t$ denotes the Hadamard product of two power series with respect to variable t. From now on we will drop the subscript t unless it is needed. Here we may write

$$Q_i(a_i, x_i; t) = \sum_{k=0}^{\infty} t^k (\Gamma_i S_i)^k \tag{3.63}$$

and by reversing the iteration, we reach the functional equations $(1 - t\Gamma_i S_i) Q_i = 1$ for each i. The

functions Q_i are solutions to one-dimensional problems with step sets S_i and Q is expressed as

$$Q(a_1, a_2, x_1, x_2; t) = Q_1(a_1, x_1; t) \odot_t Q_2(a_2, x_2; t) \tag{3.64}$$

Example 3.25. Once we know the form of the solution, we can prove that it works by a method that does not use the series solution. Suppose that Q_i is a solution to the functional equation

$$(L_i^{-1} - tE_iS_i)Q_i = \overline{a}_i \tag{3.65}$$

for $i = 1$ and $i = 2$. Then we claim that $Q = Q_1 \odot Q_2$ is a solution to the functional equation

$$(L^{-1} - tES)Q = \overline{a}_1\overline{a}_2\,. \tag{3.66}$$

To this end, we compute

$$\begin{aligned}(L^{-1} - tES)Q &= L^{-1}(Q_1 \odot Q_2) - tES(Q_1 \odot Q_2)\\ &= (L_1^{-1}Q_1 \odot L_2^{-1}Q_2) - t((E_1S_1Q_1) \odot (E_2S_2Q_2))\,.\end{aligned} \tag{3.67}$$

Then we substitute $L_i^{-1}Q_i = \overline{a}_i + tE_iS_iQ_i$ into the first term and expand it to find

$$\begin{aligned}(L_1^{-1}Q_1 \odot L_2^{-1}Q_2) &= (\overline{a}_1 + tE_1S_1Q_1) \odot (\overline{a}_2 + tE_2S_2Q_2)\\ &= \overline{a}_1 \odot \overline{a}_2 + \overline{a}_1 \odot (tE_2S_2Q_2) + \overline{a}_2 \odot (tE_1S_1Q_1)\\ &\quad + (tE_1S_1Q_1) \odot (tE_2S_2Q_2))\,.\end{aligned} \tag{3.68}$$

Since we are working with power series,

$$\overline{a}_i \odot (tE_jS_jQ_j) = \overline{a}_i[t^0]tE_jS_jQ_j = 0\,, \tag{3.69}$$

and thus,

$$(L_1^{-1}Q_1 \odot L_2^{-1}Q_2) = \overline{a}_1\overline{a}_2 + t((E_1S_1Q_1) \odot (E_2S_2Q_2))\,, \tag{3.70}$$

establishing that Q is a solution to the equation.

Lemma 3.26. *Let Q be the generating function for a CLASSIC two-dimensional interacting lattice walk model, with $S = S_1(x_1)S_2(x_2)$. Let Q_i be the generating function for a CLASSIC one-dimensional lattice walk model with step polynomial S_i. Then*

$$Q(x_1, x_2, a_1, a_2; t) = Q_1(x_1, a_1; t) \odot Q_2(x_2, a_2; t)\,. \tag{3.71}$$

With this identity, we can prove the following result via induction.

Theorem 3.27. *Consider a CLASSIC interacting d-dimensional lattice walk model. Suppose that $S(x_1, \ldots, x_d) = S_1 \cdots S_r$, where S_i and S_j do not share variables when $i \neq j$. Then the full generating function Q is given by*

$$Q = Q_1 \odot \cdots \odot Q_r \,, \tag{3.72}$$

where Q_k is the full generating function for the CLASSIC interacting sub-d-dimensional problem for the variables present in S_k.

Example 3.28. Consider the two-dimensional model with diagonal steps, that is, $S = (x_1 + \overline{x}_1)(x_2 + \overline{x}_2)$. We will look at this case in more detail in Chapter 5. The one-dimensional problems are the same and have solutions

$$\widetilde{Q}_0 = \frac{\sigma}{at - b\sigma} \quad \text{and} \quad \widetilde{Q} = \frac{t(x - \sigma)}{x(1 - t(x + \overline{x})(at - b\sigma)} \,, \tag{3.73}$$

where

$$\sigma = \frac{1 - \sqrt{1 - 4t^2}}{2t} \,. \tag{3.74}$$

We can expand and simplify $\widetilde{Q}_0$ and $\widetilde{Q}$ to find

$$\widetilde{Q}_0 = \frac{2 - a - a\sqrt{1 - 4t^2}}{2(1 - a + a^2t^2)} \,, \tag{3.75}$$

$$\widetilde{Q} = \frac{((a-2)t + (1-a)x + 2axt^2) + (at + (1-a)x)\sqrt{1 - 4t^2}}{2(1 - a + a^2t^2)(x - t - tx^2)} \,. \tag{3.76}$$

We have the expansion

$$\widetilde{Q} = 1 + xt + (a + x^2)t^2 + x(1 + a + x^2)t^3 + \cdots \tag{3.77}$$

and finally,

$$\begin{aligned} Q(x_1, x_2, a_1, a_2; t) &= \widetilde{Q}(x_1, a_1; t) \odot \widetilde{Q}(x_1, a_2; t) \\ &= 1 + x_1x_2t + (a_1 + x_1^2)(a_2 + x_2^2)t^2 \\ &\quad + x_1x_2(1 + a_1 + x_1^2)(1 + a_2 + x_2^2)t^3 + \cdots \,. \end{aligned} \tag{3.78}$$

However, obtaining an explicit function via the Hadamard product seems to be rather messy. The function is not algebraic since the unmarked variant is not algebraic. However, the function Q is D-finite since both functions in the product are.

3.3.3 Reducible Models and Groups For Walks With Small Steps

Here is another observation that will allow us to find some solutions to models in an easier way. If the model has some steps that cannot be used in the construction of lattice walks, then we can safely remove them from the step set, and we would still have the same generating function enumerating the lattice walks. This procedure works for any step sets and any regional constraints.

Definition 3.29. *Given a model $\mathcal{M}$ with step set $\mathcal{S}$, we define the minimal step set $\mathcal{S}'$ for the model to be the set of steps that can eventually be used by lattice walks in the model. We call a model reducible if $\mathcal{S}'$ is a strict subset of S.*

Lemma 3.30. *The full generating function for a lattice walk model with any step set and constraints is the same whether we use the a step set or its minimal version.*

It is easier to find the generating function for a model if we have more functional equations that will help us cancel out some of the terms. One way to find more equations is to apply a transformation on tuple $(x_1, \ldots, x_d)$ that leaves intact as many parts of the functional equation as possible but still adds some information in the form of a non-equivalent equation. We could map each $x_1, \ldots, x_d$ to some functions separately and then examine the group generated by such maps. We often eliminate the kernel in front of Q in order to find equations for the extracted forms of Q, so it is useful to consider transformations that preserve the kernel. This is a fruitful approach in the non-interacting case and for two-dimensional models, see Section 4.2 [8, p. 11]. We will call this group $G(K)$, although it is not the full group of rational transformations preserving the kernel. For example, a lot of the time $(x_1, x_2) \mapsto (x_2, x_1)$ is not in the group, but it is a valid transformation preserving the kernel. If a transformation on $(x_1, \ldots, x_d)$ preserves the kernel $K = 1 - tS$, then it must also preserve the step polynomial S. We may write S as

$$S = S_k^{-1}\overline{x} + S_k^0 + S_k^1 x \tag{3.79}$$

for some k from 1 to d. We consider sending $x_k \mapsto y_k$ and solving $S(x_k) = S(y_k)$ for y_k to determine the possible transformations we can perform. After some simplification we have the equation

$$(S_k^{-1} - S_k^1 x_k y_k)(y_k - x_k) = 0\,. \tag{3.80}$$

We have three cases depending on whether S_k^{-1} and S_k^1 are zero.

1. If both are zero, the equation is trivially-true and all transformations preserve the kernel.

2. If only one of them is zero, then $y_k = x_k$, so only the identity transformation preserves the kernel.

3. If neither one is zero, then we have the solutions $y_k = x_k$ and

$$y_k = \overline{x}_k \frac{S_k^{-1}}{S_k^1} \,. \tag{3.81}$$

Note that the transformation $g_k(x_k) = y_k$, in the latter case, is an involution since

$$g_k^2(x_k) = g_k\left(S_k^{-1} x_k S_k^1\right) = x_k \,. \tag{3.82}$$

The first two cases are somewhat trivial and we may say something about the generating function Q right away.

1. If $S_k^{-1} = 0$ and $S_k^1 = 0$, then $S = S_k^0$ and the model takes all of its steps inside the $x_k = 0$ hyperplane. Therefore, the model is equivalent to a lower-dimensional model with no x_k coordinate. Suppose that the generating function in the latter case is $q(t)$. Then the generating function in the former case is given by $Q(x_k, a_k; t) = q(a_k t)$, where the rest of the variables have been suppressed from the notation. We can also find this solution from the functional equation directly. Let $M = M' M_k$ and $E = E' E_k$

$$\begin{aligned} [x_k^0](M - tES)Q &= \overline{a}_{[d]} \\ (a_k M'[x_k^0] - tE'[x_k^0]S)Q &= \overline{a}_{[d]} \\ (M' - a_k t E' S_k^0)Q_k^0 &= \overline{a}_{[d]\setminus\{k\}} \\ (M' - a_k t E' S)Q &= \overline{a}_{[d]\setminus\{k\}} \,, \end{aligned} \tag{3.83}$$

where the last equations follows, since Q and S do not depend on x_k. This is just an equation for a $(d-1)$-dimensional model with an extra scaling of t by a_k.

2. If $S_k^{-1} \neq 0$ and $S_k^1 = 0$, then the model is reducible to one with $S = S_k^0$, that is, the model reduces to case 1.

3. If $S_k^{-1} = 0$, $S_k^0 = 0$ and $S_k^1 \neq 0$, then the walks in model leave the $x_k = 0$ hyperplane and never return. Here we an essentially lower-dimensional problem as well, since we have $S = S_k^1 x_k$. If the generating function for the lower-dimensional problem is $q(t)$, then the generating function for this model is $Q(x_k, a_k; t) = q(x_k t)$, where the rest of the variables

have been suppressed from the notation. Similarly to case 1, we may derive the result from the functional equation directly.

All the other cases are more complicated and require more careful analysis. We can define a group G_S of the nontrivial transformations.

Definition 3.31. *For a specific step polynomial S, let $G = G_S = G(K)$ be the group generated by the nontrivial transformations g_k.*

It is easy to see that there are infinitely many such groups, just consider $S = \prod_{k=1}^{d}(x_k + \overline{x}_k)$ with $g_k(x_k) = \overline{x}_k$. The group here has order 2^d. The orders of the finite groups set in 2 dimensions are 1, 2, 4, 6 and 8, from Theorem 3 in [8, p. 8]. If we were to consider walks with multiple copies of some steps, we may obtain even larger group orders as seen in Section 7 of [11, p. 11]. The d-dimensional groups defined above are Coxeter groups — a generalization of the dihedral groups encountered in dimension 2 lattice walk models. Coxeter groups are defined as follows.

Definition 3.32. *The group*

$$G = \langle r_1, \ldots, r_d | (r_i r_j)^{m_{i,j}} = 1 \rangle \tag{3.84}$$

is known as a Coxeter group. Here $m_{i,i} = 1$ and $m_{i,j} \geqslant 2$ for all $i \neq j$. Also $m_{i,j}$ could be infinite.

We call these r's reflections and $m_{i,j} = 2$, they commute. Also, $m_{i,j} = m_{j,i}$. The *rank* of G is the number of generators in this specific representation of G. Often G has more than one representation with different numbers of generators. We can draw a graph for each Coxeter group by the following procedure.

- Label vertices with the generators of G.
- Connect the vertices g_i and g_j if $m_{i,j} \geqslant 3$.
- Label the edges with $m_{i,j}$ if $m_{i,j} \geqslant 4$.

If the graph has different components, the Coxeter group factors as a direct product of Coxeter groups, represented by the individual components. We can see this phenomenon in terms of the step polynomial S.

Proposition 3.33. *If the step polynomia S factors into two disjoint step polynomials T and R as*

$$S = T(x_1, \ldots, x_k)R(x_{k+1}, \ldots, x_d)\,, \tag{3.85}$$

then the group G_S factors as

$$G_S = G_T \times G_R\,, \tag{3.86}$$

where G_S, G_T and G_R are the groups for the step polynomials S, T and R, respectively.

Proof. For a fixed integer $1 \leqslant i \leqslant k$,

$$S_i^j = [x_i^j]S = R[x_i^j]T = RT_i^j \tag{3.87}$$

and therefore,

$$g_i(x_i) = \overline{x}_i \frac{T_i^{-1}R}{T_i^1 R} = \overline{x}_i \frac{T_i^{-1}}{T_i^1}\,. \tag{3.88}$$

Similarly, if $k+1 \leqslant j \leqslant d$, then $g_j(x_j) = \overline{x}_j(R_i^{-1})(R_i^1)$. Furthermore, g_i and g_j commute since they do not affect each other's variables. Since i and j were arbitrary, the generators from different sets commute and the Coxeter graph of the group factors into two pieces, that is, the group itself factors. We can write $G_S = G_T \times G_R$. □

3.4 Specializing to Two Dimensions

Until the end of the chapter we will focus on the two-dimensional models with interactions. We can set $d = 2$ in Equation 3.21 and Equation 3.40 to find the two-dimensional master functional equations.

Corollary 3.34. *The full generating function Q for the CLASSIC interacting two-dimensional models satisfies the master functional equation*

$$KQ - K_1Q_1^0 - K_2Q_2^0 + K_{1,2}Q_{1,2}^{0,0} = \overline{a}_1\overline{a}_2\,. \tag{3.89}$$

Corollary 3.35. *The full generating function Q for the CLASSIC interacting two-dimensional models satisfies the additional functional equations*

$$\begin{aligned}
(1 - ta_1S_1^0)Q_1^0 - (c_2 - ta_1\overline{x}_2S_{1,2}^{0,-1})Q_{1,2}^{0,0} - a_1t(S_1^{-1}Q_1^1 - \overline{x}_2S_{1,2}^{-1,-1}Q_{1,2}^{1,0}) &= \overline{a}_2\,,\\
(1 - ta_2S_2^0)Q_2^0 - (c_1 - ta_2\overline{x}_1S_{1,2}^{-1,0})Q_{1,2}^{0,0} - a_2t(S_2^{-1}Q_2^1 - \overline{x}_1S_{1,2}^{-1,-1}Q_{1,2}^{0,1}) &= \overline{a}_1\,,\\
(1 - a_1a_2tS_{1,2}^{0,0})Q_{1,2}^{0,0} - ta_1a_2t(S_{1,2}^{-1,-1}Q_{1,2}^{1,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{0,-1}Q_{1,2}^{0,1}) &= 1\,.
\end{aligned} \tag{3.90}$$

3.4.1 First Examples in the Plane

The CLASSIC two-dimensional lattice walk models have been investigated thoroughly but the interacting version has not. We want to extend the CLASSIC model methods to the interacting setting or create new ones. Since the generating functions for the interacting models need to reduce to

those for the non-interacting models, we do not expect to find solutions for every model. In fact, some of the CLASSIC models lead to generating functions that are not even D-finite, so their interacting counterparts will not be D-finite either. For the rest of the problems, we hope to find the generating functions Q explicitly, or at least the functions $Q_{1,2}^{0,0}$. This appears to be difficult in most nontrivial cases. Recent research [30] has shown that there are models, for which we have an explicit relationship between Q or $Q_{1,2}^{0,0}$ and unmarked variants. As we saw in the previous chapter, the one-dimensional cases admit a linear relationship with polynomial coefficients. There are other two-dimensional models that exhibit a rational relationship. The way things have turned with regards to the CLASSIC models, it would not be surprising if there are two-dimensional interacting models for which neither a linear nor a rational relationship would exist between Q, $Q_{1,2}^{0,0}$ and various unmarked variants. In this section we will cover the first few examples of quarter plane models with interactions.

Our first step is to find all the reducible models and we can do this by looking at which of the steps $\{$N, NE, E$\}$ they have in their step set.

1. None of the three steps are present. We have 2^5 such models, in which walks cannot leave the origin and they are reducible to ones with $\mathcal{S} = \varnothing$.



Figure 3.1: Set of steps allowed in models that do not leave the origin. The models are equivalent to a model without steps.

2. There is a N step or an E step. The models are equivalent to one-dimensional ones. There are $2 \cdot 2^4 = 32$ such models.

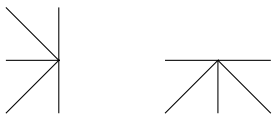


Figure 3.2: Sets of steps allowed in models with an N step or an E step. The models are essentially one-dimensional.

3. There are two or more steps from {N, NE, E} or NE $\subseteq \mathcal{S}$. In this case, the models feature walks that leave both axes and once that happens, such walks may use any step from the step set. Therefore, these models are never reducible.

Cases 1 and 2 contain all of the quarter plane models reducible to one-dimensional models. We will look at the case of a model equivalent to a one-dimensional model along the x_2 axis. From the analysis in the previous section, since both $S_1^{-1} = 0$ and $S_1^1 = 0$, we have the first situation. If the solution to the one-dimensional problem is $q(x, a; t) = q(x_2, a_2; t)$, then the solution to the models in cases 1 and 2 is $Q(x_1, x_2, a_1, a_2; t) = q(x_2, a_2; a_1 t)$. To see it in another way, consider the polynomial S itself in the reduced models. Since S factors, the function Q is a Hadamard product between two one-dimensional models.

Since the models are essentially one-dimensional, we also have a linear relationship using the operators B_1 and B_2. If $f(a, t)$ is the function that goes together with $q(x, a, t)$, then the function $F(a_1, a_2, t)$ that goes together with $Q(x_1, x_2, a_1, a_2, t)$ is given by

$$F(a_1, a_2, t) = f(a_2, a_1 t)\,. \tag{3.91}$$

Then we have $B_2(FQ) = 0$ and consequently, $B_1 B_2(FQ) = 0$. We wrap this section up with the following proposition.

Proposition 3.36. *Each of the 64 reducible models is equivalent to a one-dimensional model confined on the x_1 or x_2 axis. Their solutions are $Q = q(x_1, a_1; a_2 t)$ or $Q = q(x_2, a_2; a_1 t)$, respectively, for a generating function $q(x, a; t)$ for a one-dimensional model. Furthermore, they all satisfy a linear relation $B(FQ) = 0$ for operator $B = B_1 B_2$ and function F, corresponding to q. That is, $F = f(a_2, a_1 t)$ or $f(a_1, a_2 t)$, respectively.*

3.4.2 Kernel Group Considerations

We can partition the two-dimensional lattice walk models in another way — with respect to the nature of the group operations in G_S. We will combine cases for analysis in the form "case $i.j$": we have case i for x_1 and case j for x_2, from the section on groups. Of course, case $i.j$ is equivalent to case $j.i$. From the analysis in the previous sections, cases 1.1, 1.2 and 1.3 only feature reducible models, so we may skip them. We are left with the following cases

Case 2.2: In this case, both x_1 and x_2 can only be transformed via the identity transformation. Therefore, exactly one of S_1^{-1} and S_1^1 is zero and exactly one of S_2^{-1} and S_2^1 is zero. The models where $S_1^1 = 0$ or $S_2^1 = 0$ already have solutions since they are reducible. The only remaining

unsolved case is when $S_1^{-1} = 0$ and $S_2^{-1} = 0$. The possible models are the ones with steps in {N, E, NE} except the ones going either only in the N direction or only in the E direction. Here are the models in the positive quadrant which we have not solved yet.



Figure 3.3: Models from case 2.2.

Case 2.3: In this case, there is only one transformation for x_1 and two for x_2. Therefore, $S_1^{-1} = 0$ (positive case) or $S_1^1 = 0$ (negative case) but not both. The negative case is reducible, so we only seek solutions for the positive one. Furthermore, $S_2^{-1} \neq 0$ and $S_2^1 \neq 0$, so there must be at least one step going in the positive x_2 direction and one in the negative x_2 direction. If both NE and SE are in $\mathcal{S}$, then we have 8 cases. If one of NE and SE is in $\mathcal{S}$ but not both, then we have another 8 cases. Finally, if neither is in $\mathcal{S}$, we must have the other three steps present in order to satisfy the conditions. We have 17 cases in total, the group size is two, and none of the models are reducible. Here are all of these models.

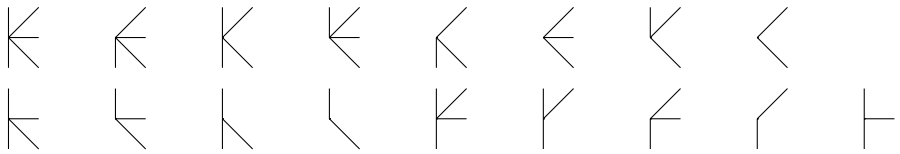


Figure 3.4: Models from case 2.3.

Case 3.3: This is the most interesting case, with two transformations for x_1 and x_2 each. Thus, there must be at least one step going in a positive and one going in a negative direction, for each of x_1 and x_2. If we look at which diagonal steps are present, similarly to case 2.2, we find that there are 145 such models. Previous results show that the group $G(K)$ has orders 4, 6, 8 and infinity, depending on the step set in question. None of these cases are reducible.

3.5 Solving Basic Two-dimensional Models

3.5.1 Case 2.2 – Group of Order One

The only models that are not reducible are those where $S_1^{-1} = 0$ and $S_2^{-1} = 0$. Here

$$S = \delta + \alpha x_1 + \beta x_2 + \gamma x_1 x_2 \,. \tag{3.92}$$

The functional equation and its zeroth term extractions are given by

$$\begin{aligned} (1 - tS)Q - c_1 Q_1^0 - c_2 Q_2^0 + c_1 c_2 Q_{1,2}^{0,0} &= \overline{a}_1 \overline{a}_2 \,, \\ (1 - ta_1(\delta + \beta x_2))Q_1^0 - c_2 Q_{1,2}^{0,0} &= \overline{a}_2 \,, \\ (1 - ta_2(\delta + \alpha x_1))Q_2^0 - c_1 Q_{1,2}^{0,0} &= \overline{a}_1 \,, \\ (1 - a_1 a_2 t\delta)Q_{1,2}^{0,0} &= 1 \,. \end{aligned} \tag{3.93}$$

This is just a linear system of four equations for the four unknowns, $Q_{1,2}^{0,0}$, Q_1^0, Q_2^0 and Q, in echelon form, and solve it by back-substitution.

$$Q_{1,2}^{0,0} = \frac{1}{1 - a_1 a_2 \delta t} \,, \tag{3.94}$$

$$Q_1^0 = \frac{\overline{a}_2}{1 - ta_2 S_1^0} \left(1 + \frac{b_2}{1 - a_1 a_2 \delta t}\right) = \frac{1 - a_1 \delta t}{(1 - ta_1 S_1^0)(1 - a_1 a_2 \delta t)} \,, \tag{3.95}$$

$$Q_2^0 = \frac{\overline{a}_1}{1 - ta_1 S_2^0} \left(1 + \frac{b_1}{1 - a_1 a_2 \delta t}\right) = \frac{1 - a_2 \delta t}{(1 - ta_2 S_2^0)(1 - a_1 a_2 \delta t)} \,, \tag{3.96}$$

$$Q = \frac{1}{1 - tS} \left(\overline{a}_1 \overline{a}_2 + \frac{1}{1 - ta_1 a_2 \delta} \left(c_1 \frac{1 - a_1 \delta t}{1 - ta_1 S_1^0} + c_2 \frac{1 - a_2 \delta t}{1 - ta_2 S_2^0} - c_1 c_2\right)\right) \tag{3.97}$$

$$= \frac{1}{1 - tS} \left(\overline{a}_1 \overline{a}_2 + \frac{1}{1 - a_1 a_2 \delta t}(c_1 f_1(a_1, t) + c_2 f_2(a_2, t) - c_1 c_2)\right) \,. \tag{3.98}$$

Note how Q has a dependency on functions f_1 and f_2 that do not depend on one of a_1 and a_2. We shall see that the linear relation $B(FQ) = 0$ is satisfied by the function

$$F(a_1, a_2, t) = 1 - a_1 a_2 \delta t \,. \tag{3.99}$$

First,

$$B((1 - a_1 a_2 \delta t)\overline{a}_1 \overline{a}_2) = \overline{a}_1 \overline{a}_2 - \overline{a}_2 - \overline{a}_1 + 1 = c_1 c_2 \,. \tag{3.100}$$

Then

$$B(c_1 f_1 + c_2 f_2 - c_1 c_2) = c_1 f_1 + c_2 f_2 - c_1 c_2 - c_2 f_2 - c_1 f_1 = -c_2 c_2 \,. \tag{3.101}$$

Putting the two quantities together, along with dividing by $1 - tS$, shows that $B(FQ) = 0$. Of course, this works for $G = Q^{0,0}_{1,2}$ too.

Example 3.37. Consider the model with $S = x_1 + x_2 + x_1x_2$. Then

$$Q^{0,0}_{1,2} = 1, \quad Q^0_1 = \frac{1}{1 - ta_1x_2}, \quad Q^0_2 = \frac{1}{1 - ta_2x_1}\,, \tag{3.102}$$

$$Q = \frac{1}{1 - tS}\left(\overline{a}_1\overline{a}_2 + \frac{c_1}{1 - ta_1x_2} + \frac{c_2}{1 - ta_2x_1}\right). \tag{3.103}$$

3.5.2 Case 2.3 – Group of Order Two

In this case, the functional equation and its extracted forms read as

$$\begin{aligned}
(1 - tS)Q - c_1Q^0_1 - (c_2 - t\overline{x}_2S_1^{-1})Q^0_2 + c_1c_2Q^{0,0}_{1,2} &= \overline{a}_1\overline{a}_2\\
(1 - ta_1S^0_1)Q^0_1 - (c_2 - ta_1\overline{x}_2S^{0,-1}_{1,2})Q^{0,0}_{1,2} &= \overline{a}_2\\
(1 - ta_2S^0_2)Q^0_2 - c_1Q^{0,0}_{1,2} - a_2tS_2^{-1}Q^1_2 &= \overline{a}_1\\
(1 - a_1a_2tS^{0,0}_{1,2})Q^{0,0}_{1,2} - ta_1a_2S^{0,-1}_{1,2}Q^{0,1}_{1,2} &= 1\,.
\end{aligned} \tag{3.104}$$

We can identify a subproblem that is essentially one-dimensional and therefore, already solved. To see this, take the second and fourth equations, and make the substitutions $a_i = a$, $c_i = c$, $x_2 = x$, $a_1t = \tau$, $S^0_1 = T$ and $Q^0_1(x_2) = q(x)$. The equations become

$$(1 - \tau T)q - (c - \tau\overline{x}T_{-1})q_0 = \overline{a}\,, \tag{3.105}$$

$$(1 - a\tau T_0)q_0 - a\tau T_{-1}q_1 = 1\,, \tag{3.106}$$

where $q_0 = q(a, \tau)$. We can see that this is a one-dimensional problem again. The solution here is $q(x, a, \tau)$, obtained by the methods in the previous chapter, so the original problem has solutions

$$Q^0_1(x_2) = q(x_2, a_2, a_1t) \quad \text{and} \quad Q^{0,0}_{1,2} = q_0(a_2, a_1t)\,. \tag{3.107}$$

The reason the method worked in this case is that we have steps that keep a lattice walk staying on the x_2 axis, steps that take it away from the x_2, but there are no steps that let it return. This is similar to the case where a lattice walk leaves the x_2 axis, never to return. Essentially, the problem splits into a one-dimensional one with regards to the x_2 axis, and a further two-dimensional one the

walks step off the axis. We rewrite the master equation as

$$(1 - tS)Q - (c_2 - t\overline{x}_2 S_2^{-1})Q_2^0 = \overline{a}_1(\overline{a}_2 + b_1 q - b_1 c_2 q_0) = \overline{a}_1 \phi(x_2)\,. \tag{3.108}$$

Since $S_1^{-1} \neq 0$ and $S_2^1 \neq 0$, we have a root $x_2 = \sigma_2$ that allows us to eliminate the $(1 - tS)Q$ term. That is, we are finding a root for x_2 again, since $x = x_2$ in the previous equations. We find that

$$Q_2^0 = -\frac{\overline{a}_1 \phi(\sigma_2)}{c_2 - t\overline{\sigma}_2 S_2^{-1}} \tag{3.109}$$

and

$$Q = \frac{1}{1 - tS}\left(\overline{a}_1 \phi(x_2) - \frac{c_2 - t\overline{x}_2 S_2^{-1}}{c_2 - t\overline{\sigma}_2 S_2^{-1}} \overline{a}_1 \phi(\sigma_2)\right)\,. \tag{3.110}$$

Of course, this form can be expanded and simplified based on what q and q_0 are from the one-dimensional setting, but the important thing is that we can build the solution in this case via one-dimensional model equations.

3.5.3 Miscellaneous Case

There is one case that is easy to solve, is not reducible, but has an infinite group. Consider the case of $S = x_1 x_2 + \overline{x}_1 \overline{x}_2$. The group operations are $g_1(x_1) = \overline{x}_1 \overline{x}_2^2$ and $g_2(x_2) = \overline{x}_2 \overline{x}_1^2$. If we apply these transformations on (x_1, x_2) in an alternating sequence, each time the absolute powers of x_1 and x_2 will increase in one of the coordinates, and therefore, the group $G(K)$ is infinite. However, this case is easy to solve, because the model is essentially one-dimensional. It is equivalent to the model on the non-negative line with $S = x + \overline{x}$ and marking visits to the origin with a, where $x = x_1 x_2$ and $a = a_1 a_2$. The solution is

$$Q(x_1, x_2, a_1, a_2; t) = q(x_1 x_2, a_1 a_2; t). \tag{3.111}$$

3.5.4 Counting Classes of Models

We can tally all the classes of lattice walk models on the quarter plane. First, we have 64 reducible models. Then we have 5 new cases with group of order one, and 34 new cases with group of order two. Also, there are 29 models with group of order four, 6 with group of order six, and 4 with group of order eight. We also have the easy-to-solve miscellaneous case. So far we have 143 cases of models we can solve for easily or models with a finite group, some of which w ecan solve easily. Next, by inspection of Table 4 in [8, p. 28], we see that there are 120 models with group of infinite

order: 12 with three steps, 40 with four steps, 44 with five steps, and 24 with six steps. Adding all of those together yields 263 cases, so we must have over-counted. There are 7 reducible models with an infinite group. The total count is 256 and we are done.

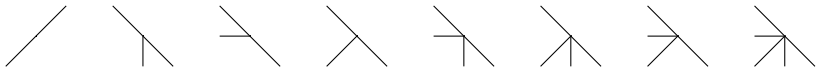


Figure 3.5: Reducible models with an infinite group.

So far we have solved the 64 reducible models, the 39 other models with group of order one or two, and the single new case of easy-to-solve model with an infinite group, for a grand total of 104 trivial or easy models. Out of the remaining 152 models, 113 have an infinite group, and their analysis will not be a part of this thesis. Finally, we have 39 cases left, with a group of size four, six or eight. The rest of the thesis will handle the analysis of those cases.

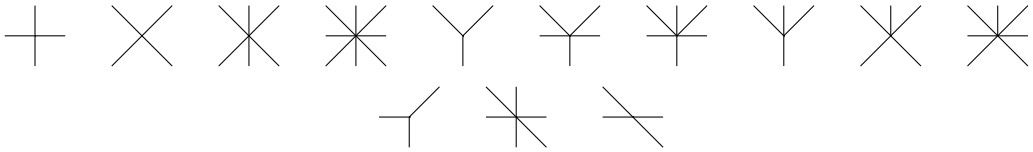


Figure 3.6: Models with a finite group.

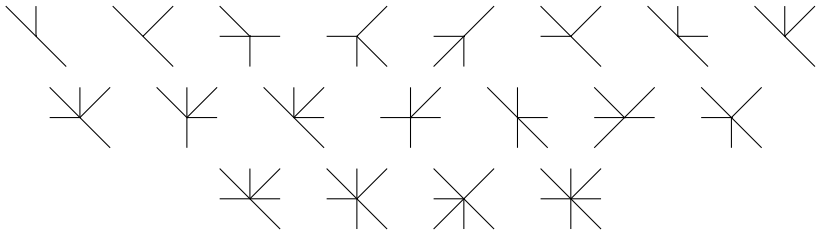


Figure 3.7: Models with an infinite group. To get the rest of them, apply one of the symmetries of the square.

Chapter 4

The Algebraic Kernel Method

In this chapter we will explore the algebraic kernel method as seen in Section 4 of [18, p. 9] and we will extend it to handle models with interactions. First, we will recast the non-interacting version of the method in a form that uses operators. Then we will generalize the operators by giving them weights, and this will allow us to investigate models with interactions.

4.1 Models Without Interactions

[FIXME: Do the results follow even if G does not have a full set of generators?] At the core of the algebraic kernel method we have the idea that we can apply transformations from a group G onto the master functional equation and acquire different equations for each $g \in G$. After we sum these equations in a nice way, we manage to eliminate the functions Q_1^0, Q_2^0, $Q_{1,2}^{0,0}$ and their orbits, only leaving the orbit of the kernel term KQ and a term on the right-hand side. What makes this possible is pre-multiplying the whole equation by x_1x_2. The analysis also works in higher dimensions, as we will show later. Although the generalization has been implemented before (for example,[8, p. 11] for the two-dimensional case and [18, p. 10] for the three-dimensional one), we will present it in a compact and easy-to-generalize way. Instead of summing equations as in [8], we will use an operator D_G that wil do the same thing. We will define D_G in a way that is analogous to the way the algebraic kernel method is currently applied.

Definition 4.1. *For a lattice walk model with small steps, let G be the group generated by the transformations $g_k : x_k \mapsto \overline{x}_k S_k^{-1}/S_k^1$ preserving the kernel, and for g in G, let $|g|$ denote the sign of g. Finally, let D_G be defined as*

$$D_G = \sum_{g \in G} (-1)^{|g|} g \,. \tag{4.1}$$

In two dimensions we can find the whole group by iterating over the two generators. That is, we apply g_1 and g_2 sequentially and the group is $\{1, g_i, g_j g_i, g_i g_j g_i, \ldots\}$. However, we shall use the equivalent orbit $\{1, g_i, g_i g_j, g_i g_j g_i, \ldots\}$, where the iteration is on the right. To declutter the notation we will use g_{iji} instead of $g_i g_j g_i$.

Definition 4.2. *For a sequence $L = (l_1, l_2, \ldots)$ of positive integers, define*

$$g_L = g_{l_1 l_2 \ldots} = g_{l_1} g_{l_2} \cdots , \tag{4.2}$$

when the list is finite or when there is convergence, in the case of an infinite list.

In the case of a finite group in two dimensions we have $(g_1 g_2)^n = 1$ for some minimal n, so we can rewrite D_G in a useful way. The following lemma is a restatement of the orbit sum method found in [8, p. 10], but it is recast in operator notation.

Lemma 4.3. *For a two-dimensional model with small steps and a group of order $2n$ with generators g_1 and g_2, we can write the operator D_G as*

$$D_G = (1 + g_{ij} + \cdots + (g_{ij})^{n-1})(1 - g_i) . \tag{4.3}$$

Proof. We can just check the identity by expanding it as

$$\begin{aligned} D_G &= (1 + g_{ij} + \cdots + (g_{ij})^{n-1})(1 - g_i) \\ &= 1 - g_i + g_{ij} - g_{iji} + \cdots - (g_{ij})^{n-1} g_i , \end{aligned} \tag{4.4}$$

fiving us the orbit summation resulting from the second form of the obit. □

Here are some examples of small groups.

Example 4.4. When the group has order four, the elements commute and the orbit is given by $\{1, g_1, g_{12} = g_{21}, g_2\}$. The operator is $D_G = (1 + g_{12})(1 - g_1)$, but it also factors in a convenient way as

$$D_G = 1 - g_1 - g_2 + g_{12} = (1 - g_1)(1 - g_2) . \tag{4.5}$$

When the group has order six, the orbit is $\{1, g_1, g_{21}, g_{121} = g_{212}, g_{12}, g_2\}$. We can write the

operator as

$$\begin{aligned} D_G &= (1 + g_{12} + g_{1212})(1 - g_1) \\ &= 1 + g_{12} + g_{1212} - g_1 - g_{121} - g_{12121} \\ &= 1 - g_1 - g_2 + g_{12} + g_{21} - g_{121} \,. \end{aligned} \tag{4.6}$$

We can generalize this result for the groups of higher-dimensional models.

Lemma 4.5. *Consider a d-dimensional CLASSIC model with group G. The set H of elements with even parity forms a subgroup.*

Proof. The proof is analogous to the proof that the set of even permutations forms a subgroup of the set of permutations of $\{1, \ldots, n\}$. First, the identity i of G is in H, because $i^2 = i$ has even parity. Then H is closed under the group operation because the product of two elements with even parity also has even parity. Finally, if x has even parity, then its inverse y has even parity, otherwise, $i = xy$ would not have even parity. Thus, H is a subgroup by the two-step subgroup test. □

Lemma 4.6. *Consider a d-dimensional CLASSIC model with group G. For a nontrivial generator g_k of G, we have $D_G = D_H(1 - g_k)$, where H is the subgroup of even elements.*

Proof. First, note that g_k is an involution that does not fix any element of G, that is $gg_k \neq g$ for any g in G. Since applying g_k on a group element changes its parity, this means that g_k induces a partition of G into two parts: a set of even elements and a set of odd elements in one-to-one correspondence. That is, we can partition G as the union of H and Hg_k. With the notation $e_h = (-1)^{|g|}$, we can immediately compute the form of D_G from the statement of the lemma:

$$D_G = \sum_{g \in H} e_g g + \sum_{g \in H} (-1) e_g g g_k = \left(\sum_{g \in H} e_g g \right) (1 - g_k) = D_H (1 - g_k) \,. \tag{4.7}$$

□

Note that we could have also computed $D_G = (1 - g_k) D_H$ if we multiplied g_k on the left. With this generalization of the operator form, we obtain the d-dimensional generalization to Proposition 5 in [8, p. 11] and Lemma 52 in [15, p. 59].

Theorem 4.7. *For a d-dimensional CLASSIC model, we have*

$$D_G(x_1 \cdots x_d K Q) = D_G(x_1 \cdots x_d) \,. \tag{4.8}$$

Proof. When we multiply the master functional equation by $x_1 \cdots x_d$, we transform almost all of the terms of the equation to ones of the form $tS_I^{-\mathbb{1}}Q_I^{\mathbb{0}}$, which are missing variables x_i for i in I. When we apply D_G these terms are eliminated and the only term on the left-hand side that survives is $x_1 \cdots x_d KQ$. □

In [8, p. 11] and [18, p. 13], the authors have found that sometimes the algebraic kernel method fails. That is, $D_G(x_1 \cdots x_d) = 0$ and the equation does not give us new information. The operator form of the method allows us to find a more general instance of when this situation occurs.

Corollary 4.8. *If an element h of G is an involution that permutes the variables $x_1, \ldots, x_d$, then the regular algebraic kernel method fails. That is,*

$$D_G(x_1 \cdots x_d KQ) = 0\,. \tag{4.9}$$

Proof. Since h is an involution, the same method applies as before and we can factor $D_G = D_H(1-h)$ for some $H \subseteq G$. The action $1 - h$ eliminates the monomial $x_1 \cdots x_d$. □

Example 4.9. The two-dimensional models for which the algebraic kernel method fails are precisely those for which we have a permutation as in the corollary. That is, $h(x_1, x_2) = h(x_2, x_1)$ and h has order two. Proposition 6 in [8, p. 12] follows.

Example 4.10. In three dimensions and models with at most 6 steps, we see that the algebraic kernel method fails in 62 three-dimensional models [18, p. 13]. For example, there is a model with group size 12 for which the algebraic kernel method fails. For that model, one of the group element is a permutation of two of the variables — in that case x_2 and x_3. Here G contain the group seen in the previous example for the variables x_2 and x_3. Therefore, every higher-dimensional model, that has group G containing the group above, will fail in the regular algebraic kernel method.

[AWKWARD]A natural question is can we take D_H instead of $D_G = D_H(1 - h)$, and check if we can perfom a half-orbit summation, and to what success. The answer is yes. The operator D_H eliminates some of the Q variables and introduces others, but the result is a function symmetric under permutation by h. One may hope to perform a method analogous to the half-orbit summation method to prove algebraicity or D-finiteness.

Example 4.11. There is another way we can prove that D_G acts as advertised. As in Chapter 3, we start with the kernel form $E(KQ) = 1$ where we make the substitution $Q = W/(x_1 \cdots x_d K)$, and rewrite the equation as $E'W = x_1 \cdots x_d$, where $E' = E'_1 \cdots E'_d$ and $E'_k = 1 - [x_k^0]$. Therefore,

$$W = x_1 \cdots x_d - (1 - E')W = x_1 \cdots x_d - \sum_{\varnothing \neq I \subseteq [d]} W_I^{\mathbb{0}}\,. \tag{4.10}$$

We immediately see that

$$D_G(x_1 \cdots x_d KQ) = D_G\left(x_1 \cdots x_d - \sum_{\varnothing \neq I \subseteq [d]} W_I^{\mathbb{0}}\right) = D_G(x_1 \cdots x_d)\,, \tag{4.11}$$

since the $W_I^{\mathbb{0}}$ terms have missing variables.

Next, we will generalize the orbit summation operator by having weights with each g in G.

Definition 4.12. *Let G be a group acting on $x_1, \ldots, x_d$. We define a weighted orbit summation operator to be an operator of the form*

$$D = \sum_{g \in G} R_g g\,, \tag{4.12}$$

where R_g are functions in the variables $x_1, \ldots, x_d$.

This is a generalization of the orbit summation operator that can be written as a weighted operator with $R_g = (-1)^{|g|}$ **[FIXME: Yeaa, don't know if I want this]**. A first glimpse of this kind of operator can be seen in [21, p. 11], where the authors use an orbit summation with weights. We can capture various possible strategies in this notation.

Example 4.13. Since the algebraic kernel method fails to give a nice result when G contains permutations, we may try to apply the operator D_G on a slightly different equation — one where we multiply the whole equation by a function $T \neq x_1 \cdots x_d$. For a function f, we have

$$D_G(Tf) = \sum_{g \in G} e_g g(Tf) = \sum_{g \in G} e_g(g(T))g(f) = \sum_{g \in G} R_g g(f)\,, \tag{4.13}$$

where $R_g = (-1)^{|g|} g(T)$. therefore, this strategy is the same as applying a weighted orbit summation operator.

Example 4.14. Another strategy is to consider applying a weighted operator several times in a row in the hopes of finding an easier equation to work with. However, this process is no different than looking at a weighted operator in the first place. This is because G is a group and the composition D_G^2 leads to another weighted operator, as we shall prove shortly. Also, since some of the operators D_G have nontrivial kernels, they are not invertible, so the set of operators is not a group. However, we can show that the set is a monoid under composition.

Proposition 4.15. *Given a group G, the set of weighted orbit summation operators forms a monoid under composition.*

Proof. Since G is a group, we have an identity element 1, which can also be written as a weighted orbit summation operator with $R_g = 0$ for all $g \neq 1$ and $R_1 = 1$. Thus, the set of operators has an identity element. Since composition of operators is associative, we only need to check that the set of operators is closed under composition. We have

$$D_1 D_2 = \sum_{g \in G} \sum_{h \in H} R_g g T_h h = \sum_{g \in G} \sum_{h \in H} R_g T_{gh} gh = \sum_{g \in G} \left(\sum_{h \in H} R_{gh_{-1}} T_g \right) g \,, \tag{4.14}$$

which is also an operator of this form once we collect terms corresponding to the same group elements. □

The main question is whether the weighted summation operators will help us in analyzing the models where the algebraic kernel method fails. We will answer this question in the negative. In general, there is no other weighted orbital operator that eliminates all the unwanted terms of the master functional equation, except the operator with weights $R_g = 1$, that does not also eliminate $x_1 \cdots x_d$. However, this setback does not diminish the usefulness of weighted operators, because their strength is in the analysis of interacting models.

Here is the strategy we will employ. If we apply the operator D on the master functional equation, we need to eliminate all terms except KQ. That is, D needs to annihilate all the Q terms that are missing variables. It is enough to work on the fundamental functional equation for $W = x_1 \cdots x_d KQ$. To go back to the master functional equation case, we pre-divide the equation by $x_1 \cdots x_d$. The resulting operator is also weighted.

Lemma 4.16. *A weighted operator D exists such that $D(KQ) = D(1)$ if and only if a weighted operator D' exists such that $D'(f_I^{\emptyset})$ for generic functions $f_I^{\emptyset}$ which are missing the variables indexed by I, for all I.*

Proof. The discussion above provides the direction from left to right for $f = W$. To prove it for generic functions f, it suffices to note that $f_I^{\emptyset}$ and $f_J^{\emptyset}$ are independent from each other if $I \neq J$. That is, $D(f_I^{\emptyset})$ and $D(f_J^{\emptyset})$ do not share terms. The converse direction is proved by setting $f = W$ and setting $D = D'\overline{x}_1 \cdots \overline{x}_d$, where $(D'\overline{x}_1 \cdots \overline{x}_d)(f) = D'(\overline{x}_1 \cdots \overline{x}_d f)$. □

This observation essentially turns the question into a linear algebra problem. We just need to apply D on generic functions $W_I^{\emptyset}$, which are missing the variables indexed by I, collect the different results and make their coefficients zero. That is, for every $f = W_I^{\emptyset}$ we calculate the orbit

$$\mathrm{Orbit}_G(f) = \{g(f) | g \in G\} \tag{4.15}$$

and then find

$$0 = D(f) = \sum_{g\in G} R_g g(f) = \sum_{f'\in Orbit(f)} R'_g f' \,, \tag{4.16}$$

where we set

$$R'_g = \sum_{\substack{g(f)=f' \\ g\in G}} R_g = 0 \,. \tag{4.17}$$

Thus, we reach a linear system of $|\mathrm{Orbit}_G(f)|$ equations in $|G|$ unknowns. Of course, we can do this for all $2^d - 1$ terms $f = W^{\mathbb{0}}_I$ and hopefully find a system with a nontrivial solution. Here is a simple example.

Example 4.17. Consider a two-dimensional model with group of order four given by

$$G = \{1, g_1, g_2, g_{12} = g_{21}\} \,. \tag{4.18}$$

In the two-dimensional problem we have the generic functions $W(0,0) = c$, a constant, and $W(0,x_2)$ and $W(x_1,0)$. We can write them as $1, f(x_1), f(x_2)$. A weighted operator for this group can be written as

$$D = R - R_1 g_1 - R_2 g_2 + R_{12} g_{12} \,. \tag{4.19}$$

Consider the different orbits. The orbit of 1 under G is trivial since the function does not change under the group actions. Next, we have

$$\mathrm{Orbit}_G(f(x_1)) = \{f(x_1), f(\overline{x}_1), f(\overline{x}_1), f(\overline{x}_1)\} = \{f(x_1), f(\overline{x}_1)\} \tag{4.20}$$

and similarly,

$$\mathrm{Orbit}_G(f(x_2)) = \{f(x_2), f(\overline{x}_2)\} \,. \tag{4.21}$$

Therefore, the equations for the weights are

$$R - R_1 + R_2 - R_{12} = 0, \quad R = R_2, \quad R_1 = R_{12}, \quad R = R_1, \quad R_2 = R_{12} \,. \tag{4.22}$$

When we solve this system of equations, we find $R = R_1 = R_2 = R_{12}$ and the operator is essentially trivial. The simplest choice is to set $R = 1$ and we find that every weighted orbit summation operator for this group G must have identical weights if it is to annihilate generic functions with missing variables.

Definition 4.18. *A trivial/unweighted, operator is a weighted operator with weights that are all*

equal.

Lemma 4.19. *Consider a d-dimensional non-interacting CLASSIC model. If its group is finite with d generators, then all weighted orbital operators are trivial.*

Proof. **[FIXME: Check proof]**It is enough to show that $D(W_k^0) = 0$ for all k implies that the operator is trivial. To this end, suppose that $D(f) = 0$ where $f = W_k^0$. Then we can partition the group G as before into sets H and Hg_k. We see that $hg_k(f) = h(f)$, so the orbit of f contains at most $|G|/2$ elements. However, these elements must all be distinct otherwise they would come from the same operations, but the group elements in H are all distinct. Therefore, $|\mathrm{Orbit}(f) = |G|/2$. Therefore, we have the equations $R_h = R_{hg_k}$ for every $h \in H$. Once we go through every k we have overlaps $R = R_1 = \cdots = R_d$ and $R_1 = R_{12} = R_{13}$, and so on. This means that all the R's are the same and the operator is trivial. □

This analysis shows that, in general, there is no clever weighted summation operator that also satisfies the condition $D(x_1 \cdots x_d) \neq 0$. However, the idea will be fruitful in that we may find nontrivial partial solution in the interacting case.

4.2 Models With Interactions

It is natural to consider extending the algebraic kernel method to one working for interacting models, but we quickly see that the regular orbit summation operator is not sufficient. This is where the power of the weighted orbital summation operators starts to show. The analysis for two-dimensional models is particularly nice because we end up having the same number of equations as unknowns, and we are able to find the required weights. The linear systems can be written as matrix equations, where the determinant of the matrix tells us when we have a useful orbit summation operator. In the end, the method does not extend fully because we do not eliminate the $Q_{1,2}^{0,0}$ term. The D-finiteness of the full generating function Q depends on the D-finiteness of the function $Q_{1,2}^{0,0}$. We will only consider the models with finite groups that have two nontrivial generators. Therefore, the group G has order 4, 6 or 8, and the functional equation in two dimensions is

$$KQ - K_1Q_1^0 - K_2Q_2^0 + K_{1,2}Q_{1,2}^{0,0} = \alpha\,. \tag{4.23}$$

4.2.1 Groups of Order Four

The orbital summation operator for the non-interacting case is

$$D_G = 1 - g_1 - g_2 + g_1g_2 = (1 - g_1)(1 - g_2) = \Delta_1\Delta_2 \tag{4.24}$$

but this operator does not eliminate the unknown functions in the interacting case. Instead of using this operator, we will use a weighted one and we will determine its weights. We will find a linear system of equations by applying the group operations on the original functional equation. The unknowns will be functions Q_1^0, Q_2^0 and their orbits under G. If the associated matrix has a nonzero determinant, the system has a unique solution and unfortunately, we will not be able to eliminate variables. However, there are some cases where the determinant is actually zero, so we can eliminate some of the unknowns.

Since the transformations commute, we have $g_i Q_i^0 = Q_i^0$ and $g_i g_j Q_i^0 = g_j Q_i^0$. The variables and their orbits are given by the next table.

Group action	Effect on Q_1^0	Effect on Q_2^0
1	Q_1^0	Q_2^0
g_1	Q_1^0	$g_1 Q_2^0$
g_2	$g_2 Q_1^0$	Q_2^0
$g_{12} = g_{21}$	$g_2 Q_1^0$	$g_1 Q_2^0$

Table 4.1: Effects of group actions $g \in G$ on Q_1^0 and Q_2^0.

Therefore, the only variables that we need to eliminate are in the set

$$E = \{Q_1^0,\, Q_2^0,\, g_2 Q_1^0,\, g_1 Q_2^0\}\,. \tag{4.25}$$

The situation looks promising since we also have four equations. We rewrite the functional equation in a way that highlights those variables.

$$K_1 Q_1^0 + K_2 Q_2^0 = KQ + K_{1,2} Q_{1,2}^{0,0} - \alpha = \beta\,. \tag{4.26}$$

Then we apply the transformations to produce the system of equations

$$K_1 Q_1^0 + K_2 Q_2^0 = \beta\,, \tag{4.27}$$

$$g_1 K_1 Q_1^0 + g_1 K_2 g_1 Q_2^0 = g_1 \beta = \beta_1\,, \tag{4.28}$$

$$g_2 K_1 g_2 Q_2^0 + g_2 K_2 Q_2^0 = g_2 \beta = \beta_2\,, \tag{4.29}$$

$$g_{12} K_1 g_2 Q_1^0 + g_{12} K_2 g_1 Q_2^0 = g_{12} \beta = \beta_{12}\,. \tag{4.30}$$

Using the order of variables in E, we can write the associated matrix M of the linear system as

$$M_4 = \begin{bmatrix} K_1 & K_2 & 0 & 0 \\ g_1K_1 & 0 & 0 & g_1K_2 \\ 0 & g_2K_2 & g_2K_1 & 0 \\ 0 & 0 & g_{12}K_1 & g_{12}K_2 \end{bmatrix}. \tag{4.31}$$

Then

$$\begin{aligned} \det M_4 &= K_1g_{12}K_1g_1K_2g_2K_2 - g_1K_1g_2K_1K_2g_{12}K_2 \\ &= K_1g_{12}K_1g_1K_2g_2K_2 - g_1(g_1K_1g_{12}K_1g_1K_2g_2K_2) \\ &= \Delta_1(K_1g_{12}K_1g_1K_2g_2K_2) \end{aligned} \tag{4.32}$$

where the last equation is obtained by using g_2 instead of g_1 in the factoring.

The determinant must vanish if we are to eliminate variables, and that only happens when the expression in Δ_1 is in the operator's kernel. The models with groups of order four exhibit at least one symmetry across a line — either vertical or horizontal symmetry, or both. Without loss of generality, let the model have symmetry across the x_2 axis. That is, $g_1(x_1) = x_1$, $g_1(S_2^i) = S_2^i$ and $g_1(S_1^i) = S_1^{-i}$, and we can compute some of the quantities. For example, $g_1K_2 = K_2$ and $g_1g_2K_2 = g_2g_1K_2$. Therefore, Δ_1 fixes both K_2 and g_2K_2 and

$$\det M_4 = \Delta_1(K_1K_2g_{12}K_1g_2K_2) = K_2g_2K_2\Delta_1(K_1g_{12}K_1)\,. \tag{4.33}$$

This quantity is zero if and only if $\Delta_1(K_1g_{12}K_1) = 0$, so we have

$$\begin{aligned} \Delta_1(K_1g_{12}K_1) &= K_1g_{12}K_1 - g_1K_1g_2K_2 \\ &= c_1t(\overline{x}_1 - x_1)(g_2S_1^{-1} - S_1^{-1}) \\ &= -c_1t(\overline{x}_1 - x_1)\Delta_2(S_1^{-1})\,. \end{aligned} \tag{4.34}$$

Therefore, this equation is true if and only if $g_2S_1^{-1} = S_1^{-1}$. We have

$$g_2(S_{1,2}^{-1,-1}\overline{x}_2 + S_{1,2}^{-1,0} + S_{1,2}^{-1,1}x_2) = S_{1,2}^{-1,-1}x_2\overline{\mu}_2 + S_{1,2}^{-1,0} + S_{1,2}^{-1,1}\overline{x}_2\mu_2 \tag{4.35}$$

$$S_{1,2}^{-1,-1}x_2\overline{\mu}_2 + S_{1,2}^{-1,1}\overline{x}_2\mu_2 = S_{1,2}^{-1,-1}\overline{x}_2 + S_{1,2}^{-1,1}x_2\,, \tag{4.36}$$

and we can write it succinctly as $S_{1,2}^{-1,-1} = \mu_2S_{1,2}^{-1,1}$.

If one of $S_{1,2}^{-1,-1}$ and $S_{1,2}^{-1,1}$ is zero, then so is the other, and since the step set has vertical symmetry, $S_{1,2}^{1,1}$ and $S_{1,2}^{1,-1}$ are also zero. That is, the model has no diagonal steps. However, since the group has two nontrivial generators, we must have both steps with monomials $\overline{x}_2$ and x_2, and the model is

highly-symmetric. If both of the S values are equal to one, then $S_2^{-1} = S_2^1$ and again, the model is highly-symmetric.

Conversely, if we started with a highly-symmetric model, we would have obtained $\det M_4 = 0$, and this leads to the following theorem.

Theorem 4.20. *The determinant* $\det M_4 = 0$ *if and only if the model is highly-symmetric.*

Corollary 4.21. *The algebraic kernel method extends to the non-interacting case for* $|G| = 4$ *if and only if the model is highly-symmetric.*

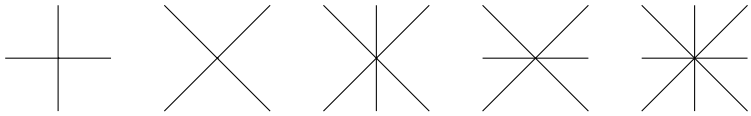


Figure 4.1: The highly-symmetric models on the plane with two nontrivial generators for the group G.

We are finally ready to use the orbit summation method. There are 8 highly-symmetric two-dimensional models, but only five of them have two nontrivial generators for the group G. Furthermore, two of the models are reflections of each other, so we are really left with four distinct cases. From left to right in the figure above, they are the Cartesian model, the Diagonal Cartesian model, the model with two horizontal steps missing, and the full model. If we are able to eliminate all the variables from the set E in the beginning of this section, then we will have a linear relation like

$$\gamma\beta - \gamma_1\beta_1 - \gamma_2\beta_2 + \gamma_{12}\beta_{12} = 0\,. \tag{4.37}$$

Here the choice of signs is inspired by the orbit summation operators. Let $\bullet$ and $*$ stand for applications of g_1 and g_2, respectively. Note that we have $K_2^\bullet = K_2$ and $K_1^* = K_1$. Next, we let

$$x = Q_1^0\,, \quad y = Q_2^0\,, \quad u = Q_1^{0,*}\,, \quad v = Q_2^{0,\bullet}\,, \tag{4.38}$$

$$a = K_1\,, \quad b = K_1^\bullet\,, \quad c = K_2\,, \quad d = K_2^*\,. \tag{4.39}$$

The original equations attain a simplified format given by

$$\beta = ax + cy\,, \quad \beta_1 = bx + cv\,, \tag{4.40}$$

$$\beta_2 = dy + au\,, \quad \beta_{12} = bu + dv\,, \tag{4.41}$$

Pluggin these into the linear relation above yields

$$\begin{aligned} 0 &= (ax+cy)\gamma - (bx+cv)\gamma_1 - (dy+au)\gamma_2 + (bu+dv)\gamma_{12} \\ &= (a\gamma - b\gamma_1)x + (c\gamma - d\gamma_2)y + (b\gamma_{12} - a\gamma_2)u + (d\gamma_{12} - c\gamma_1)v\,, \end{aligned} \tag{4.42}$$

and in order to eliminate the variables x, y, u and v, we need all their coefficients to vanish. We obtain the equations

$$a\gamma = b\gamma_1\,, \quad c\gamma = d\gamma_2\,, \quad b\gamma_{12} = a\gamma_2,\,, \quad d\gamma_{12} = c\gamma_1\,. \tag{4.43}$$

If we set $\gamma = 1$, then $\gamma_1 = a/b$ and $\gamma_2 = c/d$ and $\gamma_{12} = (ac)/(bd)$. So we can multiply them out by bd to find the relation

$$bd\beta - ad\beta_1 - cb\beta_2 + ac\beta_{12} = 0\,, \tag{4.44}$$

where we have successfully eliminated Q_1^0, Q_2^0, $Q_1^{0,*}$ and $Q_2^{0,\bullet}$. In terms of the K's we have

$$K_1^\bullet K_2^* \beta - K_1 K_2^* \beta^\bullet - K_1^\bullet K_2 \beta^* + K_1 K_2 \beta^{\bullet *} = 0\,. \tag{4.45}$$

We can rewrite this equation in the following nicer way.

Theorem 4.22. *For a two-dimensional highly-symmetric model with a group with two nontrivial generators, we have the functional equation $\Delta(K_1^\bullet K_2^* \beta) = 0$, where $\beta = KQ + K_{1,2}Q_{1,2}^{0,0} - \alpha$.*

This is a generalization of the result in Section 4.2 of [21, p. 16], where the authors use a weighted orbit sum to eliminate boundary terms. We can expand the functional equation to produce a different form. Let $K_0 = K_1^\bullet K_2^*$. We find that

$$\begin{aligned} \Delta(K_0\beta) &= \Delta(K_0(KQ + K_{1,2}Q_{1,2}^{0,0} - \alpha)) \\ &= K\Delta(K_0 Q) + \Delta(K_0 K_{1,2})Q_{1,2}^{0,0} - \Delta(K_0)\alpha\,, \end{aligned} \tag{4.46}$$

where the second equality follows since Δ commutes with K and functions with some variables x_i missing. All that remains is to expand the various quantities involved. Then we have

$$\Delta g_i K_i = -\Delta K_i = -(c_i - t x_i S_i^{-1} - c_i + t\overline{x}_i S_i^{-1}) = -t(\overline{x}_i - x_i)S_i^{-1}\,, \tag{4.47}$$

$$\Delta K_0 = \Delta_1(K_1^\bullet)\Delta_2(K_2^*) = t^2(\overline{x}_1 - x_1)(\overline{x}_2 - x_2)S_1^{-1}S_2^{-1}\,. \tag{4.48}$$

Next,

$$\Delta(K_0 K_{1,2}) = \Delta_1 K_1^\bullet \Delta_2 K_2^* K_{1,2} = \Delta_1 K_1^\bullet (K_2^* K_{1,2} - K_2 K_{1,2}^*)\,, \tag{4.49}$$

where, after expanding, the quantity in parentheses is

$$\begin{aligned} K_2^* K_{1,2} - K_2 K_{1,2}^* =& c_1 c_2 t S_2^{-1}(\overline{x}_2 - x_2) - c_2 t \overline{x}_2 S_{1,2}^{-1,-1}(\overline{x}_2 - x_2) \\ =& c_2 t(\overline{x}_2 - x_2)(c_1 S_2^{-1} - \overline{x}_1 S_{1,2}^{-1,-1}) \,. \end{aligned} \tag{4.50}$$

Finally,

$$\Delta(K_0 K_{1,2}) = c_2 t(\overline{x}_2 - x_2)\Delta_1 K_1^\bullet (c_1 S_2^{-1} - \overline{x}_1 S_{1,2}^{-1,-1}) \,. \tag{4.51}$$

To simplify this quantity, note that

$$\Delta_1 K_1^\bullet c_1 S_2^{-1} = c_1 S_2^{-1} \Delta_1 K_1^\bullet = -c_1 t S_2^{-1}(\overline{x}_1 - x_1) \,, \tag{4.52}$$

$$\Delta_1 K_1^\bullet \overline{x}_1 S_{1,2}^{-1,-1} = S_{1,2}^{-1,-1} \Delta_1 K_1^\bullet \overline{x}_1 = c_1(\overline{x}_1 - x_1) S_{1,2}^{-1,-1} \,. \tag{4.53}$$

Therefore, we find that

$$\Delta(K_0 K_{1,2}) = c_1 c_2 t(\overline{x}_1 - x_1)(\overline{x}_2 - x_2)(t S_1^{-1} S_2^{-2} - S_{1,2}^{-1,-1}) \,, \tag{4.54}$$

and after we combine terms and simplify, we can write down the following corollary.

Corollary 4.23. *For a two-dimensional highly-symmetric model with a group with two nontrivial generators and $K_0 = K_1^\bullet K_2^*$, we have the functional equation*

$$K\Delta(K_0 Q) = \alpha t(\overline{x}_1 - x_1)(\overline{x}_2 - x_2)\left(t S_1^{-1} S_2^{-1}(1 - b_1 b_2 Q_{1,2}^{0,0}) + S_{1,2}^{-1,-1} Q_{1,2}^{0,0}\right) \,. \tag{4.55}$$

4.2.2 Groups of Order Six

We will go through this case faster because a lot of the work is similar. First, we tabulate all the unknowns Q_i^0 and their orbits under the actions of G. In the following table we write a sequence L for g_L and a reduced sequence for each unknown. For example, $g_1 Q_1^0 = Q_1^0$, so we write $\varnothing$ in the corresponding entry. For another example,

$$g_{2121} Q_1^0 = g_{212} Q_1^0 = g_{121} Q_1^0 = g_{12} Q_1^0 \,, \tag{4.56}$$

since $g_{212} = g_{121}$. The corresponding entry in the table is 12.

Sequence L	Alternative	Q_1^0	Q_2^0	K_1	K_2
1	21212	$\varnothing$	1	1	1
21	1212	2	21	21	21
121	212	12	21	121	212
2121	12	12	1	12	12
12121	2	2	$\varnothing$	2	2

Table 4.2: Reductions of group actions $g \in G$ on Q_i^0 and K_i.

After we apply the group actions, we find six different functions that need to be eliminated. Let

$$u_1 = Q_1^0\,, \quad u_2 = g_2 Q_1^0\,, \quad u_3 = g_{12} Q_1^0\,, \tag{4.57}$$

$$v_1 = Q_2^0\,, \quad v_2 = g_1 Q_2^0\,, \quad v_3 = g_{21} Q_2^0\,. \tag{4.58}$$

There are twelve different K_1's and K_2's that we label sequentially as $A_1, \ldots, A_6$ and $B_1, \ldots, B_6$. The matrix associated with the linear system of equations is

$$M_6 = \begin{bmatrix} A_1 & & & B_1 & & \\ A_2 & & & & B_2 & \\ & A_3 & & & & B_3 \\ & & A_4 & & & B_4 \\ & & A_5 & & B_5 & \\ & A_6 & & B_6 & & \end{bmatrix} \tag{4.59}$$

with determinant

$$\det M_6 = A_2 A_4 A_6 B_1 B_3 B_5 - A_1 A_3 A_5 B_2 B_4 B_6\,. \tag{4.60}$$

The even index A's are found by applying g_1 on the odd index A's, and vice versa for the B's. Then

$$\det M_6 = \Delta_1(A_2 A_4 A_6 B_1 B_3 B_5) = \Delta_1(g_1(K_1) g_{121}(K_1) g_{12}(K_1) K_2 g_{21}(K_2) g_{12}(K_2))\,. \tag{4.61}$$

In the following table we show the relations between A's and B's for the six individual models with one of the two groups of order 6. $S11$, $S12$ and $S13$ have the one group, and $S21$, $S22$ and $S23$, have the other.

Model	Name	A-relations	B-relations
	S11	$A_1 = A_6, A_2 = A_5, A_3 = A_4$	$B_1 = B_4, B_2 = B_3, B_5 = B_6$
	S12	$A_1 = A_4, A_2 = A_3, A_5 = A_6$	$B_1 = B_2, B_3 = B_6, B_4 = B_5$
	S13	None	None
	S21	$A_1 = A_6, A_2 = A_5, A_3 = A_4$	$B_1 = B_2, B_3 = B_6, B_4 = B_5$
	S22	$A_1 = A_4, A_2 = A_3, A_5 = A_6$	$B_1 = B_4, B_2 = B_3, B_5 = B_6$
	S23	None	None

Table 4.3: Relations for $A = K_1$ and $B = K_2$ for groups of order six.

It is easy to check that when a model has nontrivial relation, $\det M_6 = 0$. Also, using a a computer, we can check that the determinant is nonzero in the last two cases where there are no relations. For the models with zero determinant, we seek the same type of relation on β and its orbit as in the case of groups of order four. We have the following linear relation

$$\begin{aligned} 0 =& \gamma\beta - \gamma_1\beta_1 + \gamma_{21}\beta_{21} - \gamma_{121}\beta_{121} + \gamma_{12}\beta_{12} - \gamma_2\beta_2 \\ =& \gamma(A_1u_1 + B_1v_1) - \gamma_2(A_6u_2 + B_6v_1) - \gamma_1A_2u_1 + B_2v_2) \\ &+ \gamma_{12}(A_5u_3 + B_5v_2) + \gamma_{21}(A_3u_2 + B_3v_3) - \gamma_{121}(A_4u_3 + B_4v_3)\,. \end{aligned} \tag{4.62}$$

Expanding out and collecting terms produces the system of equations

$$A_1\gamma = A_2\gamma_1\,, \qquad A_3\gamma_{21} = A_6\gamma_2\,, \qquad A_5\gamma_{12} = A_4\gamma_{121}\,, \tag{4.63}$$

$$B_1\gamma = B_6\gamma_2\,, \qquad B_5\gamma_{12} = B_2\gamma_1\,, \qquad B_3\gamma_{21} = B_4\gamma_{121}\,. \tag{4.64}$$

It does not have a nontrivial symbolic solution, so we need to use the relations here. There is a rational solution in each case, where γ is a free variable. We set γ to be the least common multiple of the denominators. We find that $K_0 = \gamma$ in each of the four cases with relations, as illustrated in the following table.

Model	S-polynomial	Weight $\gamma = K_0$	K-form
	$x_1\overline{x}_2 + \overline{x}_1 + x_2$	$A_3A_5B_6$	$g_{21}(K_1)g_{12}(K_1)g_2(K_2)$
	$\overline{x}_1x_2 + x_1 + \overline{x}_2$	$A_2A_5B_6$	$g_1(K_1)g_{12}(K_1)g_2(K_2)$
	$x_1x_2 + \overline{x}_1 + \overline{x}_2$	$A_2A_3B_5B_6$	$g_1(K_1)g_{21}(K_1)g_{12}(K_2)g_2(K_2)$
	$\overline{x}_1\overline{x}_2 + x_1 + x_2$	A_2B_6	$g_1(K_1)g_2(K_2)$

Table 4.4: Weight K_0 for the Orbit Summation operators.

Here we have picked the respresentations of A's and B's with the least number of transformations to reach the resulting expression. We also tried finding a possible K_0 by brute force for $S13$ and $S23$, but we did not find any among products of distinct A's and B's. It is likely that such a weight does not exist for these cases, and that the algebraic kernel method does not extend for them.

Theorem 4.24. *The algebraic kernel method extends for all interacting lattice walk models with small steps and group of size six if and only if* $\det M_6 = 0$*. If we let* $\beta = KQ + K_{1,2}Q_{1,2}^{0,0} - \alpha$*, then we have the functional equation* $D_G(K_0\beta) = 0$*, where* K_0 *depends on the step set.*

4.2.3 Groups of Order Eight

The analysis for groups of order 8 is quite similar, so we omit most of it. Here are the results.

Model	S-polynomial
	$x_1 + \overline{x}_1 + x_1\overline{x}_2 + \overline{x}_1x_2$
	$x_1 + \overline{x}_1 + x_1x_2 + \overline{x}_1\overline{x}_2$

Table 4.5: Models with group of size 8.

We name the $A_1 = K_1$ and $B_1 = K_2$ iterates as before; for example, $A_2 = g_1A_1$ and $A_3 = g_2g_1A_1$.

Sequence L	Alternative	Q_1^0	Q_2^0
1	2121212	$\varnothing$	1
21	121212	2	21
121	21212	12	121
2121	1212	212	121
12121	212	212	21
212121	12	12	1
1212121	2	2	$\varnothing$

Table 4.6: Reductions of group actions $g \in G$ on Q_i^0 and K_i.

The relations are the same in both models

$$A_1 = A_6\,, \quad A_2 = A_5\,, \quad A_3 = A_4\,, \quad A_7 = A_8\,, \tag{4.65}$$

$$B_1 = B_4\,, \quad B_2 = B_3\,, \quad B_4 = B_8\,, \quad B_6 = B_7\,. \tag{4.66}$$

In both cases we retrieved the same K_0.

Theorem 4.25. *The algebraic kernel method extends for all interacting lattice walk models with small steps and group of size eight. The functional equation is $D_G(K_0\beta) = 0$, where $\beta = KQ + K_{1,2}Q_{1,2}^{0,0} - \alpha$ and*

$$K_0 = A_2A_3B_5B_6 = g_1(K_1)g_{21}(K_1)g_{212}(K_2)g_{1212}(K_2). \tag{4.67}$$

Chapter 5

The Obstinate Kernel Method

The next tool that we will explore is the obstinate kernel method. For work on a two-dimensional model with interactions, see Section 4.2 of [30, p. 16]. The method is similar to the algebraic kernel method with a weighted operator, where underneath the operator notation, all we are doing is finding more valid functional equations. In both methods we apply kernel-preserving transformations on the master functional equation. However, in the obstinate kernel method we also make a substitution with one of the roots $x_k = \sigma_k(t)$ of the kernel, where σ_k may not always be a power series in t. This is done in order to annihilate the KQ term from the resulting equations. However, some transformations, taken together with the substitutions, do not lead to valid functional equations because some of the generating functions fail to remain power series in t after the substitution takes place. That is, one or more of the terms $g(Q_i^0)|_{x_k=\sigma_k}$ is not a power series in t for some transformation g from G and some i and k. This inspires use the following language.

Definition 5.1. *A function $f(x_1, \ldots, x_d; t)$ is admissible if it is a valid power series in the variable t, that is, there are only non-negative orders of t with nonzero coefficients. A transformation g preserves the admissibility of f with respect to a root σ_k of the kernel K if $g(f)|_{x_k=\sigma_k}$ is admissible.*

To recap, it is necessary that when we use a substition $x_k = \sigma_k$ we only use transformations g from G that preserve the admissibility of functions Q_i^0 for all i in the master functional equation.

We will focus our work on modes with groups of order four, six and eight, where we have two nontrivial generators g_1 and g_2 for the group preserving the kernel. In this case $S_i^{\pm 1} \neq 0$ for any i and choice of sign. The roots of the kernel are

$$\sigma_k^{\pm} = \frac{1 - tS_k^0 \pm \sqrt{(1 - tS_k^0)^2 - 4t^2 S_k^1 S_k^{-1}}}{2tS_1^1} \tag{5.1}$$

and they share the properties

$$\sigma_k^+ \sigma_k^- = \frac{S_k^{-1}}{S_k^1} \quad \text{and} \quad \sigma_k^+ + \sigma_k^- = \frac{1 - tS_k^0}{tS_k^1} \,. \tag{5.2}$$

If we expand the roots in series of t, we find that

$$\sigma_k^- = S_k^{-1} t + O(t^2) \quad \text{and} \quad \sigma_k^+ = \frac{1}{S_k^1 t} - \frac{S_k^0}{S_k^1} - S_k^{-1} t + O(t^2) \,, \tag{5.3}$$

where the higher order terms are the same.

Generally, it is better if we use the $O(t)$ root, because this guarantees that the Q_k^0 terms are admissible in the master functional equation. However, sometimes even if we use the $O(1/t)$ root, we may find that the Q_k^0 terms are admissible. For example, see [30, p. 11]. In this thesis we will focus on the first case and leave the $O(1/t)$ root analysis for future work. It may happen that the $O(t)$ term does not lead to enough enough valid equations and in those cases we would have to look to the $O(1/t)$ root to find out if we can obtain enough equations. From now on let $\sigma_k = \sigma_k^-$.

It is sufficient to show that transformations from G preserve the admissibility of the functions x_1 and x_2, and thus the pair (x_1, x_2), with respect to a root, to show that the functions Q_k^0 are also admissible under those transformations and that substitution. That is, we only need to check when $g(x_j)|_{x_k=\sigma_k}$ is admissible some $j$and k, and for g in G. Luckily, a lot of the finite groups pass this test and we obtain enough different valid functional equations, so that we can use the obstinate kernel method. For those cases where we find less equations, it possible that we will be able to show that the Q_k^0 terms are power series in t via another method like in [30].

Proposition 5.2. *Consider a two-dimensional CLASSIC model with group of order four, six or eight. Models with the same group have the same admissibility properties for x_1 and x_2, and the pair (x_1, x_2). Models with group of order four all have the same admissibility properties.*

Proof. All we need to do is compute the orbit under group G of the functions and check if substituting σ_k^- leads to admissible functions. This is tedious to do by hand but straight-forward to do on a computer. The results are summarized in tables in the following sections. However, we can perform the analysis for groups of order four without any difficulty.

Consider the function x_1. The transformations 1 and g_2 leave it unchanged, but the transformations g_1 and $g_1 g_2$ change it to $g_1(x_1) = \overline{x}_1 S_1^{-1}/S_1^1$. Note that $\overline{\sigma}_1 = \sigma_1^+ S_1^1/S_1^{-1}$, so $g_1(x_1)|_{x_1=\sigma_1} = \sigma_1^+$ and this function is not admissible. Therefore x_1's admissibility with respect to σ_1 is preserved only by the transformations $\{1, g_2\}$.

Since models with groups of order four need to have one symmetry across an axis, without

loss of generality, we can assume the symmetry is across the x_1 axis. That is, $g_2(x_2) = \overline{x}_2$, and the $x_1 = \sigma_1$ substitution trivially leads to an admissible function. Therefore, all transformation from G preserve the admissibility of x_2 with respect to σ_1. Therefore, only 1 and g_2 preserve the admissibility for both x_1 and x_2 with respect to σ_1. □

5.1 Initial Computations

This classification shows that we can treat all of the models with group of order four with the same method. The situation is somewhat more complicated when the order of the group is six. The models with $S = S_{1k}$ for some k are amenable to the obstinate kernel method but the ones with $S = S_{2k}$ would be more difficult to analyze, sine we have less group elements that keep the pair (x_1, x_2) admissible. The situation is similar when the group has size eight. Next, suppose that we have a set of valid equations for a substitution $x_k = \sigma_k$. They are a linear system in the unknown functions $Q^{0,0}_{1,2}$, Q^0_1, Q^0_2, and their orbits under the set of valid group actions. We hope for a high level of redundancy in these equations, because it will allow us to eliminate some of the unknowns by taking a weighted sum over the valid equations. That is, we attain a weighted orbital summation operator. It will be implicit in the analysis that we have made the substitution $x_k = \sigma_k$.

It turns out that it is enough to have $|G|/2$ transformations preserving the admissibility for both x_1 and x_2.

5.1.1 Groups of Order Four

We start with the admissibility table for groups of order four.

Root	$x_1(t)$	$x_2(t)$	$x_1(t)$ and $x_2(t)$
$\sigma_1(t)$	1, g_2	All	1, g_2
$\sigma_2(t)$	All	1, g_1	1, g_1

Table 5.1: Admissibility-preserving transformations with respect to roots of the kernel with group of order four.

We will work with the σ_1 substitution and group actions 1 and g_2, where we use the notation $g_2(F) = F^*$ to reduce clutter. The analysis for the σ_2 substitution is analogous. The functional

equations corresponding the the group actions are

$$KQ - K_1Q_1^0 - K_2Q_2^0 + K_{12}Q_{1,2}^{0,0} = \alpha\,, \tag{5.4}$$

$$KQ^* - K_1^*Q_1^{0,*} - K_2^*Q_2^0 - K_{12}^*Q_{1,2}^{0,0} = \alpha\,. \tag{5.5}$$

Once we make the substitution, the function Q_2^0 will be difficult to expand, so we would like to eliminate it first. We multiply the equations by K_2^* and K_2, respectively, and subtract the second one from the first one. The resulting equation is

$$K_2^*KQ - K_2KQ^* - (K_2^*K_1Q_1^0 - K_2K_1^*Q_1^{0,*}) + (K_2^*K_{12} - K_2K_{12}^*)Q_{1,2}^{0,0} - \alpha(K_2^* - K_2) = 0\,, \tag{5.6}$$

where we can set $x_1 = \sigma_1$ to eliminate the K term. The substitution will be implicit in the notation from now on. We have

$$(K_2^*K_1Q_1^0 - K_2K_1^*Q_1^{0,*}) - (K_2^*K_{12} - K_2K_{12}^*)Q_{1,2}^{0,0} + \alpha(K_2^* - K_2) = 0. \tag{5.7}$$

We can recast this elimination via the operator

$$D_2 = g_2(K_2)\cdot 1 - K_2g_2 = (1 - g_2)K_2^*\,, \tag{5.8}$$

where we have the application of transformations and multiplications by weights. If we apply it on the master functional equation, we obtain

$$\begin{aligned} D_2(KQ - K_1Q_1^0 - K_2Q_2^0 + K_{12}Q_{1,2}^{0,0}) &= 0 \\ D_2(KQ) - D_2(K_1Q_1^0) + D_2(K_{12}Q_{1,2}^{0,0}) &= 0\,, \end{aligned} \tag{5.9}$$

and once we make the σ_1 substitution, we find that

$$D_2(K_1Q_1^0) - D_2(K_{12})Q_{1,2}^{0,0} + D(\alpha) = 0\,. \tag{5.10}$$

The following notation will be useful in the next few sections.

Definition 5.3. *Let L be a concatenated list, $\Delta_L = 1 - g_L$ and let X be some function. The operator $D_L[X]$ is defined by its action on functions F as*

$$D_L[X](F) = \Delta_L(XF) = (1 - g_L)(XF) = XF - g_L(XF)\,. \tag{5.11}$$

We will drop the X from the notation if it is clear from context. In particular, $D_L[1] = \Delta_L$.

For example, $D_2 = D_2[K_2^*] =]D_2[g_2(K_2)]$.

Example 5.4. If we use $g_L(X)$ instead of X in the definition above, we find that

$$D_L[g_L(X)] = g_L(X) - Xg_L \tag{5.12}$$

is essentially an Ore operator.

Now, we can expand some of the terms in the functional equation. First,

$$\begin{aligned} D_2(\alpha) &= \alpha(K_2^* - K_2) \\ &= \alpha((c_2 - tx_2\overline{\mu}_2 S_2^{-1}) - (c_2 - t\overline{x}_2 S_2^{-1})) \\ &= \alpha t(\overline{x}_2 - x_2\overline{\mu}_2(\sigma_1))S_2^{-1}(\sigma_1)\,. \end{aligned} \tag{5.13}$$

Then,

$$\begin{aligned} D_2(K_{12}) &= K_2^* K_{12} - K_2 K_{12}^* \\ &= c_1 c_2 t(\overline{x}_2 - x_2\overline{\mu}_2)S_2^{-1} - c_2 t\overline{\sigma}_1(\overline{x}_2 - x_2\overline{\mu}_2)S_{1,2}^{-1,-1} \\ &= c_2 t(\overline{x}_2 - x_2\overline{\mu}_2(\sigma_1))(c_1 S_2^{-1}(\sigma_1) - \overline{\sigma}_1 S_{1,2}^{-1,-1})\,. \end{aligned} \tag{5.14}$$

If the model is highly-symmetric, we see that $g_2 S_1^{-1} = S_1^{-1}$ and therefore, $g_2 K_1 = K_1$, and we may bring that factor outside of the operator, so $D_2(K_1^* F) = K_1 D_2(F)$ for functions F. It turns out that this happens for both substitutions if and only if the model is highly-symmetric, meaning that the analysis will be easier only in those cases.

Proposition 5.5. *Consider a CLASSIC model with group of order four. If $i \neq j$, then $D_i[g_i K_j] = K_j\Delta_i$ in the σ_j substitutions if and only if the model has a symmetry around the x_i axis. Thus, we find that the K functions factor out of the operators in both substitutions if and only if the model is highly-symmetric.*

Proof. **[FIXME: This mess.]** Since the models must have some symmetry, assume that the model is symmetric across the x_2 axis.

This implies that $S_2^{-1} = S_2^1$ and $g_2(S_1^k) = S_1^k$ for all k, and this immediately tells us that $g_2(K_1) = K_1$ and $D_2[g_2 K_1] = K_1\Delta_2$. The equalities remain true when we substitute $x_1 = \sigma_1$. Thus, we only need to show the forward direction of the statement.

We complete the proof via the following computation. We will show that if the model is not highly-symmetric, the factoring will not happen. Assume that the model is symmetric across the x_2 axis. It is enough to show that $g_2(K_1) \neq K_1$ even after the $x_1 = \sigma_1$ substitution is made. Due to

the symmetry, $\overline{\sigma}_1 = \sigma_1^+$. In other words, if $\sigma_1 = u - v\sqrt{w}$ the $\overline{\sigma}_1 = u + v\sqrt{w}$, and $\sigma_1 + \overline{\sigma}_1 = 2u$. Now,

$$g_2 K_1 = c_2 - t\overline{x}_1 g_2(S_1^{-1}(x_2)) = c_2 - t\overline{x}_1 S_1^{-1}(\overline{x}_2 \mu_2(x_1)) \tag{5.15}$$

and $g_2 K_1 = K_1$ if and only if

$$S_1^{-1}(\overline{x}_2 \mu_2(x_1)) = S_1^{-1}(x_2) \tag{5.16}$$

upon making the substitution $x_1 = \sigma_1$. Note that

$$S_2^k(\sigma_1) = S_{1,2}^{0,k} + S_{1,2}^{1,k}(\overline{\sigma}_1 + \sigma_1) = S_{1,2}^{0,k} + S_{1,2}^{1,k} \frac{1 - tS_1^0}{tS_1^1} \tag{5.17}$$

and therefore,

$$\mu_2(\sigma_1) = \frac{S_2^{-1}}{S_2^1} = \frac{S_{1,2}^{0,-1} + S_{1,2}^{1,-1}(1 - tS_1^0)/(tS_1^1)}{S_{1,2}^{0,1} + S_{1,2}^{1,1}(1 - tS_1^0)/(tS_1^1)} = \frac{S_{1,2}^{1,-1} + t\left(S_{1,2}^{0,-1} S_1^1 - S_{1,2}^{1,-1} S_1^0\right)}{S_{1,2}^{1,1} + t\left(S_{1,2}^{0,1} S_1^1 - S_{1,2}^{1,1} S_1^0\right)}. \tag{5.18}$$

This is a rational function in x_2. Using a computer, we can compute $S_1^{-1}(\overline{x}_2 \mu_2(x_1)) - S_1^{-1}(x_2)$ for groups of order four and find that it is zero only when the model is highly-symmetric. □

We will investigate the highly-symmetric models in more detail in Section 5.2.

5.1.2 Groups of Order Six

We start with the admissibility tables for the two groups of order six.**[FIXME: Check tables]**

Root	x_1	x_2	(x_1, x_2)
σ_1	$1, g_2, g_{21}, g_{121}$	$1, g_1, g_2, g_{21}$	$1, g_2, g_{21}$
σ_2	$1, g_1, g_2, g_{12}$	$1, g_1, g_{12}, g_{121}$	$1, g_1, g_{12}$

Table 5.2: Admissibility-preserving transformations with respect to roots of the kernel with group of order six for the models $S11$, $S12$, $S13$.

Root	x_1	x_2	(x_1, x_2)
σ_1	$1, g_2, g_{21}, g_{121}$	$1, g_1, g_{12}, g_{121}$	$1, g_{121}$
σ_2	$1, g_2, g_{21}, g_{121}$	$1, g_1, g_{12}, g_{121}$	$1, g_{121}$

Table 5.3: Admissibility-preserving transformations with respect to roots of the kernel with group of order six for the models $S21$, $S22$, $S33$.

We can perform the obstinate kernel method on the three models in the first group. They have the following step sets.

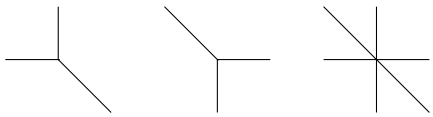


Figure 5.1: Steps for models $S11$, $S12$, $S13$.

The admissibility of (x_1, x_2) is preserved by the transformations 1, g_j and g_{jk} for $j \neq k$. For consistency, we will work on the admissibility with respect to σ_1. The actions are 1, g_2, g_{21}. We have the table from the algebraic kernel method.

Sequence L	Q_1^0	Q_2^0
$\varnothing$	$\varnothing$	$\varnothing$
2	2	$\varnothing$
21	2	21

Table 5.4: Reductions of group actions $g \in G$ on Q_1^0 and Q_2^0.

We wish to eliminate $X_2 = g_2 Q_1^0$ and $Y = Q_2^0$. Let

$$\beta = K_{12} Q_{1,2}^{0,0} - \alpha = \alpha(b_1 b_2 Q_{1,2}^{0,0} - 1)\,, \tag{5.19}$$

so that we have

$$AX + BY = \beta\,, \tag{5.20}$$

$$A_2 X_2 + B_2 Y = \beta_2\,, \tag{5.21}$$

$$A_{21} X + B_{21} Y_{21} = \beta_{21}\,. \tag{5.22}$$

We want a linear combination of β, β_2 and β_{21} that will eliminate X_2 and Y. In this case, we may write

$$a\beta - b\beta_2 + c\beta_{21} = aAX + (aB - bB_2)Y + (cA_{21} - bA_2) + cB_{21}Y_{21} \tag{5.23}$$
$$= aAX + cB_{21}Y_{21}\,, \tag{5.24}$$

and for the elimination to work, we need $aB = bB_2$ and $cA_{21} = bA_2$. One solution for a, b and c without denominators is

$$a = A_{21}B_2\,, \quad b = A_{21}B\,, \quad c = A_2B\,, \tag{5.25}$$

and these functions may have common factors. In other words, the elimination procedure yields

$$\begin{aligned} AA_{21}B_2Q_1^0 + A_2BB_{21}g_{21}Q_2^0 &= A_{21}B_2\beta - A_{21}B\beta_2 + A_2B\beta_{21} \\ &= (A_{21}B_2 - A_{21}B + A_2B)\beta\,. \end{aligned} \tag{5.26}$$

We can recast this equation into operator notation.

Proposition 5.6. *Let*

$$D(F) = A_{21}B_2F - A_{21}Bg_2(F) + A_2Bg_{21}(F)\,. \tag{5.27}$$

Then we have

$$D(AX + BY) = AA_{21}B_2Q_1^0 + A_2BB_{21}g_{21}Q_2^0 = \beta D(1)\,. \tag{5.28}$$

We may compute $D(1)$ for each case, but the expressions are somewhat complicated. $D(1)$ for each of $S11$, $S12$ and $S13$ will contain a square root. So do the terms $AA_{21}B_2$ and A_2BB_{21}.

5.1.3 Groups of Order Eight

We start with the admissibility tables for the two groups of order eight.**[FIXME: Check tables]**

Root	x_1	x_2	(x_1, x_2)
σ_1	1, g_2, g_{21}, g_{212}	all $g \in G$	1, g_2, g_{21}, g_{212}
σ_2	1, g_1, g_2, g_{12}, g_{121}, g_{2121}	1, g_1, g_{12}, g_{121}	1, g_1, g_{12}, g_{121}

Table 5.5: Admissibility-preserving transformations with respect to roots of the kernel with group of order eight for model $S1$.

Root	x_1	x_2	(x_1, x_2)
σ_1	1, g_2, g_{21}, g_{212}	all $g \in G$	1, g_2, g_{21}, g_{212}
σ_2	1, g_2, g_{21}, g_{121}, g_{212}, g_{2121}	1, g_1, g_{12}, g_{121}	1, g_{121}

Table 5.6: Admissibility-preserving transformations with respect to roots of the kernel with group of order eight for model $S2$.

Since the group G has order eight, we only have two non-isomorphic models, but we will work with the one with four group actions that preserve the admissibility of (x_1, x_2). The actions are 1, g_2, g_{21}, g_{212} and the step polynomial is

$$S = \overline{x}_1 + x_1 + \overline{x}_1 x_2 + x_1 \overline{x}_2 \tag{5.29}$$

This is the same step polynomial as in [30]. We will go through some of the derivation using this new perspective.

Sequence L	Q_1^0	Q_2^0
$\varnothing$	$\varnothing$	$\varnothing$
2	2	$\varnothing$
21	2	21
212	212	21

Table 5.7: Reductions of group actions $g \in G$ on Q_1^0 and Q_2^0 that preserve admissibility with respect to σ_1.

As before, let $\beta = K_{12} Q_{1,2}^{0,0} - \alpha = \alpha(b_1 b_2 Q_{1,2}^{0,0} - 1)$, $X = Q_1^0$ and $Y = Q_2^0$. Note that β is left unchanged by group actions. We have the four equations

$$AX + BY = \beta\,, \tag{5.30}$$

$$A_2 X_2 + B_2 Y = \beta\,, \tag{5.31}$$

$$A_{21} X_2 + B_{21} Y_{21} = \beta\,, \tag{5.32}$$

$$A_{212} X_{212} + B_{212} Y_{21} = \beta\,, \tag{5.33}$$

where we wish to eliminate the unknown X and Y functions that contain σ_1 in their input. Those are X_2, Y and Y_{21}, and we can eliminate them by first eliminating Y and Y_{21} from the first two and second two equations, and then eliminating X_2 from the resulting two equations. That is, we

compute B_2 times the first equation minus B times the second equation, and B_{212} times the third equation minus B_{21} from the last equation. The procedure is the same as applying the following two operators

$$D_2[B_2] = B_2 - Bg_2\,, \tag{5.34}$$

$$-g_{212}D_2[B_2] = g_{212}\,Bg_2 - g_{212}\,B_2 = B_{212}\,g_{21} - B_{21}\,g_{212}\,, \tag{5.35}$$

to both sides of the master functional equation and substituting $x_1 = \sigma_1$. We obtain the equations

$$AB_2X - A_2BX_2 = \beta(B_2 - B)\,, \tag{5.36}$$

$$A_{21}B_{212}X_2 - A_{212}B_{21}X_{212} = \beta(B_{212} - B_{21})\,. \tag{5.37}$$

To eliminate the X_2 terms, we take $A_{21}B_{212}$ times the first equation and add A_2B times the second equation. We have

$$A_{21}B_{212}AB_2X - A_2BA_{212}B_{21}X_{212} = \beta(A_{21}B_{212}(B - B_2) + A_2B(B_{212} - B_{21})) \tag{5.38}$$

We could recover the same result by applying the following operator D to both sides of the master functional equation.

$$\begin{aligned} D &= A_{21}B_{212}D_2[B_2] - A_2Bg_{212}D_2[B_2] \\ &= (A_{21}B_{212} - A_2Bg_{212})D_2[B_2] \\ &= (A_{21}B_{212} - g_{212}A_{21}B_{212})D_2[B_2] \\ &= D_{212}[A_{21}B_{212}]D_2[B_2]\,. \end{aligned} \tag{5.39}$$

From the master functional equation, we have

$$D(\beta) = \beta D_{212}[A_{21}B_{212}](B_2 - B)\,, \tag{5.40}$$

$$\begin{aligned} D(AX + BY) &= D(AX) \\ &= D_{212}[A_{21}B_{212}]D_2[B_2](AX) \\ &= D_{212}[A_{21}B_{212}](AB_2X - A_2BX_2) \\ &= D_{212}[A_{21}B_{212}](AB_2X) \\ &= D_{212}[AA_{21}B_2B_{212}](X)\,. \end{aligned} \tag{5.41}$$

Note that $A_{212} = A$, so we can factor out A from the operator. We have the following result.

Proposition 5.7. *For the first model with group of order eight, we have*

$$AD_{212}[A_{21}B_2B_{212}]\left(Q_1^0\right) = \beta D_{212}[A_{21}B_{212}](B_2 - B)\,, \tag{5.42}$$

where $A_L = g_L(K_1)$, $B_L = g_L(K_2)$ and $\beta = \alpha(b_1b_2Q_{1,2}^{0,0} - 1)$, and the equality is valid only once we set $x_1 = \sigma_1$ in this expression.

For the statement for the σ_2 substitutions, switch the A's and the B's, and the 1's and the 2's.

The interesting fact, as found in [30], is that the coefficients in front of the Q's become Laurent polynomials once we take into consideration the σ_k-substitutions, even though the σ_k's contain square roots. The coefficient in front of β can be reduced to a rational function plus a rational function times a square root after properly conjugating. The calculations are exceedingly complicated and would be difficult to do by hand. The square root part does not depend on a_2 in the first case, and a_1 in the second case. This observation allows the authors to find relationships between the interacting and non-interacting versions of $Q_{1,2}^{0,0}$.

5.2 Highly-symmetric Two-dimensional Models

We will continue further with the obstinate kernel method for highly-symmetric models where we have

$$x_1 = \sigma_1: \quad \Delta_2(K2^*Q_1^0) = \frac{1}{K_1}\left(\Delta_2(K_2^*K_{12})Q_{1,2}^{0,0} - \alpha(K_2^* - K_2)\right)\,, \tag{5.43}$$

$$x_2 = \sigma_2: \quad \Delta_1(K1^\bullet Q_2^0) = \frac{1}{K_2}\left(\Delta_1(K_1^\bullet K_{12})Q_{1,2}^{0,0} - \alpha(K_1^\bullet - K_1)\right)\,. \tag{5.44}$$

It is enough to perform the analysis for one of the equations. Only the model with two horizontal or vertical steps removed requires working with both equations. In highly-symmetric models $\mu_2 = S_2^{-1}/S_2^1 = 1$ and

$$\alpha(K_2^* - K_2) = \alpha t(\overline{x}_2 - x_2)S_2^{-1}(\sigma_1)\,, \tag{5.45}$$

$$\Delta_2(K_2^*K_{12}) = c_2t(\overline{x}_2 - x_2)(c_1S_2^{-1}(\sigma_1) - \overline{\sigma}_1S_{1,2}^{-1,-1})\,. \tag{5.46}$$

To compute $\Delta_2(K2^*Q_1^0)$, we need to compute $K_2(\sigma_1, x_2)$ and $K_2^*(\sigma_1, x_2)$, which are given by

$$K_2 = c_2 - t\overline{x}_2S_2^{-1}(\sigma_1) \quad \text{and} \quad K_2^* = c_2 - tx_2S_2^{-1}(\sigma_1)\,, \tag{5.47}$$

where $S_2^{-1}(\sigma)$ is a rational function given by

$$S_2^{-1}(\sigma_1) = \frac{S_{1,2}^{1,1} + tS_{1,2}^{0,1}S_{1,2}^{1,0}}{t\left(S_{1,2}^{1,0} + S_{1,2}^{1,1}(\overline{x}_2 + x_2)\right)}\,. \tag{5.48}$$

Here is a table sumarizing the values of this function for the different models. **[FIXME: Do we need the middle columns?]**

Model	$S_{1,2}^{0,1}$	$S_{1,2}^{1,0}$	$S_{1,2}^{1,1}$	$S_2^{-1}(\sigma_1)$
	1	1	0	1
	0	0	1	$\overline{t}/(\overline{x}_2 + x_2)$
	1	0	1	$\overline{t}/(\overline{x}_2 + x_2)$
	0	1	1	$\overline{t}/(\overline{x}_2 + 1 + x_2)$
	1	1	1	$(1 + \overline{t})/(\overline{x}_2 + 1 + x_2)$

Table 5.8: Function $S_2^{-1}(\sigma_1)$ for highly-symmetric models.

Lemma 5.8. *For highly-symmetric models with $e = S_{1,2}^{1,1}$, $f = S_{1,2}^{1,0}$ and $g = S_{1,2}^{0,1}$, we have*

$$\frac{\alpha(K_2^* - K_2)}{K_1} = \frac{\alpha(\overline{x}_2 - x_2)(e + tfg)}{S_1^1 K_1}\,, \tag{5.49}$$

$$\frac{\Delta_2(K_2^* K_{12})}{K_1} = \frac{c_2(\overline{x}_2 - x_2)(c_1(e + tfg) - et\overline{\sigma}_1 S_1^1)}{S_1^1(c_1 - t\overline{\sigma}_1 S_1^1)}\,. \tag{5.50}$$

5.2.1 Cartesian Model – Steps N, S, E, W

Since $S_2^{-1}(\sigma_1) = 1$, this is the simplest case to solve. What follows is essentially one long computation similar to the one we did after Proposition 2.6 in Chapter 2. However, this time we will not obtain a linear relationship between the quantities. Afterwards, we will solve for Q in terms of various version of itself with partial interactions, and $Q_{1,2}^{0,0}$.

Here $K_{12} = c_1 c_2$, so it behaves like α and we have

$$\Delta_2(K2^* Q_1^0) + \frac{\alpha\beta(K_2^* - K_2)}{K_1} = 0\,, \tag{5.51}$$

where $\beta = 1 - b_1 b_2 Q^{0,0}_{1,2}$ and

$$\frac{\alpha\beta(K_2^* - K_2)}{K_1} = \frac{\alpha\beta t(\overline{x}_2 - x_2)}{K_1} = \alpha\beta f(c_1, x_2)\,, \tag{5.52}$$

since K_1 does not depend on c_2 or x_1. The equation finally becomes

$$(c_2 - t x_2) Q^0_1(x_2) - (c_2 - t\overline{x}_2) Q^0_1(\overline{x}_2) + \alpha\beta f = 0\,. \tag{5.53}$$

We extract the x_2 term to find

$$c_2 Q^{0,1}_{1,2} - t Q^{0,0}_{1,2} + \alpha\beta f^1_2 = 0\,, \tag{5.54}$$

and then the b_2^0 term to find

$$-t\check{Q}^{0,0}_{1,2} + \overline{a}_1 f^1_2 = 0\,. \tag{5.55}$$

We can substitute the latter equation in the first one to obtain the identity

$$b_2 Q^{0,1}_{1,2} = t(a_2 Q^{0,0}_{1,2} - \beta\check{Q}^{0,0}_{1,2})\,. \tag{5.56}$$

Similarly, if we went with the other σ substitutions, we would have obtained the identity

$$b_1 Q^{1,0}_{1,2} = t(a_1 Q^{0,0}_{1,2} - \beta\widehat{Q}^{0,0}_{1,2})\,. \tag{5.57}$$

Using the same methods, we may also find that

$$b_2 Q^{1,1}_{1,2} = t(a_2 Q^{1,0}_{1,2} - \beta\check{Q}^{1,0}_{1,2})\,. \tag{5.58}$$

Multiplying by b_1 and using the relations above, leads to

$$\begin{aligned} b_1 b_2 Q^{1,1}_{1,2} &= t a_2 t(a_1 Q^{0,0}_{1,2} - \beta\widehat{Q}^{0,0}_{1,2}) - t\beta t(a_1\check{Q}^{0,0}_{1,2} - \beta\check{\widehat{Q}}^{0,0}_{1,2}) \\ &= t^2(a_1 a_2 Q^{0,0}_{1,2} - a_2\beta\widehat{Q}^{0,0}_{1,2} - a_1\beta\check{Q}^{0,0}_{1,2} + \beta^2\check{\widehat{Q}}^{0,0}_{1,2})\,. \end{aligned} \tag{5.59}$$

Let $G = Q^{0,0}_{1,2}$, $\widehat{G} = G_1$ and $\check{G} = G_2$. From the $x_1^0 x_2^0$ term of the master equation, we have

$$Q^{0,0}_{1,2} = 1 + t a_1 a_2 (Q^{1,0}_{1,2} + Q^{0,1}_{1,2})\,. \tag{5.60}$$

Multiplying by $b_1 b_2$ and using the relations above, leads to

$$\begin{aligned} b_1 b_2 G &= b_1 b_2 + t a_1 a_2 (b_2 t (a_1 G - \beta G_1) + b_1 t (a_2 G - \beta G_2) \\ &= b_1 b_2 + t^2 a_1 a_2 (b_2 a_1 G + b_1 a_2 G - b_2 \beta G_1 - b_1 \beta G_2) \,. \end{aligned} \tag{5.61}$$

Using the relation for β and simplifying leads to

$$G = \frac{b_1 b - 2 - t^2 a_1 a_2 (b_2 G_1 + b_1 G_2)}{b_1 b_2 - t^2 a_1 a_2 (a_2 b_1 + a_1 b_2 + b_1 b_2 (b_2 G_1 + b_1 G_2))} \,. \tag{5.62}$$

If we let $F = (b_2 G_1 + b_1 G_2)$ and perform the long division with respect to F, we find that

$$Q_{1,2}^{0,0} = \frac{1}{b_1 b_2} \left(1 - \frac{b_1 b_2 (1 - b_1 b_2) - t^2 a_1 a_2 (a_1 b_2 + a_2 b_1)}{b_1 b_2 + t^2 a_1 a_2 (a_1 b_2 + a_2 b_1 + b_1 b_2 F)} \right) \,. \tag{5.63}$$

Using arguments similar to those in the later section on diagonal steps, we can find out more about the various Q's.

Lemma 5.9. *For non-negative integers m and n,*

$$b_2 Q_{1,2}^{m,n+1} = t (a_2 Q_{1,2}^{m,n} - \beta \breve{Q}_{1,2}^{m,n}) \,, \tag{5.64}$$

$$b_1 Q_{1,2}^{m+1,n} = t (a_1 Q_{1,2}^{m,n} - \beta \widehat{Q}_{1,2}^{m,n}) \,. \tag{5.65}$$

If we sum the second set of equations multiplied by x_2^n, we find the following equations

$$b_1 Q_1^{m+1} = t (a_1 Q_1^m - \beta \widehat{Q}_1^m) \,. \tag{5.66}$$

Now, we sum these equations with respect to x_1 to find

$$b_1 (Q - Q_1^0) = x_1 t (a_1 Q - \beta \widehat{Q}) \,, \tag{5.67}$$

and we can solve this equation for Q_1^0 as follows

$$b_1 Q_1^0 = (b_1 - a_1 x_1 t) Q + x_1 t \beta \widehat{Q} \,. \tag{5.68}$$

We would have found an analogous expression for $b_2 Q_2^0$ if we performed the procedure on the first set of equations by summing with respect to x_1 first and then with respect to x_2. We substitute these

two expressions into the original functional equation, and after some algebra we find that

$$Q = \frac{(1 - b_1 b_2 G)(\alpha + t x_1 \overline{b}_1 K_1 \widehat{Q} + t x_2 \overline{b}_2 K_2 \breve{Q})}{K - (1 - \overline{c}_1 x_1 t) K_1 - (1 - \overline{c}_2 x_2 t) K_2} \tag{5.69}$$

We recognize some of the quantities as modified K_1 and K_2's, so we can finally write down the full solution for Q.

Theorem 5.10. *The Cartesian model in two dimensions has generating functions satisfying the equations*

$$Q^{0,0}_{1,2} = G = \frac{1}{b_1 b_2} \left(1 - \frac{b_1 b_2 (1 - b_1 b_2) - t^2 a_1 a_2 (a_1 b_2 + a_2 b_1)}{b_1 b_2 + t^2 a_1 a_2 (a_1 b_2 + a_2 b_1 + b_1 b_2 F)} \right) , \tag{5.70}$$

where $F = (b_2 \widehat{G} + b_1 \breve{G})$*, and*

$$Q = \frac{(1 - b_1 b_2 Q^{0,0}_{1,2})(\alpha - \overline{b}_1 K_1 \widehat{K}^{\bullet}_1 \widehat{Q} - \overline{b}_2 K_2 \breve{K}^{*}_2 \breve{Q})}{K - \overline{c}_1 K_1 K^{\bullet}_1 - \overline{c}_2 K_2 K^{*}_2} . \tag{5.71}$$

This shows that Q is built from versions of itself, each with a single interaction, and $Q^{0,0}_{1,2}$.

5.2.2 Diagonal Cartesian Model – Steps NE, NW, SE, SW

The next model in the progression of difficulty is the one with the diagonal step set. This model is similar to the one covered in [21], where we can find a model of two interacting walks confined in a semi-infinite strip. The models are different because there we have an extra term in the functional equations. Here $S^{-1,-1}_{1,2} = 1$, so K_{12} is nontrivial.

First, we will perform a calculation similar to the one we did after Proposition 2.6 in Chapter 2. Then we will find relations like those in Lemma 5.9 in the previous subsection. Finally we will use these relation to find a linear relation for $Q^{0,0}_{1,2}$ like its one-dimensional analogue in 2.8.

In this case,

$$\frac{\alpha(K^{*}_2 - K_2)}{K_1} = \frac{\alpha(\overline{x}_2 - x_2)}{K_1(\overline{x}_2 + x_2)} , \tag{5.72}$$

$$\frac{\Delta_2(K^{*}_2 K_{12})}{K_1} = \frac{c_2(\overline{x}_2 - x_2)(c_1 - t\overline{\sigma}_1 S^1_1)}{S^1_1(c_1 - t\overline{\sigma}_1 S^1_1)} = \frac{c_2(\overline{x}_2 - x_2)}{\overline{x}_2 + x_2} , \tag{5.73}$$

and the functional equation becomes

$$K_2^* Q_1 - K_2 Q_1^* - \frac{c_2(\overline{x}_2 - x_2)}{\overline{x}_2 + x_2} Q_{1,2} + \alpha \frac{\overline{x}_2 - x_2}{(\overline{x}_2 + x_2)K_1} = 0\,. \tag{5.74}$$

Surprisingly enough, the only remaining term depending on σ_1 is K_1, so it will be of no consequence to the rest of the analysis. We just need to take care of the remaining terms and we do this by multiplying the equation by $\overline{x}_2 + x_2$. We have

$$R^* = (\overline{x}_2 + x_2)K_2^* = c_2(\overline{x}_2 + x_2) - x_2 = c_2\overline{x}_2 - \overline{a}_2 x_2\,, \tag{5.75}$$

$$R = (\overline{x}_2 + x_2)K_2 = c_2(\overline{x}_2 + x_2) - \overline{x}_2 = c_2 x_2 - \overline{a}_2\overline{x}_2\,. \tag{5.76}$$

There are a couple of relations here

$$R^* - R = v(K_2^* - K_2) = \overline{x}_2 - x_2\,, \tag{5.77}$$

$$\overline{x}_2 R^* - x_2 R = \overline{x}_2 c_2 v - 1 - x_2 c_2 v + 1 = c_2 v(\overline{x}_2 - x_2)\,. \tag{5.78}$$

The functional equation becomes

$$(c_2\overline{x}_2 - \overline{a}_2 x_2)Q_1^0(x_2) + (\overline{a}_2\overline{x}_2 - c_2 x_2)Q_1^0(\overline{x}_2) + (\alpha\overline{K}_1 - c_2 Q_{1,2}^{0,0})(\overline{x}_2 - x_2) = 0 \tag{5.79}$$

and this form finally lets us perform coefficient extractions. Let $\delta_0 = \overline{a}_1(\overline{x}_2 - x_2)\overline{K}_1$. We rewrite the equation as

$$R^* Q_1^0 - R Q_1^{0,*} - c_2(\overline{x}_2 - x_2)Q_{1,2}^{0,0} + \overline{a}_2\delta_0 = 0\,. \tag{5.80}$$

We will use the master equation

$$(1 - tS)Q - (c_1 - t\overline{x}_1 S_1^{-1})Q_1^0 - (c_2 - t\overline{x}_2 S_2^{-1})Q_2^0 + (c_1 c_2 - t\overline{x}_1\overline{x}_2 S_{1,2}^{-1,-1})Q_{1,2}^{0,0} = \overline{a}_1\overline{a}_2 \tag{5.81}$$

to further our work here. For diagonal steps, it is

$$(1 - tS)Q - (c_1 - tv\overline{x}_1)Q_1^0 - (c_2 - t\overline{x}_2 S_2^{-1})Q_2^0 + (c_1 c_2 - t\overline{x}_1\overline{x}_2)Q_{1,2}^{0,0} = \overline{a}_1\overline{a}_2 \tag{5.82}$$

and upon extracting the x_1^0 coefficient we find that

$$Q_1^0 = \overline{a}_2 + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^1\,. \tag{5.83}$$

We can extract higher order coefficients too. Let $k \geqslant 1$. Then the x_2^k coefficient is

$$\begin{aligned} 0 &= Q_1^k - tS_1^{-1}Q_1^{k+1} - tS_1^0Q_1^k - tS_1^1Q_1^{k-1} - c_2Q_{1,2}^{k,0} \\ &\quad + t\overline{x}_2(S_{1,2}^{-1,-1}Q_{1,2}^{k+1,0} + S_{1,2}^{0,-1}Q_{1,2}^{k,0} + S_{1,2}^{1,-1}Q_{1,2}^{k-1,0}) \\ &= Q_1^k - tvQ_1^{k+1} - tvQ_1^{k-1} - c_2Q_{1,2}^{k,0} + t\overline{x}_2(Q_{1,2}^{k+1,0} + Q_{1,2}^{k-1,0})\,. \end{aligned} \tag{5.84}$$

We can rewrite the last one in a form more convenient for the next few steps

$$Q_1^k = tv(Q_1^{k+1} + Q_1^{k-1}) + c_2Q_{1,2}^{k,0} - t\overline{x}_2(Q_{1,2}^{k+1,0} + Q_{1,2}^{k-1,0})\,. \tag{5.85}$$

We will use induction to prove a nice formula, but for that, we first need to establish another base case because of this splitting of cases for the coefficient extraction. We have

$$Q_1^0 = \overline{a}_2 + c_2Q_{1,2}^{0,0} + a_1tvQ_1^1\,, \tag{5.86}$$

$$Q_1^{0,*} = \overline{a}_2 + c_2Q_{1,2}^{0,0} + a_1tvQ_1^{1,*}\,. \tag{5.87}$$

Combining them with equation (5.80) gives

$$\begin{aligned} 0 &= R^*Q_1^0 - RQ_1^{0,*} - c_2(\overline{x}_2 - x_2)Q_{1,2}^{0,0} + \overline{a}_2\delta_0 \\ &= R^*(\overline{a}_2 + c_2Q_{1,2}^{0,0} + a_1tvQ_1^1) - R(\overline{a}_2 + c_2Q_{1,2}^{0,0} + a_1tvQ_1^{1,*}) \\ &\quad - c_2(\overline{x}_2 - x_2)Q_{1,2}^{0,0} + \overline{a}_2\delta_0 \\ &= a_1tv(R^*Q_1^1 - RQ_1^{1,*}) + (R^* - R)(\overline{a}_2 + c_2Q_{1,2}^{0,0}) \\ &\quad - c_2(\overline{x}_2 - x_2)Q_{1,2}^{0,0} + \overline{a}_2\delta_0 \\ &= a_1tv(R^*Q_1^1 - RQ_1^{1,*}) + \overline{a}_2((\overline{x}_2 - x_2) + \delta_0)\,. \end{aligned} \tag{5.88}$$

We have the equation

$$R^*Q_1^1 - RQ_1^{1,*} + \overline{a}_2\delta_1 = 0\,, \tag{5.89}$$

where

$$\delta_1 = \frac{(\overline{x}_2 - x_2) + \delta_0}{a_1tv}\,. \tag{5.90}$$

This equation together with equation (5.80) form the base cases for the following lemma.

Lemma 5.11. *Let m be an integer greater than zero. Then*

$$R^*Q_1^m - RQ_1^{m,*} - c_2(\overline{x}_2 - x_2)Q_{1,2}^{m,0} + \overline{a}_2\delta_m = 0\,, \tag{5.91}$$

where δ_m is not a function of a_2.

Proof. We prove this by induction. We have the base cases $m = 0$ and $m = 1$. The latter one fits the pattern because $Q_{1,2}^{1,0} = 0$, since we cannot have walks ending on $(1, 0)$. Assume that the formula is true for all cases up to and including m. Then we use formula 5.85 to calculate

$$\begin{aligned}
&R^*Q_1^m - RQ_1^{m,*} \\
=&tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) + tv(R^*Q_1^{m-1} - RQ_1^{m-1,*}) \\
&+ c_2(R^* - R)Q_{1,2}^{m,0} - t(\overline{x}_2R^* - x_2R)(Q_{1,2}^{m+1,0} + Q_{1,2}^{m-1,0}) \\
=&tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) + tv(c_2(\overline{x}_2 - x_2)Q_{1,2}^{m-1,0} - \overline{a}_2\delta_{m-1}) \\
&+ c_2(\overline{x}_2 - x_2)Q_{1,2}^{m,0} - c_2tv(\overline{x}_2 - x_2)(Q_{1,2}^{m+1,0} + Q_{1,2}^{m-1,0}) \\
=&tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) + c_2(\overline{x}_2 - x_2)Q_{1,2}^{m,0} \\
&- c_2tv(\overline{x}_2 - x_2)Q_{1,2}^{m+1,0}) - \overline{a}_2tv\delta_{m-1}) \,.
\end{aligned} \tag{5.92}$$

Combining this expression with the equation from the lemma leads to

$$0 = R^*Q_1^{m+1} - RQ_1^{m+1,*} - c_2(\overline{x}_2 - x_2)Q_{1,2}^{m+1,0} + \overline{a}_2\delta_{m+1} \,, \tag{5.93}$$

where

$$\delta_{m+1} = \frac{\delta_m}{tv} - \delta_{m-1} \tag{5.94}$$

and it is independent from a_2. □

Of course, there is an analogous statement for Q_2^n if we follow the same analysis with the variable names switched. We can expand R^* and R to a form we can use:

$$\begin{aligned}
0 = &(c_2\overline{x}_2 - \overline{a}_2x_2)Q_1^m(x_2) - (c_2x_2 - \overline{a}_2\overline{x}_2)Q_1^m(\overline{x}_2) \\
&- c_2(\overline{x}_2 - x_2)Q_{1,2}^{m,0} + \overline{a}_2\delta_m \,.
\end{aligned} \tag{5.95}$$

We extract the x_2 coefficient

$$\begin{aligned}
c_2Q_{1,2}^{m,2} - \overline{a}_2Q_{1,2}^{m,0} - c_2Q_{1,2}^{m,0} + c_2Q_{1,2}^{m,0} + [x_2^1]\overline{a}_2\delta_m = 0 \\
c_2Q_{1,2}^{m,2} - \overline{a}_2Q_{1,2}^{m,0} + [x_2^1]\overline{a}_2\delta_m = 0 \,.
\end{aligned} \tag{5.96}$$

We can also extract the $[x_2^{n+1}]$ coefficient for $n \geqslant 1$.

$$c_2Q_{1,2}^{m,n+2} - \overline{a}_2Q_{1,2}^{m,n} + [x_2^{n+1}]\overline{a}_2\delta_m = 0 \,. \tag{5.97}$$

This formula also contains the $n = 0$ case, so we can group them together. Set $a_2 = 1$ to find

$$-\breve{Q}^{m,n}_{1,2} + [x_2^{n+1}]\delta_m = 0\,, \tag{5.98}$$

and substitute the resulting expression into the equation above to obtain

$$b_2 Q^{m,n+2}_{1,2} = Q^{m,n}_{1,2} - \breve{Q}^{m,n}_{1,2}\,. \tag{5.99}$$

We finally have the following result.

Corollary 5.12. *For integers m and n greater than zero, we have the following relations*

$$b_1 Q^{m+2,n}_{1,2} = Q^{m,n}_{1,2} - \widehat{Q}^{m,n}_{1,2}\,, \tag{5.100}$$
$$b_2 Q^{m,n+2}_{1,2} = Q^{m,n}_{1,2} - \breve{Q}^{m,n}_{1,2}\,. \tag{5.101}$$

We can sum up those equations to retrieve the following functional equations.

Corollary 5.13. *For integers m greater than zero, we have the following identities*

$$Q^m_1 - \widehat{Q}^m_1 = b_1 Q^{m+2}_1\,, \tag{5.102}$$
$$x_1^2(Q^n_2 - \widehat{Q}^n_2) = b_1(Q^n_2 - Q^{1,n}_{1,2}x_1 - Q^{0,n}_{1,2})\,, \tag{5.103}$$
$$x_1^2(Q - \widehat{Q}) = b_1(Q - x_1 Q^1_1 - Q^0_1)\,, \tag{5.104}$$

where we may also set $a_2 = 1$ to obtain further identities.

Now we can move onto the main result about the diagonal walks — the relation between $Q^{0,0}_{1,2}$ and its non-interacting versions. We use the master equation to extract the $x_1^0 x_2^0$ and $x_1^1 x_2^1$ coefficients and obtain the following identities:

$$Q^{0,0}_{1,2} = 1 + t a_1 a_2 Q^{1,1}_{1,2} \tag{5.105}$$
$$Q^{1,1}_{1,2} = t(Q^{0,0}_{1,2} + Q^{2,0}_{1,2} + Q^{0,2}_{1,2} + Q^{2,2}_{1,2})\,. \tag{5.106}$$

We also have

$$b_1 b_2 Q^{2,2}_{1,2} = b_1(Q^{2,0}_{1,2} - \breve{Q}^{2,0}_{1,2}) = Q^{0,0}_{1,2} - \widehat{Q}^{0,0}_{1,2} - \breve{Q}^{0,0}_{1,2} + \breve{\widehat{Q}}^{0,0}_{1,2}\,, \tag{5.107}$$

together with

$$b_1 b_2 Q^{2,0}_{1,2} = b_2 Q^{0,0}_{1,2} - b_2 \widehat{Q}^{0,0}_{1,2}\,, \tag{5.108}$$
$$b_1 b_2 Q^{0,2}_{1,2} = b_1 Q^{0,0}_{1,2} - b_1 \breve{Q}^{0,0}_{1,2}\,. \tag{5.109}$$

Furthermore,

$$\begin{aligned} b_1b_2Q_{1,2}^{1,1} &= t(b_1b_2Q_{1,2}^{0,0} + b_1b_2Q_{1,2}^{2,0} + b_1b_2Q_{1,2}^{0,2} + b_1b_2Q_{1,2}^{2,2}) \\ &= t((b_1b_2 + b_1 + b_2 + 1)Q_{1,2}^{0,0} - (b_2+1)\widehat{Q}_{1,2}^{0,0} - (b_1+1)\breve{Q}_{1,2}^{0,0} + \breve{\widehat{Q}}_{1,2}^{0,0}) \\ &= t(a_1a_2Q_{1,2}^{0,0} - a_2\widehat{Q}_{1,2}^{0,0} - a_1\breve{Q}_{1,2}^{0,0} + \breve{\widehat{Q}}_{1,2}^{0,0})\,. \end{aligned} \tag{5.110}$$

Our final identity is

$$\begin{aligned} b_1b_2Q_{1,2}^{0,0} &= b_1b_2 + ta_1a_2b_1b_2Q_{1,2}^{1,1} \\ &= b_1b_2 + t^2a_1a_2(a_1a_2Q_{1,2}^{0,0} - a_2\widehat{Q}_{1,2}^{0,0} - a_1\breve{Q}_{1,2}^{0,0} + \breve{\widehat{Q}}_{1,2}^{0,0})\,. \end{aligned} \tag{5.111}$$

As before, let G stand for $Q_{1,2}^{0,0}$ and G_1, G_2 and $G_{1,2}$ for the various extractions of b_1^0 and b_2^0 coefficients. The relation becomes

$$b_1b_2G = b_1b_2 + t^2a_1a_2(a_1a_2G - a_2G_1 - a_1G_2 + G_{1,2})\,, \tag{5.112}$$

or written as a linear combination,

$$(b_1b_2 - t^2a_1^2a_2^2)G + t^2a_1a_2^2G_1 + t^2a_1^2a_2G_2 - t^2a_1a_2G_{1,2} = b_1b_2\,. \tag{5.113}$$

We can divide by a_1a_2 and obtain the equation

$$(c_1c_2 - a_1a_2t^2)G + a_2t^2G_1 + a_1t^2G_2 - t^2G_{1,2} = c_1c_2\,, \tag{5.114}$$

Theorem 5.14. *$B(fQ_{1,2}^{0,0}) = c_1c_2$, where $f(a_1,a_2,t) = c_1c_2 - a_1a_2t^2$ and $Q_{1,2}^{0,0}$ is the generating function for walks ending on the origin in the Diagonal Cartesian model.*

It may be possible to find a similar combination for the function Q but we will not do this here.

5.2.3 The Full Model – All Steps

We will finally take a look at the model featuring all the steps. **[FIXME: Establish the outcome from the start]** Here

$$\frac{\alpha(K_2^* - K_2)}{K_1} = \frac{\alpha(\overline{x}_2 - x_2)(1+t)}{S_1^1 K_1}\,, \tag{5.115}$$

$$\frac{\Delta_2(K_2^* K_{12})}{K_1} = \frac{c_2(\overline{x}_2 - x_2)(c_1(1+t) - t\overline{\sigma}_1 S_1^1)}{S_1^1(c_1 - t\overline{\sigma}_1 S_1^1)} \tag{5.116}$$

$$= \frac{c_2(\overline{x}_2 - x_2))}{S_1^1}\left(1 - \frac{t}{c_1 - t\overline{\sigma}_1 S_1^1}\right)\,. \tag{5.117}$$

We will need to do some more work to find a representation for that last fraction. For that, we need the σ_1 roots given by

$$\sigma_1 = \frac{u - \sqrt{u^2 - 4w^2}}{2w} \quad \text{and} \quad \overline{\sigma}_1 = \frac{u + \sqrt{u^2 - 4w^2}}{2w} \tag{5.118}$$

for $u = 1 - tv$ and $w = t(1+v)$. Then we have

$$\begin{aligned} \frac{t}{c_1 - \overline{\sigma}_1 w} &= \frac{2t}{2c_1 - u - \sqrt{u^2 - 4w^2)}} \\ &= \frac{t(2c_1 - u + \sqrt{u^2 - 4w^2})}{2(c_1^2 - c_1 u + w^2)}\,. \end{aligned} \tag{5.119}$$

Let $v = \overline{x}_2 + x_2$. Then the denominator has the term

$$c_1^2 - c_1 u + w^2 = c_1^2 - c_1(1 - tS_1^0) + t^2(S_1^1)^2 \tag{5.120}$$

$$= \overline{a}_1^2 b_1 - \overline{a}_1 b_1 v t + (1+v)^2 t^2 \tag{5.121}$$

and the expression under the square root factors as

$$u^2 - 4w^2 = (1 - (3v+2)t)(1 + (v+2)t)\,. \tag{5.122}$$

The first few terms of the square root are

$$\sqrt{u^2 - 4w^2} = 1 - vt - 2(1+v)^2 t^2 - 2v(1+v)^2 t^3 + \cdots \tag{5.123}$$

We can give a series representation of the square root via the series

$$\sqrt{1-z} = \sum_{n=0}^{\infty} \frac{(2n)!}{(1-2n)(n!)^2 4^n} z^n \,. \tag{5.124}$$

That is,

$$\begin{aligned} &\sqrt{(1+(v+2)t)(1-(3v+2)t)} \\ &= \left(\sum_{n=0}^{\infty} \frac{(-1)^n (2n)! (v+2)^n t^n}{(1-2n)(n!)^2 4^n} \right) \left(\sum_{n=0}^{\infty} \frac{(2n)!(3v+2)^n t^n}{(1-2n)(n!)^2 4^n} \right) \\ &= \sum_{n=0}^{\infty} \frac{t^n}{4^n} \sum_{k=0}^{n} \frac{(-1)^k (2k)!(2n-2k)!}{(1-2k)(1-2n+2k)k!^2 (n-k)!^2} (v+2)^k (3v+2)^{n-k} \,, \end{aligned} \tag{5.125}$$

However, we are not sure if a closed form exists for these coefficients. Extracting x_2 coefficients from the overall expression is difficult because we would first need to expand it in series of t. Now, K_2 and K_2^* have $1+v$ in their denominators, so we can multiply the whole functional equation by $1+v$ to clear out these denominators. We have

$$(1+v)K_2 = c_2(1+v) - \overline{x}_2(1+t) = (c_2 - 1 - t)\overline{x}_2 + c_2 + c_2 x_2 \,, \tag{5.126}$$

$$(1+v)K_2^* = c_2(1+v) - x_2(1+t) = c_2\overline{x}_2 + c_2 + (c_2 - 1 - t)x_2 \,, \tag{5.127}$$

and

$$(1+v)\frac{\alpha(K_2^* - K_2)}{K_1} = \frac{\alpha(\overline{x}_2 - x_2)(1+t)}{K_1} = \alpha g(a_1, x_2, t) \,, \tag{5.128}$$

$$(1+v)\frac{\Delta_2(K_2^* K_{12})}{K_1} = c_2(\overline{x}_2 - x_2) f(a_1, x_2, t) \,, \tag{5.129}$$

where

$$f(a_1, x_2, t) = 1 - t\frac{c_1 - 1 + tv + \sqrt{(1+(v+2)t)(1-(3v+2)t)}}{c_1^2 - c_1 + c_1 tv + (1+v)^2 t^2} \,. \tag{5.130}$$

Then if we let $A = c_2 - (1+t)$, we can write the funtional equation as

$$(c_2\overline{x}_2 + c_2 + Ax_2)Q_1^0(x_2) - (A\overline{x}_2 + c_2 + c_2 x_2)Q_1^0(\overline{x}_2) = c_2(\overline{x}_2 - x_2)f - \alpha g \,. \tag{5.131}$$

As before, we try to find a relation for the walks returning to the origin via coefficients extractions. We extract the x_2 coefficient from both sides to find that

$$c_2 Q_{1,2}^{0,2} + c_2 Q_{1,2}^{0,1} - (1+t)Q_{1,2}^{0,0} = -\alpha g_2^1 + c_2(f_2^2 - f_2^0)Q_{1,2}^{0,0} \,, \tag{5.132}$$

and then we set $a_2 = 1$ on both sides to find that

$$-(1+t)\breve{Q}^{0,0}_{1,2} = -\overline{a}_1 g^1_2 \,. \tag{5.133}$$

When we finally substitute this quantity back into the equation, it simplifies to

$$b_2(Q^{0,2}_{1,2} + Q^{0,1}_{1,2}) = (b_2(f^2_2 - f^0_2) - a_2(1+t))Q^{0,0}_{1,2} - (1+t)\breve{Q}^{0,0}_{1,2} \,. \tag{5.134}$$

The question is can we find the quantity $f^2_2 - f^0_2$. It appears that

$$-(f^2_2 - f^0_2) = 1 + a_1 t + \frac{a_1^2}{b_1}(a_1^2 - a_1 - 4)t^3 + \frac{4a_1^2}{b_1^2}(a_1^3 - 3a_1^2 + 1)t^4 + \cdots \,, \tag{5.135}$$

which is not a series that would appear to have a closed form. Unfortunately, we have not been able to go further with this case. One could try making some assumptions for the form of equations like these for $Q^{m,n}_{1,2}$, $\breve{Q}^{m,n}_{1,2}$ and $\widehat{Q}^{m,n}_{1,2}$, but even then it is not clear if we can find a relation between $Q^{0,0}_{1,2}$, $\breve{Q}^{0,0}_{1,2}$, $\widehat{Q}^{0,0}_{1,2}$ and $\breve{\widehat{Q}}^{0,0}_{1,2}$. We would have six equations like the one above plus six more, where $b_i = 0$ for the b_i's not present. Then one may use the four regular coefficient extractions from the master functional equation for $Q^{m,n}_{1,2}$ and $m, n \leqslant 1$ plus twelve more, where we set one or more of b_1 and b_2 to zero. We have a grand total of 28 equations in 36 unknowns, of which we need to eliminate 32. A quick attempt to do this via computer was not sufficient.

Chapter 6

Generalizations

6.1 Constraint Generalizations

In this final chapter we will finally analyze the kinds of generalizations that we can explore with the help of the operator perspective. There is already some work that hints at those more general situations. For example, Definition 1.1 of [7, p. 2] has a functional equation for the non-interacting lattice walks on the quarter plane, which is valid for models with $\mathcal{S}$ containing negative coordinates of size at most -1. Similarly, [11] covers walks in the quarter plane with multiple copies of small steps. Also, we may consider the region of a semi-infinite strip on the plane as in [21, 28, 29]. In [27] the author considers lattice walks in an octant with interactions with the diagonal.

The analysis is analogous to the one in Chapter 2 and Chapter 3 — we start with the initial condition $\mathcal{Q}$, that is represented by the function q, and grow lattice walks from it by multiplying by tS. Those walks may venture outside of $\mathcal{D}$ and may interact with $\mathcal{D}_0$. Just like in the previous analysis, we will need operators to help us remove or mark walks.

Definition 6.1. *Let $\mathcal{M}$ be a given model set in d dimensions. Let E be an operator that eliminates monomials $x^P_{[d]}$ when P is not in $\mathcal{D}$. Similarly, let L be an operator that marks monomials $x^P_{[d]}$ for $P \in \mathcal{D}_0$ in the way specified by the model. Let $K = 1 - tS$ be the kernel of the lattice walk model.*

From what we have seen in previous chapters, it should be clear that the operator L is invertible — just take an operator marking the same locations in $\mathcal{D}_0$ but with inverted labels. That is, if the operator marks a monomial with the label a, then the inverse operator is one marking the same monomial with the label $1/a$. Also, the caution in the definition is there because it may be possible to imagine such convoluted constraints that an operator may no exist. It is not difficult to come up with constraints that lead to unwieldy operators. To make the analysis work, we need the operators

to act term-wise on series with respect to the variable t.

We can dive into the main result concerning the general models delineated so far.

Theorem 6.2. *Let $\mathcal{M}$ be a lattice walk model and Q its associated generating function. Then Q satisfies the following functional equations*

$$(1 - tLES)Q = q\,, \tag{6.1}$$

$$(L^{-1} - tES)Q = L^{-1}(q) = \alpha\,, \tag{6.2}$$

and its initial term is $q_0 = q$.

Proof. The second functional equation is obtained by applying L^{-1} to the first one, so it only remains to show that the first functional equation holds. We show this via iteration. We start with Q and multiply it with tS. This gives tSQ, an enumeration of all walks extended by a single step. Here the power of t is raised by one to keep track of the extension. Then we apply E to the result to eliminate all lattice walks that venture outside of the region $\mathcal{D}$. Finally, we mark those walks that have arrived at locations in $\mathcal{D}_0$ by applying L to $E(tSQ)$. The resulting expression is almost Q, but one term is missing — namely, $q_0 = q$. That is, we have transformed Q to itself in a nontrivial way and we have the equation

$$Q = q + LE(tSQ) = q + tLE(SQ)\,, \tag{6.3}$$

since the operators do not interact with t. □

Example 6.3. Also, we can consider a different approach. Instead of removing walks that venture outside of $\mathcal{D}$, we just mark them with some variable z for later deletion. Suppose that we are given some model $\mathcal{M}$ and we want to frame the analysis in this new way. We have a new lattice walk model $\widetilde{\mathcal{M}}$ with $\widetilde{\mathcal{D}} = \mathbb{Z}^d$ and $\widetilde{\mathcal{D}}_0 = \mathcal{D}_0 \cup \mathcal{D}^c$, where we mark monomials in $\mathcal{D}^c$ with z and those in $\mathcal{D}_0$ in the same way as in model $\mathcal{M}$. Here $\widetilde{E}$ is the identity operator since we do not remove any walks. We have the functional equations

$$(1 - t\widetilde{L}S)\widetilde{Q} = q\,, \tag{6.4}$$

$$(\widetilde{L}^{-1} - tS)Q = \widetilde{\alpha}\,. \tag{6.5}$$

It is not clear how one would proceed and if this method will be a better way to handle the problem. The upside is that the operators are invertible, but the downside is that we now have more extracted terms in the functional equation, since the lattice walks can now go anywhere.

Example 6.4. Conside the lattice walk models in two dimensions from previous chapters. Another way we can look at the quarter plane models is to us the idea from the previous example and also symmetrize the problem by also marking parts on the other side of the origin. Instead of just marking walks on the axes and on lines, where one coordinate is -1, we can also mark walks on lines with one coordinate $+1$. This process leads to a symmetrical arrangement of marking operations on both sides of the origin. We have the equation

$$(1 - tLS)\widetilde{Q} = (1 - tL_{\text{-}1}L_0L_1S)\widetilde{Q} = 1\,. \tag{6.6}$$

We can skip the marking of the axes with L_0 if we do not wish to investigate wall interactions. In general, we may mark more locations in space so we can combine all of these operators into one called L. Then we have

$$(1 - tLS)\widetilde{Q} = 1\,, \tag{6.7}$$

or alternatively,

$$K\widetilde{Q} = L^{-1}(1) + (1 - L^{-1})\widetilde{Q}\,. \tag{6.8}$$

Before the end of this section, we will take a moment to talk about some of the abstract properties of the operators L and E. The operator E has the eigenvalues 1 or 0 for monomials $x^P_{[d]}$, where $P \in \mathcal{D}$ or $P \in \mathcal{D}^c$, respectively. The latter eigenvalue characterizes the monomials contained in $\ker E$. Similarly, L has eigenvalue 1 for when $P \in \mathcal{D}_0^c$, and the marking monomial a for when $P \in \mathcal{D}_0$, where we keep in mind that a is a multi-index variable and it varies from point to point. Also, L^{-1} has the reciprocal eigenvalues of L.

The task for the rest of the chapter is to find representations for the operators involved. In general, this should will not be an easy task because if we could express arbitrary operators L and E with a small number of coefficient extractions, we would also be able to use this scheme to compress data without loss, and this is impossible in general.

6.1.1 Marking Operators

We start with the observation that the operator E is just a marking operator that applies the label $a = 0$ to all monomials coming from points outside of $\mathcal{D}$. Therefore, we can combine the discussions for operators L and E into one. Whenever invertibility is concerned, the discussion will apply to operators L only. First, we show one form of a marking operator.

Lemma 6.5. *Let P be a subset of $\mathbb{Z}^d$ and suppose that we have to mark each point p in that set with*

some label a_p, with some marking operator Γ_P. Then the operator can be written as

$$\Gamma_P = 1 + \sum_{p \in P} (a_p - 1)[[x^p_{[d]}]] \,. \tag{6.9}$$

Proof. It is enough to confirm the operator's action on single monomials. If p is not in P, then Γ_P returns the monomial itself, since all the term extractions return zero. If p is in P, then the only extraction that returns a nonzero term is the one associated with p. In that case we obtain

$$\Gamma_P C x^p_{[d]} = C x^p_{[d]} + (a_p - 1) C x^p_{[d]} = a_p C x^p_{[d]} \,. \tag{6.10}$$

□

Corollary 6.6. ***[FIXME: Maybe move it somewhere more useful.]*** *Marking operators commute.*

Proof. Since marking operators act on monomials by multiplication with marking variables and multiplication is commutative, then the application of marking operators is also commutative. □

This form of the operator is rather inefficient — just consider one-dimensional models. There, both L and E contain a single term extraction, whereas the representation here would need infinitely many. In other words, a lot of models allow for smaller marking operators that do essentially the same job. The following example illustrates this notion.

Example 6.7. If we consider lattice walks with small steps on the half-line in one dimension, we have the operator

$$E = 1 - \sum_{n=1}^{\infty} \overline{x}_1^n [\overline{x}_1^n] \,, \tag{6.11}$$

but since we only have small steps, there are no walks that go past $\overline{x}_1$ in the negative direction. Therefore, we can use the operator $E' = 1 - \overline{x}_1[\overline{x}_1]$ to the same effect instead. Although the operators are different in form, they are essentially the same with regards to the problem at hand. We will say the two operators are model-equivalent.

To make the equivalence work, we need to consider the whole model. With that in mind, we may think of the lattice walk models as living completely within the sumset $\mathcal{S} + \mathcal{D}$ of points that are combinations of points in $\mathcal{D}$ and steps $\mathcal{S}$. This is how far walks from the model can reach from within $\mathcal{D}$ by taking a single step. We will call this sumset the range of the model.

Definition 6.8. *Given sets A and B, we let their sumset be*

$$A + B = \{a + b \,|\, a \in A,\, b \in B\}\,. \tag{6.12}$$

Definition 6.9. *Given a lattice walk model $\mathcal{M}$ with domain $\mathcal{D}$ and step set $\mathcal{S}$, define the range of model $\mathcal{M}$ to be the sumset $\mathcal{D} + \mathcal{S}$.*

With this in mind, we can define a notion of equivalence for operators.

Definition 6.10. *Let $\mathcal{M}$ be a lattice walk model in $\mathbb{Z}^d$ and let Γ and Γ' be two operators acting on $\mathbb{Z}^d$. Then $\Gamma \equiv_{\mathcal{M}} \Gamma'$ if and only if Γ and Γ' act identically on monomials from the range of $\mathcal{M}$. That is, the operators are model-equivalent if and only if $\Gamma x^p = \Gamma' x^p$ for all $p \in \mathcal{D} + \mathcal{S}$.*

In the preceding example E is equivalent to E', since both operators act the same way on monomials from the range of $\mathcal{M}$, that is, monomials x_1^n for $n \geqslant -1$. Similarly, in the two-dimensional problem we have an equivalence between the operator E and the operator $E' = (1 - \overline{x}_1[\overline{x}_1])(1 - \overline{x}_2[\overline{x}_2])$.

When the boundary of $\mathcal{D}$ is complicated, we may not find nice equivalent operators, but there are some cases where we can say something. In problems featuring half-planes, we have an operator E with infinitely many terms, and we find that there is a model-equivalent operator E' that is only comprised of finitely many terms. We will investigate which operators are equivalent to finite ones.

Definition 6.11. *Given a model $\mathcal{M}$, a coefficient extraction operator Γ in the variables $x_1, \ldots, x_d$ is finite if it is equivalent to an operator Γ' comprised of finitely many coefficient extraction terms. That is,*

$$\Gamma \equiv_{\mathcal{M}} \Gamma' = \sum_{\substack{I \subseteq [d] \\ J \subseteq \mathbb{Z}^d}} a_{I,J} x^J [x_I^J]\,, \tag{6.13}$$

where each $|J| < \infty$. Also, Γ' is infinite if it is not finite.

For most purposes, we will not distinguish between Γ and Γ'. A natural question is to consider which models $\mathcal{M}$ lead to finite operators and this very much depends on the region $\mathcal{D}$ and the nature of the marking of region $\mathcal{D}_0$. We have some cases where it is easy to say.

Lemma 6.12. *Let $\mathcal{D}$ be a d-dimensional box in $\mathbb{Z}^d$ and $|\mathcal{S}| < \infty$. Then the operator $\Gamma = LE$ is finite.*

Proof. The operator L is finite since it only needs to mark finitely many points. We can find E explicitly. Let the box have x_k coordinates $[a_k, b_k]$. Suppose that the steps in $\mathcal{S}$ have x_k coordinates

of absolute size at most n_k. If $n_k = 0$, the walks do not change their x_k coordinate and the walks do not venture past the boundaries in that coordinate. When $n_k > 0$ we need to remove all monomials x^P with $P_k \in [a_k - n_k, a_k - 1] \cup [b_k + 1, b_k + n_k]$.

We can then write the operator as

$$E_k = 1 - \sum_{j=1}^{n_k} x_k^{a_k - j}[x_k^{a_k - j}] + x_k^{b_k + j}[x_k^{b_k + j}], \tag{6.14}$$

which is equal to $E_k = 1$ when $n = 0$. Applying $E_1, \ldots, E_d$ sequentially ensures that we remove all monomials outside of $\mathcal{D}$. That is, $E = E_1 \cdots E_d$. Since each of the operators is finite, their composition $\Gamma = LE$ is finite as well. □

Theorem 6.13. *Let $\mathcal{D}$ be a finite region in $\mathbb{Z}^d$ and $|\mathcal{S}| < \infty$. Then the operator $\Gamma = LE$ is finite.*

Proof. The region $\mathcal{D}$ lies within a bounding box B, so we may write $E = E'E''$, where $E' = E_1 \cdots E_d$ from the lemma and E'' is the operator removing all monomials outside of $\mathcal{D}$ but inside the interior of B. E'' is finite since there are only finitely many points in that region. Since we are marking only finitely many points $\mathcal{D}_0 \subseteq \mathcal{D}$, L is finite, and finally, Γ is finite. □

Also, we may handle semi-infinite regions if we are marking only finitely many points. The analysis is the same as the one above, except now one or more of a_k, b_k may be infinite, and in this case, we actually lose parts of Γ_{B^c} and the operator is still finite.

Definition 6.14. *A lattice walk model with finitely many steps is finite if its extraction and marking operators E and L are finite, and its starting condition q has a finite degree. Otherwise, we say the model is infinite.*

Example 6.15. All the models so far has been finite, but not all models are finite. Consider a one-dimensional model on the whole line where for each integer i, we mark point x^i with parameter a_i using operator L. Then all equivalent marking operators L' would have to have a term like $(a_i - 1)[[x^i]]$; otherwise, it would not act the same way on monomials in the model's range $\mathbb{Z}$. Therefore, L is not equivalent to a finite operator.

Finite models are nicer because they have functional equations with finitely many terms and there is a chance that they will be amenable to the sort of analysis we have performed for quarter plane walk models in the past. What follows is a rather easy application of the theory so far.

6.1.2 Examples of Regions

Example 6.16. Here is an example of an infinite strip. The region $\mathcal{D}$ is comprised of all the points with coordinates (i, j) in $\mathbb{Z}^2$ with $0 \leqslant j \leqslant n$. Maybe we have interactions with the lines $x_2 = 0$ and $x_2 = n$. The operators are

$$E = E_{-1}E_{n+1} = (1 - [[\overline{x}_2]])(1 - [[x_2^{n+1}]]) = 1 - [[\overline{x}_2]] - [[x_2^{n+1}]]\,, \tag{6.15}$$

$$L = L_0 L_n = (1 + b_0[[x_2^0]])(1 + b_n[[x_2^n]]) = 1 + b_0[[x_2^0]] + b_n[[x_2^n]]\,, \tag{6.16}$$

and the functional equation for Q is given by

$$KQ - (c_0 - t\overline{x}_2 S_2^{-1})Q_2^0 - x_2^n(c_n - tx_2 S_2^1)Q_2^n = \overline{a}_0\,. \tag{6.17}$$

If we extract the $[x_2^0]$ term, we find that

$$(1 - a_0 t S_2^0)Q_2^0 - tS_2^{-1}Q_2^1 = 1\,. \tag{6.18}$$

If we extract the $[x_2^n]$ term, we find that

$$(1 - a_n t S_2^0)Q_2^n = tS_2^1 Q_2^{n-1}\,. \tag{6.19}$$

Similarly, we can extract the $[x_2^k]$ terms for $1 \leqslant k \leqslant n-1$. We will have $n+1$ equations in $n+1$ unknown functions $Q_2^0, \ldots, Q_2^n$, and we would be able to solve them to find out what all those functions are. Of course, we only need Q_2^0 and Q_2^n to obtain a solution for Q. In the simplest case, $n = 1$ and the term extraction equations become

$$(1 - a_0 t S_2^0)Q_2^0 = 1 + tS_2^{-1}Q_2^1\,, \tag{6.20}$$

$$(1 - a_1 t S_2^0)Q_2^1 = tS_2^1 Q_2^0\,. \tag{6.21}$$

These solutions are

$$Q_2^0 = \frac{1 - a_1 t S_2^0}{(1 - a_0 t S_2^0)(1 - a_1 t S_2^0) - S_2^1 S_2^{-1} t^2}\,, \tag{6.22}$$

$$Q_2^1 = \frac{S_2^1 t}{(1 - a_0 t S_2^0)(1 - a_1 t S_2^0) - S_2^1 S_2^{-1} t^2}\,, \tag{6.23}$$

and we can find Q explicitly from

$$Q = \frac{1}{K}(\overline{a}_0 + (c_0 - t\overline{x}_2 S_2^{-1})Q_2^0 + x_2(c_1 - tx_2 S_2^1)Q_2^1)\,. \tag{6.24}$$

Example 6.17. Another example is a finite box. The region $\mathcal{D}$ is comprised of all the points with coordinates (i, j) in $\mathbb{Z}^2$ with $0 \leqslant i \leqslant m$ and $0 \leqslant j \leqslant n$, for some positive integers m and n. The lattice walks interact with all four sides of the box, each with its own label. As in the previous example, we may write a functional equation directly via the following operators

$$E_{i,j} = 1 - [[x_i^j]] \quad \text{and} \quad L_{i,j} = 1 + b_{i,j}[[x_i^j]] \,. \tag{6.25}$$

For the model itself, we have

$$E = E_{1,\text{-}1} E_{1,m+1} E_{2,\text{-}1} E_{2,n} \,, \tag{6.26}$$

$$L = L_{1,0} L_{1,m} L_{2,0} L_{2,n} \,. \tag{6.27}$$

We may attempt a solution similar to that of the previous example. The function Q should be rational here as well.

Example 6.18. Suppose that $\mathcal{D}$ is a finite region in $\mathbb{Z}^2$. If we set $x_1 = 0$ for points in $\mathcal{D}$, we obtain the projection of the set onto the x_2 axis, and this tells us how many x_2 coordinates we have in functions in the functional equation. Suppose that the number of points is m_1 and define m_2 analogously. Lastly, let there be m lattice points in $\mathcal{D}$. Next, we examine the number of unknown functions Q_i^j and $Q_{1,2}^{i,j}$ that we have in the functional equation and its various extractions. The master functional equation may only have m_1 functions Q_2^i and m_2 functions Q_1^i, together with m functions $Q_{1,2}^{i,j}$, because those are the possible nonzero functions Q that we may find through various extractions. Similarly, when we extract terms like $[x_i^j]$ and $[x_1^i x_2^j]$, we only find a subset of the functions listed.

In general, we may even find less of those functions as we perform more extractions. In the end, we have the same number of equations as unknown functions Q and various extractions of Q. Therefore, the problem reduces to the solution of a linear system of $1 + m_1 + m_2 + m$ equations and that many unknowns Q. Note that, because we have the term KQ in the master functional equation, the $[x_i^j]$ extraction will have the function Q_i^j with a nonzero coefficient. Because there is a unique function Q for the model, this forces the uniqueness of the Q_i^j and $Q_{1,2}^{i,j}$ terms. Therefore, we have a unique solution to the system of equations. Furthermore, because all the the coefficients are rational functions, so is Q. We will generalize this observation in the following theorem.

Definition 6.19. *Given a d-dimensional region $\mathcal{D}$ and a finite set I from $\{1, \ldots, d\}$, let $\mathcal{D}_I$ be defined as follows*

$$\mathcal{D}_I = \{(x_1, \ldots, x_d)_{x_I=0} | (x_1, \ldots, x_d \in \mathcal{D})\} \,, \tag{6.28}$$

that is, $\mathcal{D}_I$ is the projection of $\mathcal{D}$ onto the space generated by coordinates x_i for i in $[d]\backslash I$. Let $m_I = |\mathcal{D}_I|$.

Note that $I = \varnothing$ corresponds to the set of points in $\mathcal{D}$, and $I = [d]$ corresponds to exactly one point and the master functional equation.

Theorem 6.20. *If a d-dimensional model $\mathcal{M}$ is set in a finite region $\mathcal{D}$, then the full generating function Q is rational in its variables. To obtain the solution, we solve a system of $N = \sum_{I\subseteq[d]} m_I$ equations and unknown functions Q_I^J, with a unique solution.*

Proof. The master functional equation may at most contain all the possible nonzero functions Q_I^J. Each possible extraction $[x_K^P]$ may contain at most a portion of these variables — all terms of the form $Q_{I,K}^{J,P}$. We just need to count the number of equations and unknowns. The projections $\mathcal{D}_I$ helps us in determining these numbers. The set $\mathcal{D}_I$, with m_I points corresponds to all the functions $Q_{[d]\backslash I}^P$ for $P \in \mathcal{D}_I$. In fact, we have m_I unknowns of the form $Q_{[d]\backslash I}^P$, because the master functional equation contains a term Q, and so we have all the possible extractions among the extraction equations. Thus, have N equations and N unknowns.

Next, since there is a unique generating function solution Q to the master functional equation for the model at hand, the various extractions Q_I^J are also unique, and overall, there must be a unique solution to the system of equations. Therefore, Q is a rational function in its variables because all the coefficients in the system are rational functions. □

Starting with the master functional equation, we have N unknown functions. When we perform a single extraction $[x_i^i]$ in one of $m_{1,\ldots,i-1,i+1,\ldots,d}$ ways, we no longer see the function Q or any of the functions Q_k^l for k not equal to i. This kind of pattern continues on with further extractions. We can write out a schematic version of the system of equations. For example, in two dimensions we have

$$\begin{bmatrix} Q & Q_1^i & Q_2^i & Q_{1,2}^{i,j} \\ & Q_1^i & & Q_{1,2}^{i,j} \\ & & Q_2^i & Q_{1,2}^{i,j} \\ & & & Q_{1,2}^{i,j} \end{bmatrix}. \tag{6.29}$$

That is, in the master functional equation we have all of the unknown functions. In the next m_2 equations we have variables Q_1^i and $Q_{1,2}^{i,j}$ only, and so on. If we take the last set of equations only involving functions of the form $Q_{1,2}^{i,j}$, we have a system of equations that we can solve for the various functions. Then we may back-substitute these quantities in the rest of the systems of

equations. Next, we may solve for the functions Q_1^i and Q_2^i. When we do this, we may finally back-substitute all the quantities in the master functional equation. The procedure is reminiscent of the long-winded approach to finding the transformation from $(1-tLES)Q = 1$ to $(L^{-1}-tES)Q = \alpha$ that we avoided through the use of operator notation. We are hopeful that such an approach exists in this case too.

Example 6.21. Consider the one-dimensional model set on $\mathcal{D} = \{u, \dots, v\}$. Here the step set must have steps of size at most $n = v - u$ in either direction, because any larger steps will always step outside of the region, and therefore, we may discard them. We have $S = \sum_{k=-n}^{n} S_k x^k$. The operators E and L are given by

$$E = 1 - \sum_{k=1}^{n} ([[x^{u-k}]] + [[x^{v+k}]]) \,, \tag{6.30}$$

$$L = 1 + \sum_{k=u}^{v} a_k [[x^k]] \,, \tag{6.31}$$

$$L^{-1} = 1 - \sum_{k=u}^{v} c_k [[x^k]] \,. \tag{6.32}$$

We do not need to use all the variables a_k but we may as well do that. The master functional equation is $(L^{-1} - tES)Q = L^{-1}(q)$, as usual. The only extraction equations we obtain here are those coming from the coefficient extraction with $[x^i]$ with $u \leqslant i \leqslant v$. We have

$$[x^i]E = [x^i] \,, \tag{6.33}$$

$$[x^i]L^{-1} = [x^i] - c_i[x^i] = \overline{a}_i[x^i] \,, \tag{6.34}$$

for those case and the extracted functional equations become

$$[x^i](L^{-1} - tES)Q = [x^i](\overline{a}_i - tS)Q = [x^i]L^{-1}(q) = q_i \,, \tag{6.35}$$

or in other words, $[x^i](1 - a_i tS)Q = [x^i]K(a_i t)Q = a_i q_i$. The explicit equations are

$$Q_i - a_i t \sum_{j=-n}^{n} S_{i-j} Q_j \,. \tag{6.36}$$

We may write all of those equations as the system

$$IX - tHX = (I - tH)X = \alpha\,, \tag{6.37}$$

where I is the $n \times n$ identity matrix, H is the matrix with entries $H_{i,j} = a_i S_{i-j}$ for $u \leqslant i, j \leqslant v$, and α is the vector with entries q_i. Because of the uniqueness of the solution, the matrix $(I - tH)$ must be invertible. We may write the master functional equation as $KQ + F \cdot X = q$ and thus, the function Q is given by

$$Q = \frac{1}{K}(q - F \cdot X)\,. \tag{6.38}$$

Example 6.22. Consider the one-dimensional model set on $\mathcal{D} = \{0, \ldots, 1\}$. Here $X^T = [Q_0 \quad Q_1]$ and

$$H = \begin{bmatrix} a_0 & a_0 \\ a_1 & a_1 \end{bmatrix}\,, \tag{6.39}$$

and the determinant of $I - tH$ is $1 - (a_0 + a_1)t$. Therefore,

$$X = \frac{1}{1 - (a_0 + a_1)t} \begin{bmatrix} (1 - a_1 t)q_0 - a_0 t q_1 \\ (1 - a_0 t)q_1 - a_1 t q_0 \end{bmatrix} \tag{6.40}$$

and from the master functional equationg,

$$KQ - (c_0 - t\overline{x})Q_0 - x(c_1 - tx)Q_1 = q_0 + xq_1\,, \tag{6.41}$$

we find that

$$Q = \frac{1}{(1 - (a_0 + a_1)t)K}(f_0 q_0 + f_1 q_1)\,, \tag{6.42}$$

and if we expand the function into series we find that there is a stabilization of the coefficients involving x in front of t^k. We expect that such stabilization will occur for all finite region models.

6.2 Connections With Other Theories

So far we have explored using operators to simplify the derivation of functional equations pertaining to lattice walk models on $\mathbb{Z}^d$, and to simplify the first steps in solutions to these problems. It would be great to know if we can say more about the solutions. It seems that the work so far points to connections with Fredholm and Sturm-Liouville theory. For example, if we interpret coefficient

extractions from a complex analysis point of view, by using Cauchy integrals, we may obtain an equation that looks much like a Fredholm integral equation of the first kind. The connection is made stronger by noting that both theories involve a possible Neumann series solution, written differently in each case. Using the Neumann series expansion, we have been able to show that Q may be formed as a Hadamard product of generating functions for lower-dimensional lattice walk models. However, using the Neumann series expansion is unwieldy, so our other hope is to find another kind of series expansion. For example, since the marking operators have eigenvalues and eigenfunctions, we may hope to find some sort of eigenfunction expansion for Q, along the lines of Sturm-Liouville theory. For that, we would need a suitable inner product and find the adjoints of the marking operators, with respect to the inner product. Unfortunately, time was insufficient to pursue either of these approaches with effort, and we only have the following writeup in hopes of inspiring future mathematicians to work along those lines and find out if the approaches are fruitful.

6.2.1 Properties of Marking Operators

The operators we have seen so far are marking operators, like L and E, in combination of multiplication by another function, like ΓS, or a combination of operators, like $L^{-1} - tES$. In this order, we will find out their properties - eigenvalues, eigenvectors, kernels and invertibility. The building block of these operators are coefficient extractions $[x_I^J]$ and term extractions $[[x_I^J]]$.

Combinatorially speaking, we can easily find some of the eigenvalues and eigenfunctions by looking at single monomials. If a monomial is not marked by Γ, then it is an eigenfunction with eigenvalue 1, since Γ does not change it; if it is, then it is an eigenfunction with eigenvalue equal to the label given by Γ. These are all the polynomial eigenfunctions. This does not mean there cannot be more eigenfunctions among those expressible by power/Laurent series and beyond.

Proposition 6.23. *Let Γ be an operator marking a single monomial x_I^J with label a. Let λ and u_λ be an eigenpair. Then $\lambda = a$ with $u_\lambda = x_I^J$, or $\lambda = 1$ with u_λ a function not containing x_I^J, in the formal power series sense. In particular, every monomial other than x_I^J is an eigenfunction with eigenvalue 1.*

Proof. Let $u_\lambda = u' + u'' x_I^J$, where $u \neq 0$ and u'' does not share variables with x_I^J. Then

$$\Gamma(u) = u' + a u'' x_I^J = \lambda(u' + u'' x_I^J)\,, \tag{6.43}$$

which we may write as

$$(1-\lambda)u' + (a-\lambda)u'' x_I^J = 0\,. \tag{6.44}$$

We have two cases.

- If $u' = 0$, then $u'' \neq 0$ and $\lambda = a$. The eigenfunction is x_I^J.
- If $u' \neq 0$ and there is no term in $u''x_I^J$ that can compesate for this, then $\lambda = 1$ and $u'' = 0$. □

Corollary 6.24. $\ker \Gamma$ *is generated by the monomials marked with 0.*

6.2.2 Marking Operators With Multiplication

The operators we see in the theory of lattice walk models often feature application on generating functions that are pre-multiplied by a Laurent polynomial, like S. This leads us to definite such combinations of operators in the following way.

Definition 6.25. *We call an operator inhomogeneous if it consists of a homogeneous operator* Γ *and a (possibly infinite) laurent polynomial* S*, and acts in one of the following ways*

$$\Gamma S(u) = \Gamma(Su) \quad \textit{or} \quad S\,\Gamma u = S \cdot \Gamma u\,. \tag{6.45}$$

This kind of operator is important because it can be found in the functional equations for lattice walk models. To find their eigenpairs, we first show that the definitions are mostly equivalent.

Proposition 6.26. *The two operators* ΓS *and* $S\,\Gamma$ *have the same eigenvalues with eigenfunctions differing by a multiple of* S*. That is,* (λ, u) *is an eigenpair for* ΓS *if and only if* (λ, Su) *is an eigenpair for* $S\,\Gamma$*. In particular, if* $\lambda = 0$*, we have*

$$\ker(\Gamma S) = \overline{S}\ker(\Gamma)\,. \tag{6.46}$$

Proof. Note that

$$\Gamma Su = \lambda u \quad \Longleftrightarrow \quad S\,\Gamma(Su) = \lambda(Su)\,. \tag{6.47}$$

Therefore, the operators have the same eigenvalues, and they have the eigenfunctions u and Su respectively. Now, if we pick $\lambda = 0$ to find the kernels of the operators, we find that $u \in \ker(\Gamma S)$ if and only if $Su \in \ker(S\,\Gamma) = \ker(\Gamma)$, and this completes the proof. □

As the next example shows, it may not be easy to find the eigenpairs for such operators.

Example 6.27. Let Γ be a marking operator and let u be an eigenfunction of ΓS, where

$$S = x\,, \quad u = \sum_k u_k x^k\,. \tag{6.48}$$

Computing $(\Gamma S - \lambda I)u$ leads to the system of equations

$$\lambda u_n = a u_{n-1}\,, \tag{6.49}$$

$$\lambda u_k = u_{k-1}, \quad \text{if } k \neq n\,. \tag{6.50}$$

We have a doubly-infinite recursion, so we can pick an arbitrary initial condition. If $u_n = 0$, the rest of the terms will be zero as well, so we must have $u_n \neq 0$. Set $u_n = 1$. Then via induction,

$$u_{n+k} = 1/\lambda^k, \quad \text{for } k \geqslant 0\,, \tag{6.51}$$

$$u_{n-k} = \lambda^k/a^k, \quad \text{for } k \geqslant 1\,, \tag{6.52}$$

and therefore, we have an eigenfunction for each $\lambda \neq 0$. Computing $u(x)$ formally leads to

$$u(x) = -\frac{\lambda b x^{n+1}}{(\lambda - x)(\lambda - ax)}\,, \tag{6.53}$$

and the situation becomes more complicated for more complicated functions S.

6.2.3 Lattice Walks Operator

In this section we explore the operator $\mathcal{L} = 1 - t\Gamma S$ for a step Laurent polynomial S and some marking operator Γ. Let (u, λ) be an eigenpair for $\mathcal{L}$. Then we calculate

$$\begin{aligned}(\mathcal{L} - \lambda I)u &= u - \lambda u - t\Gamma S u \\ t\Gamma S u &= (1-\lambda)u = \mu u\,.\end{aligned} \tag{6.54}$$

If $\lambda = 1$, we would have to compute the kernel of ΓS, which we did in the previous section. The eigenfunctions of $\mathcal{L}$ for eigenvalue $\lambda = 1$ are the functions $u \in \overline{S} \ker \Gamma$. Let u be given by a finite order Laurent series

$$u = u_n t^n + u_{n+1} t^{n+1} + \cdots\,, \tag{6.55}$$

where $u_n \neq 0$. Then we can rewrite the functional equation as

$$t^{-n}(1 - t\Gamma S)u = \lambda t^{-n} u \tag{6.56}$$

$$(\lambda - 1)t^{-n} u = -t^{1-n}\Gamma S u\,. \tag{6.57}$$

Letting $t \to 0$ in this equation yields

$$(\lambda - 1)u_n = -\lim_{t\to 0} t^{1-n}\Gamma S u = -\Gamma S \lim_{t\to 0}(t^{1-n}u) = 0\,. \tag{6.58}$$

Since $u_n \neq 0$, then $\lambda = 1$, and again we see that $u \in \overline{S}\ker\Gamma$. Since S is a Laurent polynomial, the resulting functions will be a finite order Laurent series. The analysis shows that the only eigenfunctions of $\mathcal{L}$ among finite order Laurent series in t are those that are members of $\overline{S}\ker\Gamma$ with eigenvalue $\lambda = 1$.

Example 6.28. Let $\Gamma = LE$ in the one-dimensional problem with small steps on the non-negative line with nontrivial operator E. Then $\ker\Gamma = \ker E$ consists of functions $\alpha(t)\overline{x}$, and the eigenfunctions of $\mathcal{L}$ are $\alpha(t)\overline{x}\overline{S}$ for various $\alpha(t)$. In our case, we only look at finite order Laurent series of $\alpha(t)$. Note that (?)

$$Q = \frac{t(\overline{x} - \overline{\sigma})}{(1 - tS)(b - a\overline{\sigma}t)} = \frac{1}{xS}\frac{\sigma - x}{b\sigma - at}\frac{tS}{1 - tS}\,, \tag{6.59}$$

where we may write

$$\frac{\sigma - x}{b\sigma - at} = \alpha(t) + \beta(t)x\,, \tag{6.60}$$

where $\alpha(t)$ and $\beta(t)$ are algebraic functions of t and a since $\sigma = \sigma(t)$. This observation at least suggests the possibility of connecting the eigenfunctions of $\mathcal{L}$ to the solution Q.

If we are to have a theory similar to Sturm-Liouville theory, we still need to find a suitable inner product on multivariate Laurent series that will let us write solutions for Q in terms of those eigenfunctions with eigenvalue 1.

Chapter 7

Epilogue

[FIXME: Write up some stuff here.]

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